गणितसूत्रम्

GaṇitSūtram

THE GOLDEN EDITION

♦ ♦ ♦

Complete Self-Learning Edition

Stages 1–7 · Fifteen Chapters · The Bridge to Algebra

Jawahar R. Mallah

PrathamOne Press

"Clarity over Speed. Always."

GaṇitSūtram · The Golden Edition · Complete Self-Learning Edition

First published 2024

Author: Jawahar R. Mallah

Publisher: PrathamOne Press

ISBN: 978-93-5593-912-8

Copyright © 2024 Jawahar R. Mallah. All rights reserved.

No part of this publication may be reproduced, stored in a retrieval system, or transmitted in any form or by any means, electronic, mechanical, photocopying, recording, or otherwise, without the prior written permission of the publisher.

Typeset in Palatino Linotype.

Page size: B5 (176 mm × 250 mm).

Printed on 70–80 gsm cream paper with lay-flat binding.

For permissions and educational licensing: PrathamOne Press.

♦ ♦ ♦

For every student who was told they were bad at mathematics.

You were not bad at mathematics.

You were not yet shown what mathematics is.

♦ ♦ ♦

And for the mentors who accompany them —

with patience, with calm, with the knowledge that

your presence is the most powerful teaching tool in the room.

A Note on the Tradition

♦ ♦ ♦

The methods in this book are drawn from the Vedic mathematical tradition as presented by Swami Bharati Krishna Tirthaji (1884–1960) in his work *Vedic Mathematics* (1965). Tirthaji described sixteen sūtras — aphorisms — which he identified as encoding the complete system of Vedic arithmetic.

The historical provenance of these sūtras as ancient Vedic texts is a matter of scholarly discussion. What is not disputed is the mathematical validity and pedagogical power of the methods themselves. The crosswise multiplication method (Ūrdhva-Tiryagbhyām) is equivalent to the standard algebraic expansion of a polynomial product. The Base Method (Nīkhilaṃ) is a direct application of the algebraic identity $(a+p)(a+q) = a(a+p+q)+pq$. These identities are universally valid; the tradition through which they reached us is their biographical context, not their mathematical foundation.

GaṇitSūtram presents these methods within their broader Indian mathematical heritage: Āryabhaṭa (499 CE), Brahmagupta (628 CE), Mahāvīra (9th century), Bhāskara II (1150 CE), and the Sulbasūtra geometers. This heritage is real, documented, and extraordinary. The ambitions of ancient Indian mathematics were as large as the sky.

The sixteen sūtras are used in this book as working tools — named patterns that the mind, once shown, will always recognise. Whether they are ancient or modern, they are useful. That is why they are here.

Table of Contents

♦ ♦ ♦

FRONT MATTER

A Note on the Tradition

How to Use This Book

The Seven Stages Map

The Sixteen Sūtras — Reference Card

PART ONE
THE MIND RESET

Stage One · Chapters 0–2

Ch. 0 The Mind Reset — *Fear, Clarity, and the Learning Conditions*

Ch. 1 Seeing Patterns in Numbers — *The Pattern Engine*

Ch. 2 **Number Sense and Friendly Numbers**

PART TWO
THE LANGUAGE

Stage Three · Chapters 3–5

Ch. 3 The Sixteen Sūtras — *Logic Maps and the Sūtra as Technology*

Ch. 4 The Heritage of Indian Mathematics — *Five World-Changing Contributions*

Ch. 5 The Base Method (Nīkhilaṃ) — *Algebraic Identity and 5-Digit Extension*

PART THREE
THE BASE ENGINE

Stages Three–Four · Chapters 6–9

Ch. 6 The Sub-Base Method (Ānurūpyeṇa) — *4-Digit and 5-Digit Edge Cases*

Ch. 7 Squaring and Special Products — *Geometric Area Model Proofs*

Ch. 8 Two-Digit Crosswise — *Ūrdhva-Tiryagbhyām and Parallel Computing*

Ch. 9 Three-Digit Crosswise, Vinculum, and Verification

PART FOUR

THE UNIVERSAL ENGINE

Stages Five–Six · Chapters 10–14

Ch. 10 Percentages and Discounts — *The Four Problem Types*

Ch. 11 Time, Distance and Speed — *Relative Speed and Average Speed*

Ch. 12 Bills, Budgets and Real-World Arithmetic

Ch. 13 Word Problems — *The Four-Step Method: SEE–UNDERSTAND–SOLVE–VERIFY*

Ch. 14 The Bridge to Algebra — *Calanā-Kalanābhyām*

PART FIVE

THE MASTERY CELEBRATION

Stage Seven · Chapter 15

Ch. 15 The Mastery Drill — 100 Questions

The 30-Day Progress Journal

The Forever Tools — Ten Sūtras as Portable Intelligence

The Beauty Index

Completion Certificate

Author's Closing Message

APPENDICES

App. A The Sixteen Sūtras — Full Reference with Sanskrit

App. B The Mentor's Diagnostic Compass — Error→Page Index

App. C Complete Solutions to the Mastery Drill

How to Use This Book

♦ ♦ ♦

For the Self-Learning Student

Begin at Chapter Zero. Do not skip it. Chapter Zero is not a preamble — it is the first lesson, and in many ways the most important one. Read it slowly. Do the Stillness Protocol. Then proceed to Chapter One.

Work through the chapters in order. The book is designed as a sequence: each chapter builds on what came before. If a chapter feels too difficult, return to the previous one and complete its Mastery Lab Level 5 before proceeding.

Verify every answer. The habit of verification — checking your work before moving on — is not a precaution. It is the practice that separates fluency from guessing. Every chapter provides the Casting Out Nines method as a verification tool from Chapter Nine onward.

Use the Mastery Labs honestly. If you reach Level 3 and cannot complete it confidently, do not move to Level 4. Return to the worked examples, identify the gap, and fill it. Speed comes after clarity, not before.

For the Mentor, Parent, or Teacher

Read the Mentor's Sanctuary at the beginning of each chapter before your student begins that chapter. The Sanctuary describes the emotional landscape the student will encounter, the most common errors, and the diagnostic approach.

> *Your calm is the most powerful teaching tool in this room. A student who is accompanied by a patient, calm adult will attempt problems that a frightened student will abandon. The mathematics is secondary to the presence.*

Allow the student to struggle productively. The Mastery Lab levels are calibrated to become progressively harder. Level 5 is designed to challenge even a confident student. Do not

intervene before the student has genuinely attempted the problem. Do intervene if the student shows signs of mathematical anxiety — return to Chapter Zero's protocol.

Use the Diagnostic Flowcharts in Appendix B when errors appear. Most errors in this book fall into a small number of identifiable patterns, each with a specific remediation.

The Seven Stages

❖ **Stage One (Chapters 0–2)**: The foundation. Pattern recognition, number sense, and the mental conditions for mathematical clarity.

❖ **Stage Two** (integrated in Chs. 0–2): Addition, subtraction, friendly numbers, complements. The arithmetic that supports every method that follows.

❖ **Stage Three (Chapters 3–7)**: The core Vedic methods. Sūtras, heritage, Base Method, Sub-Base Method, Squaring.

❖ **Stage Four (Chapters 8–9)**: The crosswise method — the most general multiplication technique in the book. Two-digit and three-digit, with verification.

❖ **Stage Five (Chapters 10–12)**: Real-life mathematics. Percentages, distance-speed-time, bills, budgets.

❖ **Stage Six (Chapters 13–14)**: Word problems and the Bridge to Algebra. The Four-Step Method. Every technique unmasked as an algebraic identity.

❖ **Stage Seven (Chapter 15)**: The Mastery Drill and 30-Day Journal. The celebration of completion.

The Seven Stages Map

♦ ♦ ♦

★ 1 Stage

The Foundation

Chapters 0–2 · Pattern recognition · Number sense · The mental conditions for clarity

Building the eye that sees patterns before the hand calculates.

★ 2 Stage

Number Fluency

Integrated in Chs. 0–2 · Addition · Subtraction · Friendly numbers · Complements

The arithmetic that supports every method that follows.

★ 3 Stage

The Core Vedic Methods

Chapters 3–7 · The sixteen sūtras · Heritage · Base Method · Sub-Base · Squaring

Where the ancient patterns are named and their identities seen.

★ 4 Stage

Crosswise Mastery

Chapters 8–9 · Ūrdhva-Tiryagbhyām · Two- and three-digit · Vinculum · Verification

The most general multiplication method — parallel by design.

★ 5 Stage

Real-Life Mathematics

Chapters 10–12 · Percentages · Distance-Speed-Time · Bills and Budgets

Where arithmetic meets the world.

★ 6

The Bridge

Chapters 13–14 · Word problems · The Four-Step Method · Algebra unmasked

Stage

Every technique is an algebraic identity. The bridge is crossed.

★

7

Stage

Mastery and Excellence

Chapter 15 · The Mastery Drill · The 30-Day Journal · The Forever Tools

One hundred questions. One clear mind. That is all that is required.

The Sixteen Sūtras

Reference Card

♦ ♦ ♦

1.	**Ekādhikena Pūrveṇa** — *"By one more than the previous one"* · *Squares ending in 5. Sequences with constant differences.*
2.	**Nīkhilaṃ Navataścaramaṃ Daśataḥ** — *"All from 9, last from 10"* · *Complement calculation. Base Method. Vinculum. Any subtraction from a power of ten.*
3.	**Ūrdhva-Tiryagbhyām** — *"Vertically and crosswise"* · *General multiplication — two- and three-digit. The universal algorithm.*
4.	**Parāvartya Yojayet** — *"Transpose and apply"* · *Division. Transposing equations. D–S–T formula.*
5.	**Shūnyam Sāmya Samuccaye** — *"When the sum is the same, that sum is zero"* · *Equations where each side has the same sum.*
6.	**Ānurūpye Shūnyamanyat** — *"If one is in ratio, the other is zero"* · *Sub-Base Method. Percentage scaling. Proportional adjustment.*
7.	**Saṅkalāna-Vyavakalānabyām** — *"By addition and by subtraction"* · *Simultaneous equations. Sums and differences.*
8.	**Purānāpūraṇābhyām** — *"By the completion or non-completion"* · *Completing to a round number. Algebraic completion of the square.*
9.	**Calanā-Kalanābhyām** — *"By sequential motion; by differential calculus"* · *Successive differences. Identifying polynomial degree.*
10.	**Yāvadūnam** — *"Whatever the extent of the deficiency"* · *Squaring near a base. Higher powers near a base.*
11.	**Vyṣṭisamasamastihi** — *"Part and whole"* · *Factorisation. Sum and product relationships.*
12.	**Sheshaṇyankaṃ Caramena** — *"Remainders by the last digit"* · *Casting Out Nines. Verification by digit sums.*
13.	**Sopāntyadvayamantyam** — *"The ultimate and twice the penultimate"* · *Partial fractions. Arithmetic series.*
14.	**Ekānyūnena Pūrveṇa** — *"By one less than the previous one"* · *Multiplication by 9, 99, 999. Special products.*

15. **Guṇita Samuccayaḥ** — *"The product of the sum equals the sum of the products"* · *Verification by factor-sum method.*

16. **Guṇakasamuccayaḥ** — *"The factors of the sum equal the sum of the factors"* · *Advanced verification. Polynomial roots.*

IMPORTANT NOTE ON ORDERING

The sūtras above are numbered in the order they appear in Chapter Three of this book, which reflects their **pedagogical sequence** — the order in which they are introduced and built upon. This differs slightly from their listing in some other presentations of Vedic Mathematics. The mathematical content of each sūtra is identical regardless of numbering. See Chapter Three for the complete Logic Maps.

✶ TRADITION PROLOGUE ✶

The Lineage This Book Enters

"Gaṇita means reckoning. But it also means that which has been meditated upon."

— Root: gaṇ — to count, to reckon, to know

You are not simply learning arithmetic.

You are entering a tradition of mathematical observation that spans fifteen centuries and produced some of the most consequential ideas in the history of human thought. The scholars who built this tradition did not work in isolation — they worked in lineage. Each one received what the previous generation had discovered, refined it through their own practice, extended it through their own observation, and passed it forward.

The sūtras in this book are not shortcuts invented for competitive examinations. They are compressed descriptions of how numbers actually behave — principles that were present in every calculation long before they were named. What the lineage below contributed was the naming: the act of making visible what had always been true, so that the next generation could see it too.

When you open this book, you join that lineage. Not as a passive recipient, but as its newest practitioner.

I. The Gurukula System — How Knowledge Survived

The Gurukula system — from *guru* (one who dispels darkness) and *kula* (household, family) — was not an institution in the modern sense. It was a relationship. A student did not attend a Gurukula to receive information. They went to live in proximity with understanding.

The guru did not deliver lectures. The guru demonstrated, by the quality of their own thinking, what a trained mathematical mind looked like in action. The student's task was first to watch, then to imitate, then to diverge and discover. This pedagogical model has a

direct neurological basis: mirror neurons activate both when we perform an action and when we observe another performing it. Watching a master calculate is not passive. It is active neural mirroring.

The curriculum was oral and observational before it was written. A student might spend three years observing a single technique before being invited to attempt it. The assumption underlying this patience was radical: understanding cannot be rushed into existence, only created through sufficient time and attention.

What the Gurukula Model Taught

- Understanding before technique
- Observation before calculation
- Patience as an intellectual virtue
- Error as information, not shame
- Mastery as transmission, not possession

What the Modern Classroom Often Forgets

Procedure is taught before pattern is seen

Speed is rewarded before accuracy is consolidated

Anxiety is treated as character, not biology

The wrong answer ends the enquiry

Content is delivered, not experienced

II. The Four Scholars This Book Stands Upon

The lineage from which GaṇitSūtram draws runs through four figures, each contributing a distinct dimension to the mathematical inheritance this book teaches. They are presented not as historical footnotes but as living voices — each one represents a quality of mind this book asks you to bring to your own study.

Āryabhaṭa · *476–550 CE* · *The Astronomer-Mathematician*

Āryabhaṭa was twenty-three years old when he wrote the Āryabhaṭīyā — a text so dense with original observation that scholars are still parsing its implications fifteen centuries later. He calculated the length of the solar year to 365.2586 days; modern measurement gives 365.2422. He described the rotation of the Earth on its own axis eleven centuries before Copernicus. He calculated π to four decimal places and had the rare intellectual honesty to call the result āsanna — approximate, close enough.

His most consequential contribution for this book was conceptual: he worked with quantities as deviations from reference values. This is the intellectual ancestor of the Base Method — the insight that computation becomes simpler when you work with the distance to a landmark rather than the absolute value.

Brahmagupta · *598–668 CE* · *The Algebraist of Zero*

Brahmagupta did something extraordinary: he treated zero as a number. Before him, zero was a placeholder. Brahmagupta made zero an entity that could be added, subtracted, and multiplied. He established formal rules for negative numbers, which he called *debts*, against positive numbers, which he called *fortunes*.

This matters for this book because zero is not an absence in Vedic mathematics — it is a reference point. When we say a number is 'near base 100,' we mean the gap from that number to 100. The gap might be positive or negative. Brahmagupta gave us the arithmetic of both. His framework is the silent structure beneath every signed-gap calculation in Stage Three and beyond.

Bhāskara II · *1114–1185 CE* · *The Pedagogue of Beauty*

Bhāskara II named his mathematics text *Līlāvatī — Playful*. He addressed it to a young woman, speaking to her directly in the text, asking her to consider this problem or observe that pattern. He was, as far as the historical record shows, the first mathematician on record to believe that mathematical beauty is a pedagogical instrument — not a decoration at the end of a proof, but the mechanism by which understanding enters and stays.

His geometric proofs demonstrated ideas by diagram rather than algebraic symbol. The most famous is his proof of the Pythagorean theorem, presented with a single word: *"Behōld!"* — look at this, see what it means, no further words are needed. This visual proof

tradition anticipates the Area Model diagrams used in Part Five of this book by eight centuries.

Bharati Krishna Tirthaji · *1884–1960 CE* · *The Systematiser*

Tirthaji spent eight years in meditative study of Sanskrit texts before emerging with sixteen mathematical principles he called sūtras — threads. He worked as both scholar and practitioner, developing each sūtra through a combination of textual study and direct calculation, refining until the formulation was as compressed as possible without losing precision.

His single published volume, *Vedic Mathematics*, was delivered to press in 1965, five years after his death, from a manuscript he had rewritten entirely from memory after the original was lost in transit. That act of reconstruction is significant: genuine mathematical understanding is not a set of notes that can be lost. It is a structure of insight that can be reproduced from first principles, because it was built from first principles the first time.

A Note on Scholarly Honesty

The precise Vedic textual origins of Tirthaji's sixteen sūtras have been the subject of serious scholarly debate. Historian K.S. Shukla and others have noted that several sūtra formulations do not appear verbatim in ancient Vedic texts in the form Tirthaji presented them. This is a matter the tradition has not resolved, and this book does not pretend otherwise.

What is not in dispute: the mathematical patterns these sūtras describe are ancient, real, and algebraically verifiable. The base method corresponds to the identity $(a+p)(a+q) = a(a+p+q) + pq$, confirmed independently in multiple mathematical traditions. Crosswise multiplication is structurally identical to polynomial multiplication.

> **GaṇitSūtram honours the tradition, teaches the mathematics, and maintains intellectual honesty. These three commitments are not in conflict.**

III. The Sūtra as a Technology of Compression

The word *sūtra* comes from the same root as the English word *suture*: something that stitches together. A sūtra is a formulation so compressed that every word is load-bearing. Remove a single word and the meaning changes. Add a word and the elegance is lost.

Compare this to the way mathematics is typically presented in textbooks: as a sequence of steps, each step described in full. This format is useful for initial instruction, but it is not how mathematical knowledge is stored in an expert mind. An expert does not remember a sequence of steps. They remember a principle — a sūtra — from which the steps can be reconstructed in any context, including novel contexts that were never explicitly practised.

This is why the sūtras in Chapter Three are introduced *before* the techniques that derive from them, not after. The sūtra is the cognitive scaffolding onto which the technique is hung. When the scaffolding is in place first, each technique has somewhere to live in memory — not as an isolated procedure to be recalled, but as an expression of a principle already understood.

DIAGRAM 01

The Lineage Map

Āryabhaṭa (Gap Logic) → Brahmagupta (Signed Arithmetic) → Bhāskara (Visual Proofs) → Tirthaji (Sūtras) → GaṇitSūtram

IV. What This Lineage Asks of You

Every tradition makes demands of its students. This one makes three.

The first demand is patience. Not the passive patience of waiting for someone to explain, but the active patience of the Gurukula student who watches the master calculate for three years before attempting the first problem. Patience here means: do not move to the next level

before the current level is genuinely clear. *Clarity over Speed* is not a slogan. It is the first principle of the tradition this book stands inside.

The second demand is honesty. Tirthaji rewrote his manuscript from memory because he had understood it from the inside. Most students read mathematics from the outside — following steps without seeing the principle behind them. When you encounter a technique that you can execute but not explain, the tradition asks you to stop. To ask: what is actually happening here? Why does this work?

The third demand is engagement with beauty. Bhāskara called his text Playful because he believed that mathematical insight carries a quality that can only be called aesthetic: the moment when a pattern that was hidden becomes visible, when a calculation that seemed laborious becomes trivial, when you see that two things you thought were different are the same thing in different clothes. This book will produce those moments for you. When they arrive, attend to them. They are not decorations. They are the thing itself.

♦ ♦ ♦

★ STAGE ONE ★

The Mind Reset

Releasing mathematical fear through neuroscience, stillness, and the architecture of the learning mind

"Fear does not come from numbers. It comes from pressure. And pressure can be released."

— *GaṇitSūtram · Stage One*

This chapter contains no Vedic techniques.

What this chapter contains is more valuable than any of those things: a complete account of why mathematical anxiety exists in the human brain, how it works, and — most importantly — what can be done about it before a single calculation begins.

If you are reading this as a student who has struggled with mathematics: this chapter is for you. Every statement in it is grounded in neuroscience research. Your experience is not a character flaw. It is a biological response that is fully understandable and entirely modifiable.

If you are reading this as a parent or teacher: everything in this chapter has direct practical application. The Mentor's Sanctuary at the end provides specific language and specific interventions. Please read both.

1. Mathematics and the Brain: The Two Systems

The human brain does not process mathematical information in a single, unified system. It processes it through at least two distinct neural systems that are, in important ways, in competition with each other. Understanding this competition is the first step toward resolving it.

The system responsible for mathematical reasoning lives primarily in the *prefrontal cortex* — the large, folded region at the front of the brain. The prefrontal cortex is the seat of working memory, executive function, logical reasoning, and deliberate thought. It is the 'thinking brain' in the most direct sense.

The system responsible for threat detection lives in the *amygdala* — a small, almond-shaped structure deep in the temporal lobe. The amygdala monitors every incoming experience for signs of danger. It is extraordinarily fast — faster than conscious perception — and it does not distinguish between physical threats and social or cognitive ones. A tiger on the path and a mathematics test on the desk produce recognisably similar responses in the amygdala.

DIAGRAM 02

The Two Brain Systems

Prefrontal Cortex: The Logic Centre / Working Memory. Amygdala: The Threat Centre / Emergency Override. Arrows showing inhibitory and calming signals between them.

1.1 The Prefrontal Cortex: Where Mathematics Lives

Working memory — the brain's short-term workspace — is primarily a function of the PFC. When you hold the intermediate steps of a calculation in mind while performing the next step, you are using working memory. When you read a word problem and simultaneously track what each number represents, you are using working memory.

Working Memory

The brain's short-term computational workspace — the place where multi-step reasoning is held and processed. Typically holds 4–7 items. Anxiety reduces effective capacity by occupying resources with threat-monitoring rather than mathematics.

Working memory capacity is not fixed — it varies significantly within the same individual depending on cognitive and emotional state. This is the critical point: the PFC is a conditional resource. When conditions are compromised — by stress, threat perception, fatigue, or anxiety — its capacity reduces, sometimes dramatically.

1.2 The Amygdala: The Fear Centre

The amygdala is one of the oldest structures in the vertebrate brain. When it detects a threat signal, it triggers the stress response — releasing cortisol and adrenaline, directing blood flow toward large muscle groups for fight or flight, and simultaneously *inhibiting the prefrontal cortex*.

That inhibition is the mechanism this book is designed to work around. When the amygdala fires, it does not send a polite message to the PFC asking it to stand down. It issues a physiological override. The PFC, starved of blood flow and flooded with stress hormones, cannot perform at full capacity. The precise, multi-step processing required for mathematics becomes, in biological terms, unavailable.

> *This is not metaphor. It is not a figure of speech. It is what happens in the brain of a student who is afraid of mathematics when they are asked to calculate under pressure. The mechanism that produces incorrect answers and blank minds and frozen pens is not stupidity or laziness. It is a stress response that is doing exactly what it evolved to do.*

▣ DIAGRAM 03 ▣

The Amygdala Hijack During Mathematical Stress

Timeline: Threat signal received → Amygdala fires, cortisol released → Blood flow diverted from PFC → Working memory drops from ~7 items to ~2–3 → Multi-step calculation fails.

1.3 The Competition: Why They Cannot Both Win

The PFC and the amygdala are, in important ways, in active competition for the same neural resources. When one is dominant, the other is suppressed. In situations of genuine physical danger, full PFC function is not needed. But in situations of mathematical learning, amygdala dominance is catastrophically counterproductive.

The Research Finding That Changes Everything

Lyons and Beilock (2012, University of Chicago) found that high-maths-anxiety individuals showed activation of neural pain pathways when *anticipating* mathematics — not during it, but before it. The pain response preceded the task.

But the same study found something equally important: the pain response was significantly reduced when students used expressive writing to articulate their anxiety before the test. The act of naming and writing out the fear reduced its neural intensity. The amygdala is calmed by acknowledgement, not suppression.

This is the neurological basis for every Reflection exercise in this book. They are not optional self-help. They are calibrated amygdala management tools.

2. Mathematical Anxiety: A Clinical Picture

Mathematical anxiety is formally defined in cognitive psychology as a feeling of tension, apprehension, or fear that interferes with mathematics performance. It is distinct from general anxiety, and it is distinct from poor mathematical ability. Research consistently shows that mathematically anxious students often have the same abstract reasoning capacity as non-anxious students — what differs is their access to that capacity under the conditions in which mathematics is typically taught and assessed.

The prevalence is significant. Meta-analyses of school-age populations across India, the UK, the US, and East Asia suggest that between 25 and 40 percent of students experience mathematics anxiety at a level that measurably impairs their performance. In competitive examination contexts, the proportion rises further.

The developmental trajectory is consistent. Mathematical anxiety typically begins between ages seven and nine, coinciding with the introduction of timed arithmetic assessments and the first exposure to the social dimension of mathematical performance. Once established, it is self-reinforcing: anxiety reduces performance, reduced performance confirms the belief that mathematics is threatening, which deepens the anxiety.

2.1 How Mathematical Anxiety Is Acquired

Mathematical anxiety is *learned, not inherited.* The most common acquisition pathway is the timed test. When a student is placed under time pressure before they have achieved procedural automaticity, working memory is divided between the mathematics and the monitoring of time. The result is degraded performance. When degraded performance is then treated as evidence of mathematical inadequacy rather than as a predictable response to premature pressure, the student internalises a false identity claim: *I am bad at mathematics.*

A second pathway is social observation. Children who watch a parent express anxiety about mathematics, or who watch a teacher respond to mathematical error with frustration, acquire their anxiety by the same mirror-neuron mechanism that underpins all social learning.

A third pathway is the single traumatic classroom experience — the public correction, the laughter at an error, the teacher's visible exasperation. The amygdala encodes high-emotion experiences with unusual durability. A single episode of mathematical humiliation can establish an avoidance response that persists for decades.

▣ DIAGRAM 04 ▣

The Mathematical Anxiety Acquisition Cycle

Circular flowchart: Timed pressure → Working memory divides → Performance degrades → Negative feedback → Amygdala encodes maths as threat → Future expectation triggers fear → Cycle continues. Break-out arrows at three intervention points.

2.2 The Three Presentations of Mathematical Anxiety

Presentation 1: Avoidance

The student minimises contact with mathematics — skipping problems, leaving assessments incomplete, choosing routes through schooling that reduce mathematical exposure. Avoidance feels like relief but deepens the anxiety by preventing the corrective experiences of successful calculation.

Presentation 2: Rigidity

The student insists on a single method and becomes panicked or resistant when asked to approach a problem differently. Rigidity is an anxiety-management strategy: if there is only one acceptable approach, the number of ways to fail is minimised. But rigidity also prevents the pattern-recognition and method-selection skills that mathematical fluency requires.

Presentation 3: Catastrophisation

The student predicts total failure before beginning. "I can't do this." "I'll never understand this." These statements are not accurate assessments of ability — they are amygdala outputs. The amygdala does not make nuanced predictions. It produces threat-maximising assessments because maximum threat assessment is the safest evolutionary strategy when genuine danger is possible.

2.3 What Mathematical Anxiety Is NOT

Mathematical anxiety is not a sign of low intelligence. The correlation between mathematical anxiety and general intelligence, across multiple large studies, is approximately zero.

Mathematical anxiety is not a genetic fate. While temperament and general anxiety predisposition have some heritable component, mathematical anxiety specifically is acquired through experience and is therefore modifiable through experience.

Mathematical anxiety is not something to 'push through.' Encouragement to 'just try' when the amygdala is dominant does not increase PFC function — it often increases the threat signal by adding social pressure to mathematical pressure.

Mathematical anxiety is not the same as disliking mathematics. A student who has never had mathematics taught in a way that allowed genuine understanding may simply never have experienced mathematics as likeable. That is a pedagogical failure, not an inherent fact about the student.

3. Stillness as Biology — The Neurological Case

Stillness, as used here, refers to a specific neurological state: the state in which amygdala activity is low enough that the prefrontal cortex can operate at near-full working-memory capacity. This state is not mystical, and it is not passive. It is a biological condition that can be deliberately created through specific, learnable practices.

DIAGRAM 05

The Physiology of Stillness

Side-by-side: Amygdala-dominant state (PFC blood flow reduced, working memory 2–3 items) vs. Stillness state (PFC blood flow restored, working memory 5–7 items). Arrow: "60 seconds of deliberate slow breathing is sufficient to begin the transition."

3.1 The Neuroscience of the Breathing Protocol

Slow diaphragmatic breathing activates the parasympathetic branch of the autonomic nervous system. The mechanism is vagal: slow exhalation stimulates the vagus nerve, which carries parasympathetic signals from the brainstem to the heart, slowing heart rate and initiating a cascade of physiological changes that include reduced cortisol production and restored prefrontal blood flow.

A breathing rate of 4–6 breaths per minute produces measurable reductions in physiological stress markers within 60–120 seconds. This is not a long time. It is the cost, in seconds, of giving the brain the conditions it needs to do mathematics.

THE STILLNESS PROTOCOL — 60 Seconds Before Every Study Session

Step 1 Sit in an upright position. Place both feet flat on the floor.

Step 2 Close your eyes, or soften your gaze to a point approximately one metre ahead.

Step 3 Inhale through the nose for a count of four. Feel the abdomen expand, not the chest.

Step 4 Hold gently for a count of two.

Step 5 Exhale slowly through the nose for a count of six. Let the abdomen fall.

Step 6 Repeat five times. This takes approximately 60 seconds.

Step 7 Open your eyes. Notice that the room is the same as it was. The only thing that has changed is the state of your brain.

3.2 Attentional Focus and the Default Mode Network

When the mind is not deliberately focused on a specific task, it activates the default mode network (DMN) — a set of brain regions associated with self-referential thought and the rehearsal of past experiences and future concerns. For mathematically anxious students, the DMN in a mathematics context is frequently occupied with threat-related content: memories of past failures, anticipations of future ones.

Deliberate attentional focus — directing attention to a specific, neutral anchor before beginning mathematics — reduces DMN activity and increases activity in task-relevant networks. In Gurukula terms, this practice has a name: *dhāraṇā* — the holding of attention on a single point. It precedes study in every classical account of the Gurukula curriculum. The modern neuroscience confirms what ancient pedagogy discovered empirically: a directed, anchored mind learns more than a distracted, anxious one.

3.3 Expressive Writing and the Amygdala Reduction Effect

The Lyons and Beilock study identified a specific intervention that reduced mathematical anxiety responses in the brain: expressive writing before a mathematical task. Students who spent ten minutes writing about their anxieties before a mathematics test showed lower activation of pain-related neural regions and higher mathematics performance than control groups.

The mechanism: articulating an emotional state reduces its neural intensity. The amygdala's threat signal is partly self-sustaining through its resistance to verbal labelling — fear that is unnamed is fear that cannot be evaluated, contextualised, or moderated. When a student writes 'I am afraid I will fail this,' they are performing a naming operation that the PFC can process. This is the neurological basis for the Reflection Pages at the end of every chapter in this book.

4. The Learning Window: When the Brain Receives

The three primary conditions for the learning window are: moderate positive affect (the student is engaged and comfortable, not bored and not anxious), active processing (the student is constructing understanding rather than passively receiving information), and spaced repetition. All three are engineered into the structure of this book.

DIAGRAM 06

The Dopamine Learning Curve

Three curves: (A) Understanding-first — begins lower but rises steeply after insight moment; (B) Procedure-first drilling — rises quickly but plateaus and decays; (C) Anxious drilling — minimal acquisition, high decay. Annotation: "The insight moment produces a dopamine encoding event."

4.1 The Role of Dopamine in Mathematical Learning

Dopamine is widely misunderstood as a 'pleasure chemical.' Its actual function is more specific: it is a prediction-error signal. Dopamine neurons fire when an outcome is *better than predicted.* They are suppressed when an outcome is worse than predicted.

This means that dopamine is most active in the exact condition that good mathematics teaching deliberately creates: the moment when a student realises that a pattern they thought was complex is actually simple, or that a technique they thought was isolated is actually an instance of a general principle they have already learned. The moment of insight — the 'Behōld!' moment Bhāskara knew so well — produces a dopamine response that encodes the insight durably.

Routine drilling of known procedures produces minimal dopamine response and therefore minimal memory encoding. This is one of the reasons why repeated practice of a misunderstood technique fails to produce genuine learning: the student is not encountering prediction errors, so dopamine is not marking the material for long-term retention.

4.2 Working Memory, Chunking, and the Sūtra

Working memory capacity is limited — approximately four to seven items for most adults in optimal conditions, significantly fewer under stress. This creates an apparent problem for mathematics: many procedures require holding more information in mind than working memory can typically accommodate.

The solution the brain evolved is chunking: grouping multiple items of information into a single, named unit that occupies only one working-memory slot. A chess grandmaster looking at a board does not see 32 individual pieces — they see three or four named positional patterns.

> *The sūtra is a chunking device. When a student has genuinely internalised a sūtra, it occupies one working-memory slot and encodes the complete logical structure of the calculation. A student who has only memorised the procedure must hold each step separately, exhausting working memory before the calculation is complete. The sūtra is not decorative heritage. It is a working-memory optimisation tool that is fifteen centuries old.*

5. The Fear/Clarity Map — Your Starting Point

Before proceeding to any calculation in this book, every student is asked to complete the Fear/Clarity Map. This is not a test. It is a diagnostic tool designed to create an accurate picture of your current emotional relationship with different areas of mathematics.

The Map serves two functions. First, it gives you specific language for what has been vague: rather than 'I am bad at mathematics,' you will be able to say 'I am anxious about multiplication but confident about addition.' Specificity is itself a therapeutic act.

Second, it creates a baseline against which to measure your progress. By the time you reach Chapter Fifteen, you will have a record of where you started. The distance between starting point and current position is the actual measure of what this book has accomplished for you.

▣ DIAGRAM 07 ▣

The Fear/Clarity Map (Student Worksheet)

Two-axis grid. X-axis: FEAR (left) to CLARITY (right). Y-axis: 12 mathematical topic areas. Student places a mark on the spectrum for each. Space for date and a one-line note per row.

Instructions for the Fear/Clarity Map

1. Work in silence. Do not consult anyone or refer to past grades. This is your current felt experience, not an objective assessment.
2. For each topic, mark where you honestly are right now, not where you think you should be.
3. Notice any physical sensations that arise as you mark certain topics. Write them in the notes column.
4. There is no correct answer. The student who marks every topic at maximum fear has completed the exercise as correctly as the student who marks maximum clarity.
5. Date and sign the Map. Keep it. You will return to it at the end of Stage Seven.

6. The Speed Illusion — Why Rushing Is the Enemy

Mathematical education has, for most of the twentieth century and the first decades of the twenty-first, been structured around speed. Timed tests, mental arithmetic races, and competitive classroom formats all prioritise the student who reaches the answer first. The implicit message is that speed equals ability. This equation is **false**, and its falseness has measurable consequences.

Speed is a product of automaticity, and automaticity is a product of genuine understanding consolidated through repetition. A student who understands the base method will, after sufficient practice, execute it faster than a student who has only memorised the procedure — because the student with genuine understanding can reconstruct any step they have forgotten. Fast mathematicians are not born fast; they became fast after becoming accurate, which they became after achieving genuine understanding.

THE SEQUENCE THAT PRODUCES LASTING MATHEMATICAL ABILITY

1. Understanding → The pattern is seen. Why this works is clear.

2. Accuracy → The procedure is executed correctly, without rushing.

3. Fluency → Through repetition, correct execution becomes effortless.

4. Speed → Speed arrives automatically once fluency is secure. It cannot be installed before then.

"Clarity over Speed" is not a preference. It is the only sequence that actually works.

7. The Four Conditions of Mathematical Readiness

These are not prerequisites that must be met before learning can begin. They are ongoing states that this book is designed to help you cultivate as the learning progresses.

Condition One: Neurological Safety

The amygdala signal is low enough that the prefrontal cortex can operate at near-full capacity. This is achieved, and maintained, through the Stillness Protocol, the absence of time pressure during initial learning, and the presence of a mentor whose calm is genuine.

Condition Two: Genuine Curiosity

The student is attending to the pattern, not just the procedure. They are asking 'why does this work?' alongside 'how do I do this?' Curiosity is not a personality trait — it is a practice that can be cultivated deliberately by consistently asking the why-question.

Condition Three: Permission to Be Uncertain

The student knows that not knowing is not failure. A student who has been given explicit permission to not know, to ask, to be wrong, and to try again will use their working memory for mathematics. A student without that permission will use their working memory for threat-monitoring.

Condition Four: Belief in Modifiability

Carol Dweck's research on growth mindset demonstrates that students who believe their ability is fixed respond to difficulty with avoidance. Students who believe their ability is modifiable respond to difficulty with increased persistence. This belief is not innate. It is taught.

♦ ♦ ♦

Chapter Zero · Mastery Lab

Level 1 · Exercise 1 — The Stillness Protocol

Perform the 60-second Stillness Protocol described in Section 3. Before you begin, write one sentence describing what is happening in your mind right now. After you complete it, write one sentence describing what has changed.

Before:

_____

_____

After:

_____

_____

Level 1 · Exercise 2 — The Fear/Clarity Map

Complete the Fear/Clarity Map (Diagram 07). Date it. Sign it. Fold it and place it between the pages at the end of this chapter. You will retrieve it in Stage Seven.

Level 1 · Exercise 3 — The Amygdala Interview

Write for five minutes — without stopping, without editing — about your earliest memory involving mathematics. Do not evaluate the memory. Simply describe it: what happened, who was there, what was said, what you felt.

When you have finished writing, read what you wrote once, then set it aside. You have just named something that the amygdala had been holding unnamed. That naming is the first act of this book.

Level 1 · Exercise 4 — The Letter to Your Future Self

Write a short letter (minimum one paragraph, maximum one page) addressed to yourself at the end of Stage Seven — after the 100-Question Mastery Drill. Tell them what you are afraid of right now. Tell them what you hope will be different. Sign it.

Fold this letter and place it with the Fear/Clarity Map. You will open it in Stage Seven. This letter will be the most accurate measure of your journey that this book contains.

Level 2 · Identification — Recognising the Amygdala Signal

For the following five scenarios, identify whether the described response is primarily amygdala-driven (threat response) or prefrontal-driven (reasoning response). Explain your identification in one sentence.

Scenario A: *A student is told they have 30 seconds to answer a multiplication problem in front of the class. Their mind goes blank. They cannot recall a multiplication table they have known for two years.*

My identification:

Scenario B: *A student reads a word problem twice, identifies the relationship type as a rate problem, draws a simple diagram, and begins working through the calculation step by step.*

My identification:

Scenario C: *A student receives a test paper. Before reading the first question, they say: 'I always fail these. I don't know why I bother.'*

My identification:

Scenario D: *A student gets an incorrect answer. They pause, look back at their working, identify the step where the error occurred, and redo from that step.*

My identification:

Scenario E: *A student refuses to attempt a chapter that involves fractions because 'fractions are impossible' even though they have not read the chapter.*

My identification:

Level 3 · Application — Pattern Observation

Look at the following number pairs. For each pair, simply observe and describe what you notice. Do not calculate. Do not apply any technique. This exercise is purely observational.

98 × 97 · 99 × 96 · 95 × 95

I notice:

103 × 107 · 102 × 108 · 105 × 105

I notice:

You have just used the prefrontal cortex in the way this book intends. The observations you made are the seeds of the pattern-recognition that will make every technique in Stage Three legible. The specific pattern you noticed will be named and formalised in Chapters Three through Seven. For now, simply notice that noticing is itself a form of mathematical work.

Level 4 · Reflection — What Has This Chapter Changed?

Write three sentences in response to each of the following prompts.

1. **Before reading this chapter, I thought mathematical anxiety was…**

2. **Now I understand that mathematical anxiety is actually…**

3. The specific thing I will do differently before every study session is…

4. The person in my life who would benefit most from reading Section 2 of this chapter is… because…

♦ ♦ ♦

MENTOR'S SANCTUARY

For the Parent, Teacher, or Mentor · Chapter Zero

Chapter Zero asks your student to examine their emotional relationship with mathematics before picking up a pencil. This is not a delay — it is the most important preparation available. Every hour spent on Chapter Zero saves two in every subsequent chapter.

validation, not reassurance. Rather than 'don't worry, you'll be fine,' try: 'that sounds really frustrating — let's look at it slowly together.' Validation acknowledges the experience without dismissing it, which opens the mind to new information.

Your calm is the most powerful teaching tool in this room. If you are anxious about mathematics, your student will mirror that anxiety before you say a word. Model the Stillness Protocol yourself, in your student's presence, before beginning any session.

❖ When a child says "I can't do this": hear it as an amygdala output, not a capability assessment. Respond to the emotion first, then the mathematics.

❖ When a child makes an error: ask them to say aloud what they were thinking at each step. This diagnostic approach locates the error without evaluating the child. The error is information. It is not a verdict.

❖ The Stillness Protocol takes 60 seconds. It is not optional. Begin every session with it, regardless of how little time you have. A calm 30-minute session produces more durable learning than an anxious 90-minute one.

❖ Level 5 of every Mastery Lab is intentionally difficult. When a student reaches it and struggles, that is the correct response to Level 5. Do not infer inadequacy. Infer engagement.

"Clarity over Speed. Always."

❤ *Clarity over Speed. Always.* ❤

GaṇitSūtram · Golden Edition · PrathamOne Press

STAGE ONE · THE FOUNDATION · CHAPTER ONE

Chapter One

Seeing Patterns in Numbers

The Pattern Engine · SEE → UNDERSTAND → SOLVE

"Mathematics is not a subject. It is a way of seeing.
And seeing is a trainable skill." — GaṇitSūtram · Stage One

Āryabhaṭa is twenty-three years old. The year is 499 CE. He is sitting in Kusumapura — the city we now call Patna — and he is writing 108 verses that will encode the mathematics of the cosmos in a

form that can be memorised, recited, and sung. The verses cover planetary motion, trigonometry, and the value of π to four decimal places. But the first verse is not a calculation. It is not even a definition. It is an act of orientation: a statement of where the mathematician stands before the mathematics begins.

What Āryabhaṭa understood — and what this chapter is built on — is that mathematics does not begin with a formula. It begins with a prepared mind. A mind that has learned to look before it calculates. A mind that has trained the attention to find structure before it reaches for a procedure.

The technique you are about to learn is 1,500 years old in its roots and entirely alive in your hands today. Before you touch it, do what Āryabhaṭa did: orient yourself. Notice where you are. Breathe. Look.

CHAPTER AT A GLANCE

This chapter installs the most important habit in this book: the habit of **observing before calculating**. Every Vedic technique is a pattern made visible. This chapter trains your eye to see those patterns — in sequences, in products, in numerical structure itself.

By the end of this chapter you will have:

- Named and activated the four components of your Pattern Engine
- Learned the SEE → UNDERSTAND → SOLVE framework — the operating method for every chapter that follows
- Seen the Base Method working before it has been formally introduced
- Understood why speed is a consequence of genuine understanding — not its precondition
- Encountered the tradition that produced this mathematics — not as history, but as living context

1. The Human Brain as a Pattern-Recognition Engine

A pattern is not decoration. A pattern is the **signature of understanding**. When a student learns to recognise mathematical patterns rather than memorising individual numerical facts, everything in mathematics becomes easier, lighter, and more lasting. The human brain is, at its computational core, a pattern-recognition system of extraordinary power.

The capacity to extract pattern from apparent complexity is not a special skill. It is a **universal human competency** present in every brain that has reached normal developmental maturity. Mathematical anxiety does not destroy this capacity — it ***hides it*** behind a threat-response that directs the brain's attention elsewhere. Chapter One is about removing the hiding.

> ◈ **Āryabhaṭa, 499 CE —** *composed his entire Āryabhaṭīya in verse — not because he lacked prose, but because verse activates a different kind of memory. Pattern-recognition is not only a mathematical skill; it is the operating mode of a mind trained to hold knowledge in structured form.*

The Vedic mathematical tradition understood this before neuroscience had the language to describe it. The sūtras are compressed patterns — statements so dense with structural information that unpacking one could occupy a semester. The tradition did not ask students to memorise techniques. It asked them to memorise **structural descriptions** — and let the techniques emerge from the descriptions as naturally as water flows downhill.

> *Before reading further, look at any two numbers near 100 — say, 97 and 98. Do not calculate their product. Simply look at them. Notice how close they are to 100. Notice the gap each one leaves. You have just done the first thing this entire book asks you to do: you have seen something before you calculated something. That act of seeing is where Vedic mathematics begins. Not in the formula. In the look.*

2. The Pattern Engine — Naming the Mechanism

From this point in the book, we give a name to the brain's mathematical pattern-recognition capacity: the **Pattern Engine**. This name is functional, not metaphorical. The Pattern Engine

is the neural network that, with training, automatically extracts structural regularities from numerical data and uses those regularities to simplify calculation. The Vedic tradition cultivated this engine through sūtras. This book cultivates it through structured observation.

The Pattern Engine has four components. Each one is developed through specific practices across Stage One. Each one corresponds to a quality that the Vedic tradition named — not with neuroscientific vocabulary, but with pedagogical precision.

1	**Component 1 — NOTICING** Attend to numerical structure before processing it. The Vedic student was trained to look at a problem before touching it — sometimes for minutes. "What do I see here, before I calculate?" Underdeveloped in procedurally-trained students who skip directly to the answer.
2	**Component 2 — CATEGORISING** Identify which type of pattern is present — proximity to a base, symmetry, factors, complement relationships. The sūtras are categorisation tools: each one names a structural family, allowing the trained mind to recognise which family a given calculation belongs to.
3	**Component 3 — EXTENDING** Ask: "If this is true here, where else is it true?" The Vedic tradition called this generalisation by the word 'sāmānya' — the general, the universal. Learning a technique once, at the level of structure, is worth more than drilling it a hundred times at the level of procedure.
4	**Component 4 — VERIFYING** Estimate what an answer should be before calculating, then check whether the result is consistent with the estimate. The tradition built verification into every technique — not as a separate step, but as a quality of mathematical attention. Chapter 9 will formalise this.

> *The four components above are not a model invented for this book. They are a description of what the tradition already knew. When Tirthaji wrote that a sūtra should be approached with "the eye that sees before the hand that calculates," he was describing the Noticing component. When Bhāskara wrote "Behold!" beneath a geometric proof, he was invoking the moment when Categorising and Extending happen simultaneously —*

> *when the student sees not just this proof but the family of proofs it belongs to. The Pattern Engine is old. The name is new.*

3. The SEE → UNDERSTAND → SOLVE Framework

Every technique in this book is taught through a single three-stage framework. This is not a procedural sequence to be memorised — it is a description of the mental movement that genuine mathematical understanding requires. The Gurukula tradition had no name for this movement, but it practised it rigorously: a student was never shown a procedure before they had observed the pattern that the procedure expresses.

The framework is given a name here so that the student can apply it independently — within this book and in any mathematical situation they encounter outside it.

▶ SEE

Before any calculation begins, observe the structure of the numbers. Ask: Are these close to a base? Is there a complement relationship? Are they symmetric? Do not calculate. Only see. This is the step the tradition never skipped.

Activates the Noticing component. The Vedic student looked first. Always.

▶ UNDERSTAND

Once the structure is visible, ask: which principle governs this? Which sūtra does this pattern express? The answer is always a general principle — never just "the method from page 45." Understanding means naming the structure.

"What sūtra is speaking here?" — the question Tirthaji said should precede every calculation.

▶ SOLVE

Once the governing principle is clear, the procedure follows naturally. The calculation is the final act of an understanding already in place — not the first act of a memorised sequence.

SOLVE should always be the fastest stage. If it is the hardest, the SEE and UNDERSTAND stages were not done. Return.

◈ **Tirthaji, 1958** — *wrote in the introduction to his Vedic Mathematics: "the student does not need to learn the technique first. He needs to see the pattern first. The technique will then reveal itself." This is the SEE → UNDERSTAND → SOLVE framework, stated fifty years before this book.*

4. Speed is a Consequence, Not a Cause

One of the most damaging beliefs in mathematics education is that ***fast students are good students***. Speed in mathematics is a product of automaticity. Automaticity is a product of fluency. Fluency is a product of genuine understanding consolidated through deliberate, unpressured practice. The Vedic tradition understood this chain precisely and named its final stage.

The Gurukula phrase was ***ślokatva*** — the quality of a statement or technique that has become as natural as breathing. A student who has achieved ślokatva with the Base Method does not think about it any more than a skilled reader thinks about recognising letters. The recognition is immediate, the execution is automatic, and the attention is free for the understanding that produced the automaticity in the first place.

The path to ślokatva passes through slowness. There is no shortcut. A student who is slow because they are genuinely understanding is building the neural foundation for future speed. A student who is fast because they are pattern-matching memorised procedures will fail the moment they encounter an unfamiliar variation — because the pattern they memorised was the procedure, not the structure.

THE GURUKULA PRINCIPLE · Ślokatva

"The Gurukula student watched the master for three years before touching the chalk. Not because the master was withholding knowledge — but because watching is not passive. Watching, in the Gurukula tradition, is the deepest form of learning: the student's Pattern Engine is being calibrated, one observation at a time, before a single calculation is attempted."

What this means for you: do not time yourself. Do not race through the Mastery Labs. The speed will come — it always comes — from the understanding. Attempt the understanding first.

5. Observation Lab One — Number Proximity

Every technique chapter in this book opens with an Observation Lab: a structured exercise in pure pattern-noticing before any procedure is introduced. The Observation Lab is the modern form of what the Gurukula tradition called *darśana* — "seeing, perception, the directed gaze." In the tradition, a student who had not completed their darśana of a problem was not permitted to begin their calculation.

There are no correct answers in an Observation Lab. This is not a calculation exercise. It is a perception exercise. The quality of your observations ***is*** the quality of your mathematical attention.

OBSERVATION LAB 1 — Number Proximity and Structure

For each number below, write three things you notice about its relationship to nearby round numbers (10, 50, 100, 1000). Do not calculate. Only observe.

The tradition calls this darśana — the directed gaze. Look before you calculate. Always.

97 — *e.g. Is it close to 100? By how much? Above or below?*

I notice:

103

I notice:

48

I notice:

52

I notice:

999

I notice:

1004

I notice:

25

I notice:

75

I notice:

✦ *After completing all observations: do any descriptions use "close to" or "gap of" or "away from"? If so, your Pattern Engine is already producing the observations that underpin the Base Method. The pattern was visible to you before the technique was named.*

6. Observation Lab Two — Products and Proximity

The five products below are given with their answers. Do not verify by calculation. Only observe: is there a consistent relationship between how far each number is from 100 and the structure of the answer? You are looking for a pattern that was first written down in the Atharvaveda appendix — a pattern that, once seen, is never unseen.

OBSERVATION LAB 2 — Multiplication and Structure

Answers are given. Do NOT calculate. Observe the relationship between the factors and the answer.

"The sūtra is in the observation. Not in the formula." — Tirthaji

97 × 96 = 9312

Hint: (97 − 4 = 93 | 3 × 4 = 12)

What I notice:

---

98 × 97 = 9506

Hint: (98 − 3 = 95 | 2 × 3 = 06)

What I notice:

---

102 × 103 = 10506

Hint: (102 + 3 = 105 | 2 × 3 = 06)

What I notice:

---

999 × 997 = 996003

Hint: (999 − 3 = 996 | 1 × 3 = 003)

What I notice:

---

48 × 47 = 2256

Hint: (These numbers are near 50, not 100. Which base are they using?)

What I notice:

✦ These five observations have just given you Nikhilaṃ Navataścaramaṃ Daśataḥ *before it has been named*. In Chapter 5, this sūtra will be formally introduced. When it is, you will recognise it — not as new information, but as the name for something your own Pattern Engine has already discovered.

SŪTRA

निखिलं नवतश्चरमं दशतः

Nikhilaṃ Navataścaramaṃ Daśataḥ

ETYMOLOGY · Root by Root

Nikhilaṃ — *all, entirely, without exception — not most, not usually. Every digit, without exception.*

Navataḥ — *from nine — the nine that precedes the base.*

Caramam — *last, final, ultimate — the digit that ends the number, which is treated differently.*

Daśataḥ — *from ten — from the base itself.*

ENGLISH MEANING

"All from nine, last from ten"

PRIMARY DOMAIN

Complement pairs to base 10 or 100 · Base method multiplication · Subtraction from round bases

IN THIS CHAPTER

The pattern you noticed in Observation Lab 2 — the gap between each number and 100 — is this sūtra operating. In Chapter 5 it will be fully deployed. Here, only notice: the sūtra was already at work in your observations before you knew its name.

✦ *Say this sūtra aloud once before continuing. Listen to what the syllables suggest before reading the translation.*

Hold the Sanskrit for a moment: Nikhilaṃ. All. Without exception. The sūtra does not say "subtract most digits from nine." It says nikhilaṃ — all of them, completely, without exception. The single exception — the last digit, subtracted from ten rather than nine — is named explicitly. The sūtra is exactly as precise as the rule it encodes. There is not one word too many and not one word too few. This compression is not a linguistic accident. It is the tradition's understanding that a well-formed statement, truly memorised, carries the entire

> *rule inside it. You do not need to remember the procedure. You need to remember the sūtra. The procedure follows.*

7. The First Glimpse — 97 × 98

One worked example before the Mastery Lab. Not to teach the method — that is Chapter Five. Only to show you what becomes possible when the Pattern Engine is trained, when SEE and UNDERSTAND are done with care, and when the sūtra has been seen operating in the observations above.

Read each stage slowly. The calculation at the end will take four seconds. The time you spend on the SEE and UNDERSTAND stages is the investment that makes those four seconds possible.

WORKED EXAMPLE

Most students seeing 97 × 98 reach for a calculator or a long-multiplication grid. What follows uses nothing but the pattern you observed in Lab 2 — applied deliberately, with full awareness of why each step works.

▶ **SEE** — *Look before you think*

```
                  -3            -2
                   ↓             ↓
———————————————————●———————————●———————————●
                  97            98          100
                   └── gap = 3 ──┘└── gap = 2 ──┘
```

```
Nikhilaṃ reads the gap: 97 is 3 below 100.
98 is 2 below 100. Both below. Both near 100.
That is everything the sūtra needs.
```

▶ **UNDERSTAND** — *Name the pattern*

The sūtra speaks: Nikhilaṃ — the gap of each number from its base is the complete information.

The structure: LEFT = either number minus the OTHER's gap. RIGHT = product of the two gaps.

The signs: Both below base → both gaps negative → product of two negatives = positive right part.

▶ **SOLVE** — *Let the pattern do the work*

LEFT	97 − 2 = 95	(or: 98 − 3 = 95)	
RIGHT	3 × 2 = 06	(two digits — base 100 has 2 zeros)	
ANSWER	95 | 06	=	**9506 ✓**

WHY DOES THIS WORK? *(100−3)(100−2) = 100(100−3−2) + (3×2) = 9500 + 6 = 9506. The Pattern Engine ran that algebra in your head, guided by the sūtra.*

VERIFY A DIFFERENT WAY 97 × 100 − 97 × 2 = 9700 − 194 = 9506. ✓ Same answer. Different path. The truth is the same from every direction.

✦ *You just multiplied two 2-digit numbers mentally. No calculator. No written working. No carrying. This is not a trick — it is Nikhilaṃ Navataścaramaṃ Daśataḥ, the pattern that has been living in numbers since before it was given a name.*

SOLVE was the fastest stage. This is always the case when SEE and UNDERSTAND have been done thoroughly. If SOLVE felt hard, return to UNDERSTAND.

> *Bhāskara II, writing in 1150 CE, placed a single word beneath his most beautiful geometric proof: Behold! He believed that a well-constructed demonstration, properly seen, needed no further explanation. The calculation above is an invitation to a Behold! moment. The numbers 97 and 98 looked difficult. The gaps were 3 and 2. One subtraction. One multiplication. Four digits assembled. The difficulty was an illusion — a product of not yet having seen the structure. Now you have seen it. The difficulty is permanently gone from these two numbers and from every pair of numbers near every base. That is what a sūtra does.*

8. Mastery Lab — Chapter One

Work through each level in order. Do not proceed to the next level until you can complete the current one confidently. The tradition calls this 'readiness' — the point at which proceeding is not courage but wisdom.

MASTERY LAB · CHAPTER 1 · LEVEL 1

★☆☆☆☆ — Sequence Recognition

Ch. 1

Sequence

For each sequence, identify the pattern, supply the next two terms, and state the principle in one sentence.

1. **2, 4, 8, 16, 32, _____, _____**

Pattern:

_____

2. **100, 99, 97, 94, 90, _____, _____**

Pattern:

_____

3. **1, 3, 6, 10, 15, _____, _____**

Pattern:

_____

4. **9, 99, 999, 9999, _____, _____**

This sequence is related to a sūtra. Which one?

Pattern + sūtra connection:

_____

5. **1, 4, 9, 16, 25, _____, _____**

Pattern:

_____

MASTERY LAB · CHAPTER 1 · LEVEL 2

★★☆☆☆ — Pattern Articulation

Ch. 1

Pattern

Write a sentence describing the pattern you observe in each group. Do not calculate.

1. **25 × 25 35 × 35 45 × 45**

Pattern I observe:

2. **11 × 11 11 × 22 11 × 33**

Pattern I observe:

3. **99 × 98 999 × 998 9999 × 9998**

What do the answers have in common? Which sūtra might be operating?

Pattern + sūtra guess:

MASTERY LAB · CHAPTER 1 · LEVEL 3

★★★☆☆ — Framework Application

Ch. 1

Framework

Apply SEE → UNDERSTAND → SOLVE to each problem. Fill in every stage before calculating. Notice whether SOLVE really is the fastest stage.

98 × 97 — Apply the Full Framework

▶ **SEE** — *Look at the structure. What do you notice?*

▶ **UNDERSTAND** — *Which principle governs this? Which sūtra speaks here?*

▶ **SOLVE**

Answer: 9506. *Check: did your SEE and UNDERSTAND stages predict this?*

103 × 104 — Both numbers above the base this time

▶ **SEE** — *How does this differ from 97 × 98?*

▶ **UNDERSTAND** — *Does the same sūtra still apply?*

▶ **SOLVE**

MASTERY LAB · CHAPTER 1 · LEVEL 4

★★★★☆ — Extension

Ch. 1

Extension

The tradition called extension sāmānya — the generalisation. For each pattern you identified, write one example not shown anywhere in these exercises. Then explain why your example works.

1. Write your own example of a number proximity pattern:

My example:

Why it works (mention the sūtra if you can):

2. Write a proximity multiplication of your own (two numbers near 1000). Apply SEE → UNDERSTAND → SOLVE.

My example and working:

MASTERY LAB · CHAPTER 1 · LEVEL 5

★★★★★ — The Teaching Test — and the Contemplation

Ch. 1

The

Level 5 has two parts. Part A is the teaching test — the tradition's ultimate measure of understanding. Part B is a contemplation — the tradition's measure of depth.

A. **Write three to five sentences explaining SEE → UNDERSTAND → SOLVE to a younger student who has never heard of it. Use an example of your own choosing.**

My explanation:

B. **After completing this chapter: write one sentence about what surprised you. Not what you learned — what surprised you. There is a difference.**

What surprised me:

♦ ♦ ♦

✦ **MENTOR'S SANCTUARY · Parent · Teacher · Mentor · Chapter One**

This Chapter's Purpose

Chapter One establishes the observational habits without which no technique in this book can be genuinely understood. A student who skips or rushes Chapter One will execute the Base Method as a memorised procedure, not as a pattern made explicit. The difference is not visible in the answer — it is visible in every subsequent chapter, when the student who understood Chapter One transfers their pattern-recognition to new contexts and the student who did not cannot. This is not hyperbole. It is a predictable consequence of how the Pattern Engine is built.

The Tradition on Teaching This

"A teacher who shows the procedure before the student has seen the pattern has taught a trick, not a principle." — attributed to the Gurukula oral tradition. The mentor's primary task in Chapter One is not instruction. It is the creation of conditions for observation. Ask questions. Create space. Resist explaining. The student's Pattern Engine builds itself through its own looking — not through being shown.

Practical Coaching Note

The Observation Labs and the worked example are designed to be done without any calculation. Resist the temptation to let your student "just find the answer." The value is entirely in the perception, not the result. When a student reads a gap instantly — without subtracting, only perceiving — something has changed in how their brain processes numbers. That change is the entire goal of this chapter.

The Story Spark and the Contemplative Pauses are not ornamental. They are pedagogical. The Gurukula tradition understood that mathematical knowledge that is disconnected from its human and cultural roots becomes thin and brittle. A student who knows that Āryabhaṭa oriented himself before calculating, that Tirthaji said the sūtra should precede the procedure, and that Bhāskara's single word was "Behold!" — that student brings a different quality of attention to the mathematics.

Diagnostic Flowchart — Chapter One

If Student skips the Story Spark and Pauses →

Gently ask: "What did Āryabhaṭa do before he calculated?" If they don't know, read the Story Spark aloud together. The tradition is not decoration. It is the context that gives the technique its weight.

If Student wants to calculate in the Observation Labs →

Gently return them: "Do not calculate yet. Just tell me what you see." This is the most important error to catch in the entire book. The habit of observation before calculation is the single most valuable thing Chapter One installs.

If Student cannot explain what the sūtra means →

Ask them to read the etymology box aloud, one root at a time. Ask: "What does nikhilaṃ mean?" "How many digits does it cover?" "Which one is the exception?" The meaning unfolds through the roots, not through the translation.

If Student completes Level 3 but Level 5B stumps them →

Good. Surprise is the sign that something real was encountered. Ask them: "What did you expect this chapter to be? And what was it instead?" The distance between expectation and experience is where understanding lives.

♦ ♦ ♦

★ STAGE ONE COMPLETE · You Have Completed The Mind Reset ★

What you now carry from Stage One:

- A neurological understanding of why mathematical fear exists — and why it is modifiable.
- The Stillness Protocol — sixty seconds of vagal nerve activation before every session.
- The Pattern Engine — the four-component system that will underpin every technique in this book.
- The SEE → UNDERSTAND → SOLVE framework — the operating method for every chapter that follows.
- The knowledge that speed is ślokatva — a consequence of genuine understanding, not its precondition.

- Your first encounter with Nikhilaṃ Navataścaramaṃ Daśataḥ — before it was formally introduced.
- One completed Behold! moment — the pattern seen before the formula was given.

"The Gurukula student watched the master for three years before touching the chalk."

You have spent two chapters preparing your mind.

Stage Two awaits.

COMING NEXT · Stage Two — Foundations for Life

Chapter 2 — Number Sense and Friendly Numbers

The gap game. Complement pairs. Left-to-right addition. Decomposition and estimation. The number sense that makes every subsequent chapter faster — and that Brahmagupta was already building in 628 CE when he defined zero and named the "fortune" and "debt" of signed arithmetic.

STAGE TWO · FOUNDATIONS FOR LIFE · CHAPTER TWO

Chapter Two

Number Sense and Friendly Numbers

The Gap Game · Complements · Left-to-Right · Decomposition · Estimation

"Before any technique, there must be number sense.
Number sense is not taught — it is grown, slowly, through attention." — GaṇitSūtram · Stage Two

Brahmagupta is sitting in Bhillamāla in the year 628 CE. He is the director of the astronomical observatory — a man whose job is to measure the distance between stars, to calculate the position of planets, and to know the precise moment when an eclipse will occur. For this work, he needs something that no mathematician before him has rigorously defined: the arithmetic of zero and negative numbers.

He writes: "A debt minus zero is a debt. A fortune minus zero is a fortune. Zero minus zero is zero." And then, twelve words later, he writes something that will change mathematics for the next fourteen centuries: "A debt subtracted from zero is a fortune." The gap between what you have and what you owe is not an absence. It is a number. It has a direction. It has a magnitude.

The number sense this chapter builds — the immediate perception of how far a number sits from a convenient round value, and in which direction — is Brahmagupta's insight, alive in every calculation you will do. When you look at 97 and perceive "three below a hundred," you are doing

exactly what Brahmagupta formalised: measuring a signed gap from a reference point. The arithmetic of fortune and debt. The arithmetic of number sense.

CHAPTER AT A GLANCE

This chapter builds the mental infrastructure that every calculation technique in this book depends on. You are not learning procedures — you are developing **number sense**: the feel for how numbers relate to each other, how close they are to round values, how they balance and complete each other. Without this sense, every technique is a rule to be memorised. With it, every technique is a pattern already almost visible.

Seven worked examples. Seven facets of the same understanding:

- Example 1 — The Gap Game: measuring distance from a base by perception, not calculation
- Example 2 — Complement Pairs: numbers that complete each other to a round base (Sūtra 2)
- Example 3 — Left-to-Right Addition: processing from the largest digit first
- Example 4 — Left-to-Right with Carrying: the running total that never loses magnitude
- Example 5 — Decomposition: any number as anchor + signed gap
- Example 6 — Estimation: knowing the shape of the answer before computing it
- Example 7 — Everything Together: 9997 × 9993 without a calculator

1. What Number Sense Actually Is

Number sense is not the ability to calculate quickly. It is the ability to **recognise numerical structure** without calculating at all. A student with strong number sense looks at 97 and immediately perceives "three below a hundred." They look at 48 and perceive "two below fifty." They look at 999 and perceive "one short of a thousand." None of these are calculations. They are perceptions — the numerical equivalent of recognising a familiar face.

The Vedic tradition built this perception deliberately, through daily practice with number relationships rather than number operations. The student was not drilled in sums. They were asked to ***look*** at numbers — to sit with them the way a sculptor sits with stone before cutting, reading the grain and the structure before the chisel moves.

◈ **Brahmagupta, 628 CE** — *was the first mathematician to systematically define arithmetic operations on zero and negative numbers. His conception of a number as a signed distance from a reference point — "fortune" for positive, "debt" for negative — is the conceptual foundation of every gap calculation in this book.*

THE FOUR QUALITIES OF NUMBER SENSE

Proximity awareness — You immediately notice how far any number is from the nearest round number (10, 100, 1000). This is perceived, not calculated — like distance in physical space.

Complement intuition — You immediately know what must be added to any number to reach the next round number. 97 needs 3. 48 needs 2. 993 needs 7. These become known facts, not sums to be performed.

Structural pattern recognition — You notice when two numbers share a structural relationship — both near the same base, both with small gaps, both amenable to the same method.

Magnitude sense — You know immediately whether an answer is in the right range. 97 × 98 cannot be 8000. It cannot be 12000. It is near 9500. This costs nothing and catches everything.

✦ *These four qualities are not separate skills. They are four expressions of the same underlying capacity: attention that has been trained to see numerical structure before it reaches for numerical procedure.*

2. The Gap Game

The gap is the single most important concept in this book. Every technique in GaṇitSūtram — without exception — begins by reading the gap between a number and its base. The gap has a magnitude and a direction. It is, in Brahmagupta's language, a signed number: a fortune (positive, above the base) or a debt (negative, below it).

The goal of this section is not to teach you how to subtract. You already know how to subtract. The goal is to train the gap-reading capacity from calculation into ***perception*** — so that seeing 97 and thinking "minus three" requires no deliberate effort, the same way seeing the word "cat" requires no deliberate decoding of three letters.

THE GAP — FORMAL DEFINITION

For a number N and a base B: **Gap = N − B**

If N < B, gap is negative (Brahmagupta's "debt" — the number is below the base).

If N > B, gap is positive (Brahmagupta's "fortune" — the number is above the base).

The sign of the gap is not optional. It is the difference between a correct answer and a wrong one.

> *Brahmagupta's genius was to see that the distance below a reference point is just as real as the distance above it. A debt is not "no fortune" — it is a fortune with a sign. When you read the gap of 97 from 100 as "−3," you are making the same conceptual move that changed mathematics in 628 CE. The minus sign is not a technicality. It is a direction. Hold it with that weight.*

EXAMPLE 1
The Gap

Gap reading: the one skill that, once automatic, makes every other skill in this book effortless. The goal is not to subtract — it is to perceive.

▶ **SEE** — *Look before you think*

```
BASE 100:
        −3                +7
         ↓                  ↓
————————●——————————————————●————————●
        97                107        100
         └── 3 below ────┘└── 7 above┘

BASE 1000:
993 → gap = −7    (3 short of 1000 — "debt of 7")
1008 → gap = +8   (8 past 1000 — "fortune of 8")
```

▶ **UNDERSTAND** — *Name the pattern*

The rule: Gap = Number − Base. Negative = below. Positive = above. Sign is mandatory.

Brahmagupta's frame: Below the base is a "debt." Above the base is a "fortune." Both are real.

The training goal: To see the gap without subtracting — as direct perception, like reading a face.

▶ **SOLVE** — *Let the pattern do the work*

97	Gap from 100	= 97 − 100	= −3 ↓
107	Gap from 100	= 107 − 100	= +7 ↑
993	Gap from 1000	= 993 − 1000	= −7 ↓
1008	Gap from 1000	= 1008 − 1000	= +8 ↑

WHY DOES THIS WORK? *The gap is a signed distance. The Base Method in Chapter 5 uses these signed gaps as the complete building material for multiplication.*

VERIFY A DIFFERENT WAY 97 + 3 = 100 ✓ 107 − 7 = 100 ✓ 993 + 7 = 1000 ✓ Every gap points correctly back to the base.

✦ *Once gap-reading becomes perception rather than subtraction, the Base Method requires almost no mental effort. The difficulty is here, in this section. Everything that follows is its natural consequence.*

SOLVE was the fastest stage. This is always the case when SEE and UNDERSTAND have been done thoroughly.

3. Sūtra 2 and Complement Pairs

A complement pair is two numbers that sum to a round base. The most important complement pairs are those that sum to 10, 100, and 1000. Knowing these pairs by immediate recall — not by calculation — is what makes the Base Method feel effortless. And the sūtra that generates every complement is Sūtra 2.

SŪTRA

निखिलं नवतश्चरमं दशतः

Nikhilaṃ Navataścaramaṃ Daśataḥ

ETYMOLOGY · Root by Root

Nikhilaṃ — *all, entirely, without exception — every digit, not most, not usually. All.*

Navataḥ — *from nine — subtract from nine, not ten.*

Caramam — *last, final, ultimate — the one digit that is treated differently.*

Daśataḥ — *from ten — the base, the reference point. The last digit alone comes from here.*

ENGLISH MEANING

"All from nine, last from ten"

PRIMARY DOMAIN

Complement pairs to base 10, 100, 1000 · Gap calculation for below-base numbers · Base method multiplication

IN THIS CHAPTER

In this chapter, Nikhilaṃ generates complement pairs. In Chapter 5, those complement pairs become gaps. The sūtra is already doing the Base Method's foundational work, quietly, before the technique is introduced.

✦ *Say this sūtra aloud once before continuing. Listen to the structure of the syllables before reading the meaning.*

> *Read the etymology again: nikhilaṃ means all, without exception. The sūtra does not say "subtract most digits from nine." It says all of them. The only exception — the last digit, subtracted from ten — is named explicitly. This precision is not accidental. The tradition understood that a rule stated with maximum clarity requires no interpretation. Every time you apply this sūtra, every digit without exception goes to nine. Only one — the last — goes to ten. The exception proves the rule, and the rule contains the exception.*

EXAMPLE 2
Complements

The complement of any number is not calculated — it is read. Sūtra 2 gives the rule. Practice until the ru becomes perception.

▶ **SEE** — *Look before you think*

`Nikhilaṃ Navataścaramaṃ Daśataḥ — applied:`

```
97  →  9-9=0, 10-7=3  →  complement = 03
       ↑ all from 9    ↑ last from 10

84  →  9-8=1, 10-4=6  →  complement = 16
37  →  9-3=6, 10-7=3  →  complement = 63

CHECK: 97+03=100 ✓  84+16=100 ✓  37+63=100 ✓
```

▶ UNDERSTAND — *Name the pattern*

The sūtra: "All from nine, last from ten" — digit by digit, left to right.

The connection: For below-base numbers: complement = gap magnitude. They are the same thing.

The goal: To know complements by heart — not by applying the rule, but by direct recall.

▶ SOLVE — *Let the pattern do the work*

97	complement to 100:	(9−9=0)(10−7=3)	**= 03**
84	complement to 100:	(9−8=1)(10−4=6)	**= 16**
993	complement to 1000:	(9−9)(9−9)(10−3)	**= 007**
876	complement to 1000:	(9−8)(9−7)(10−6)	**= 124**

WHY DOES THIS WORK? *The complement of N to base B is B − N. Sūtra 2 computes B − N digit by digit witho performing a subtraction — by working entirely within the single-digit range.*

VERIFY A DIFFERENT WAY 97 + 03 = 100 ✓ 84 + 16 = 100 ✓ 876 + 124 = 1000 ✓ The sūtra is self-verifyin

✦ *When complement pairs become instant knowledge, the "find the gap" step of every Base Method calculati disappears into a glance. This is ślokatva applied to a single operation.*

SOLVE was the fastest stage. This is always the case when SEE and UNDERSTAND have been done thoroughly.

◈ **Āryabhaṭa, 499 CE** — *was already using complement-based arithmetic in his planetary calculations, computing the deficiency of a day-count from a full cycle. The "all from nine, last from ten" pattern appears*

implicitly in his carrying procedures — though the sūtra as an explicit statement belongs to Tirthaji's recovery of the tradition.

4. Left-to-Right Addition

Standard arithmetic teaches addition right-to-left — units first, then tens, then hundreds — because carrying is mechanically simpler in that direction on paper. But mental arithmetic is easier **left-to-right**, because the left-most digits carry the most magnitude and give the shape of the answer before the exact digits are confirmed.

The tradition taught mental calculation before written calculation. The Gurukula student was expected to hold a running total in working memory — a total that began rough and became precise, rather than a total that remained zero until the final step. This running-total habit is left-to-right thinking.

EXAMPLE 3 — Left to Right **346 + 285**

Right-to-left (paper): 6+5=11 carry 1 · 4+8+1=13 carry 1 · 3+2+1=6 → 631. Correct, but the magnitude is unknown until the last step.

Left-to-right (mental): 300+200=500. Running total: 500. Then +40+80=620. Then +6+5=631. Each step gives a better approximation.

The Vedic habit: "What is the answer approximately?" is answered after step one: "around 500." Then "around 620." Then exactly 631.

✓ *Left-to-right is not faster on paper. It is faster in the mind — because magnitude sense is built at every step, not only at the end.*

EXAMPLE 4 — With Carrying **567 + 384**

Step 1 (hundreds): 500 + 300 = 800. Running total: 800.

Step 2 (tens): 800 + 60 = 860. Then + 80 = 940. (The tens sum 60+80=140 pushes the total past 900 — the carry is handled by the running total automatically.)

Step 3 (units): 940 + 7 = 947. Then + 4 = 951. Final answer: 951. ✓

The carry rule: When adding a column sends the running total past the next round number, the running total already absorbs the carry. There is no separate carry digit to track.

✓ *Practice until the running total updates smoothly without tracking separate carry values. The carry is not a problem to manage — it is a signal that the magnitude is larger than expected.*

> *In the Gurukula tradition, a student who could not compute left-to-right was not considered ready for multiplication. The reasoning was precise: if you do not know approximately what an answer should be before you calculate it, you cannot know whether your calculation is wrong. Magnitude sense is not a convenience — it is the prerequisite of mathematical integrity. The left-to-right habit builds magnitude sense into every calculation, automatically, at no additional cost.*

5. Decomposition — Anchor and Gap

Decomposition is the art of rewriting any number as a **friendly anchor** plus a **signed gap**. The anchor is always a convenient round number — a multiple of 10, 100, or 1000. The gap is as small as possible. Every Base Method calculation in this book is, at its core, a multiplication of two decomposed numbers.

The key insight is that the anchor and the gap are not arbitrary. The anchor is chosen to make the gap small. A small gap means a small product. A small product means a simpler right-side calculation. The entire power of the Base Method rests on the quality of the decomposition.

EXAMPLE 5
Decomposition

Any number can be written as a friendly anchor plus a small signed gap. See this — and the Base Meti is already half-understood.

▶ **SEE** — *Look before you think*

```
ANCHOR + GAP decomposition:
```

```
97  =  100  +  (-3)    anchor: 100   gap: -3
48  =   50  +  (-2)    anchor:  50   gap: -2
203 =  200  +  (+3)    anchor: 200   gap: +3
998 = 1000  +  (-2)    anchor: 1000  gap: -2

MULTIPLICATION by decomposition:
997 × 994 = (1000-3)(1000-6)
```

▶ **UNDERSTAND** — *Name the pattern*

Choose the anchor: The nearest power of ten, or a convenient multiple (50, 200, 500).

Find the gap: Number minus anchor. Keep the sign. Small gap = efficient calculation.

The payoff: Multiplying two decomposed numbers is the Base Method. The structure is already he

▶ **SOLVE** — *Let the pattern do the work*

LEFT	997 − 6 = 991	(or: 994 − 3 = 991)	
RIGHT	3 × 6 = 018	(three digits — base 1000)	
ANSWER	991 | 018	=	**991018** ✓

WHY DOES THIS WORK? *(1000−3)(1000−6) = 1000(1000−3−6) + (3×6) = 1000×991 + 18 = 991018. T decomposition made this algebra invisible.*

VERIFY A DIFFERENT WAY 997 × 1000 − 997 × 6 = 997000 − 5982 = 991018. ✓ Same truth, different pa

✦ *Master decomposition and the Base Method is already learned — not as a procedure, but as a natu consequence of how multiplication distributes over signed gaps.*

SOLVE was the fastest stage. This is always the case when SEE and UNDERSTAND have been done thoroughly.

6. Estimation — The Shape of the Answer

Estimation is not approximation. Approximation gives a rough answer. Estimation gives the **shape** of the answer — the range, the order of magnitude, the rough position on the number

line — before any calculation begins. A good estimation makes a calculation verifiable. A missing estimation makes a calculation unverifiable.

The tradition's word for estimation was *āsanna* — "near, approximate, close to." Āryabhaṭa used āsanna $\pi = 3.1416$, knowing it was close but not exact, and stated this explicitly. The tradition was not embarrassed by approximation. It understood that a well-stated āsanna is more honest, and more useful, than a false precision.

◈ **Āryabhaṭa, 499 CE** — *wrote: "Add four to one hundred, multiply by eight, and then add sixty-two thousand. The result is approximately the circumference of a circle of diameter twenty thousand." His $\pi \approx 3.1416$ is correct to four decimal places — and he called it āsanna: approximate. The tradition prized honest estimation over performed precision.*

EXAMPLE 6 — Estimation (Āsanna) **Estimate 97 × 98 before calculating**

Step 1 — Round to base: Both numbers ≈ 100. Rough estimate: 100 × 100 = 10,000.

Step 2 — Adjust for direction: Both are below 100. Product must be below 10,000. Adjusted estimate: ≈ 9,500.

Step 3 — Use gap magnitudes: Gaps are 3 and 2. The Base Method will subtract 300 and 200 and add 6. Refined āsanna: 10,000 − 300 − 200 + 6 ≈ 9,506.

The exact answer: 9,506. The āsanna was exact — because it followed the structure of the calculation.

✓ *When estimation is this precise, it is no longer a sanity check. It is the calculation itself, done informally. The formal calculation that follows is simply confirmation.*

Notice what just happened in Example 6. The estimation followed the same logical structure as the calculation — it used the gaps, it used the direction, it used the base. It was not a guess. It was the calculation done at low precision. This is what Āryabhaṭa meant by āsanna: not a wild approximation, but a close approach, made consciously and stated honestly. Every calculation in this book should begin with an āsanna. If your final answer contradicts your āsanna, one of them is wrong.

7. Everything Together — A 4-Digit Multiplication

The final worked example of this chapter brings all seven skills together: gap-reading, complement knowledge, left-to-right processing, decomposition, āsanna estimation, and magnitude sense. The problem is a four-digit multiplication — the kind most students reach for a calculator to solve. What follows uses nothing but number sense.

Read each stage slowly. Notice that the SEE and UNDERSTAND stages are long. The SOLVE stage is four lines. This is always the correct ratio. The difficulty is not in the calculation — it is in the seeing. Get the seeing right, and the calculation takes care of itself.

EXAMPLE 7 All Together

9997 × 9993. Four digits. No calculator. No written working except two small numbers. This is what number sense built over one chapter makes possible.

▶ **SEE** — *Look before you think*

```
Base: 10000
           -3          -7
            ↓           ↓
————————————●————————————●————————————●——
          9997        9993        10000
            └─ gap=-3 ─┘└─ gap=-7─┘

Both below 10000.  Gaps: -3 and -7.
Right part: 4 digits (10000 has 4 zeros).
Āsanna: near 10000² = 100,000,000.
Adjusted: 100,000,000 - 30,000 - 70,000 ≈ 99,900,000.
```

▶ **UNDERSTAND** — *Name the pattern*

Decomposition: 9997 = 10000 + (−3) and 9993 = 10000 + (−7).

Both below base: Two negative gaps → positive right part (Brahmagupta: debt × debt = fortune).

Left part: Either number minus the other's gap: 9997 − 7 = 9990.

Right part: Product of gap magnitudes: 3 × 7 = 0021 (four digits — pad with zeros).

▶ **SOLVE** — *Let the pattern do the work*

LEFT	9997 − 7 = 9990	(or: 9993 − 3 = 9990)	
RIGHT	3 × 7 = 0021	(four digits for base 10000)	

ANSWER	9990 | 0021	=	**99900021 ✓**

WHY DOES THIS WORK? *(10000−3)(10000−7) = 10000(10000−3−7) + (3×7) = 10000×9990 + 21 = 99900021. Two gaps. One product. One answer.*

VERIFY A DIFFERENT WAY 9997 × 10000 − 9997 × 7 = 99970000 − 69979 = 99900021. ✓ The āsanna of 99,900,000 was correct to six significant figures.

✦ *A four-digit multiplication solved in four seconds. Using nothing but the number sense built in this chapter. This is not the peak of what this book teaches — it is the foundation. Every technique that follows builds on exactly this capacity.*

SOLVE was the fastest stage. This is always the case when SEE and UNDERSTAND have been done thoroughly.

> *The calculation above used seven skills, but only two moments of actual arithmetic: 9997 − 7, and 3 × 7. Everything else was structure, perception, and knowledge that was already in place. This is the Vedic ideal: reduce every calculation to its irreducible minimum by seeing everything that can be seen before the arithmetic begins. The tradition called this "making the calculation small." A calculation made small enough needs only the simplest arithmetic. The intelligence is entirely in the seeing.*

8. Mastery Lab — Chapter Two

Five levels. The tradition calls completion 'readiness' — proceed to the next level only when the current one feels effortless, not merely correct.

MASTERY LAB · CHAPTER 2 · LEVEL 1 *Ch. 2*

★☆☆☆☆ — Gap Reading

For each number, state the base, the gap, and the sign. Do not subtract — perceive. The goal is to do this without writing intermediate steps.

1. 97 Base: _____ Gap: _____ Sign: _____
2. 103 Base: _____ Gap: _____ Sign: _____
3. 48 Base: _____ Gap: _____ Sign: _____
4. 998 Base: _____ Gap: _____ Sign: _____
5. 1006 Base: _____ Gap: _____ Sign: _____
6. 9997 Base: _____ Gap: _____ Sign: _____
7. 52 Base: _____ Gap: _____ Sign: _____
8. 10003 Base: _____ Gap: _____ Sign: _____

MASTERY LAB · CHAPTER 2 · LEVEL 2

★★☆☆☆ — Complement Pairs — Nikhilaṃ

Ch. 2

Apply Sūtra 2: all from nine, last from ten. Then verify. For Question 6, explain the sūtra in your own words.

1. **Complement of 97 to 100:** _____ **Verify: 97 + _____ = 100**
2. **Complement of 84 to 100:** _____ **Verify: 84 + _____ = 100**
3. **Complement of 63 to 100:** _____
4. **Complement of 993 to 1000:** _____ **Verify: 993 + _____ = 1000**
5. **Complement of 876 to 1000:** _____
6. **Complement of 9994 to 10000:** _____

   Then explain Nikhilaṃ Navataścaramaṃ Daśataḥ in your own words.

   My explanation of the sūtra:

   _____

MASTERY LAB · CHAPTER 2 · LEVEL 3

★★★☆☆ — Left-to-Right Addition

Ch. 2

Use left-to-right processing. Show the running total after each column. Notice when you know the approximate answer.

1. **346 + 275**

Running total:

2. **584 + 347**

Running total:

3. **1436 + 879**

Running total:

4. **9873 + 1256**

Running total:

MASTERY LAB · CHAPTER 2 · LEVEL 4

★★★★☆ — Decomposition and Āsanna

Ch. 2

For each pair: write the anchor-gap decomposition (Brahmagupta's "fortune and debt"). Then state your āsanna. Then calculate.

1. **97 × 96**

Decomposition:

Āsanna (estimate):

Working and answer:

2. **103 × 104**

Decomposition:

Āsanna:

Working and answer:

3. **9998 × 9994**

Decomposition:

Āsanna:

Working and answer:

MASTERY LAB · CHAPTER 2 · LEVEL 5

★★★★★ — Number Sense — Thinking, Not Calculating

Ch. 2

This level requires sentences, not numbers. The tradition called Level 5 the "teaching level" — the point at which understanding becomes expressible.

1. Why is 97 a more "friendly" number than 83 for the Base Method with base 100?

Because:

2. Brahmagupta described negative numbers as "debts" and positive numbers as "fortunes." How does this framing help you remember the sign rules for gaps?

My answer:

3. For the number 253, which base gives the smallest gap: 100, 250, 300, or 1000? Why is a smaller gap always better for the Base Method?

Best base and reason:

4. Design your own near-base multiplication (base 10000, both numbers below, gaps under 10). State your āsanna first. Then calculate.

My design:

Āsanna:

Working and answer:

♦ ♦ ♦

✦ **MENTOR'S SANCTUARY · Parent · Teacher · Mentor · Chapter Two**

This Chapter's Purpose

Chapter Two builds the perceptual foundation without which every subsequent technique is a procedure without understanding. The seven skills — gap reading, complements, left-to-right addition, decomposition, estimation, and the number line — are not preliminary exercises. They are the substance of fluent Vedic arithmetic. A student who has absorbed Chapter Two will make the Base Method their own; a student who has not will always execute it as a borrowed rule. Do not rush this chapter. The speed it enables is earned here.

The Tradition on Teaching This

"Number sense is not taught. It is grown, through daily attention to numerical structure." — The Gurukula tradition held that the teacher's primary task in the early months was not instruction but the creation of conditions for perception. A teacher who explains too quickly deprives the student of the observation that would have given them the same knowledge — more deeply and more lastingly. Ask more questions than you answer. The number sense will grow.

Practical Coaching Note

The Story Spark in this chapter names Brahmagupta as the father of signed arithmetic. Use this: when your student encounters a negative gap, remind them that they are doing Brahmagupta's arithmetic — the arithmetic of "fortune and debt" that was new to the world in 628 CE and has been fundamental to mathematics ever since. This context makes the sign feel earned rather than arbitrary.

The Contemplative Pauses are pedagogically essential. After Example 6 (Estimation), the pause asks the student to notice that their āsanna followed the same structure as the calculation. If your student has not noticed this, read the pause aloud together. This observation — that estimation and calculation share the same logic — is one of the deepest insights in the book.

Diagnostic Flowchart — Chapter Two

If Student reads gaps by subtracting rather than perceiving →

Expected at first. Do not correct directly. Instead, ask: "Look at 97. Without touching your pencil, how many steps to 100?" The physical/spatial question often activates the perceptual channel that subtraction blocks.

If Student forgets the sign of the gap →

Return to the Brahmagupta frame: "Is this number above the base (fortune) or below it (debt)?" The fortune/debt metaphor carries the sign information in a way that "positive/negative" sometimes does not.

If Student's left-to-right addition is slower than their right-to-left →

Expected and correct. Left-to-right is a new neural pathway — it will be slower than the practised pathway until it is itself practised. Reassure: the goal is not speed now. The goal is the correct habit. Speed comes with repetition over weeks.

If Student cannot connect Example 7 to the chapter's skills →

Work backwards: ask them to identify which of the seven skills was used in each of the four stages of Example 7. This reconstruction exercise makes explicit what was implicit in the flow of the example.

♦ ♦ ♦

GATE REVIEW · PART ONE

The Mind Reset and Foundations — Stage One and Stage Two Complete

Before proceeding, confirm that all of the following are true:

- ☐ The student can explain what the Pattern Engine is and name its four components.
- ☐ The student can apply SEE → UNDERSTAND → SOLVE to any near-base multiplication.
- ☐ The student reads gaps from base 10, 100, 1000, and 10000 without calculation.
- ☐ The student knows complement pairs to 100 by immediate recall for all numbers 91–99.
- ☐ The student can apply Nikhilaṃ Navataścaramaṃ Daśataḥ to find any complement to 1000.
- ☐ The student calculates left-to-right with a running total, not right-to-left with carries.
- ☐ The student states an āsanna before every multiplication and checks the answer against it.
- ☐ The student has completed Mastery Lab Levels 1–4 in both Chapter One and Chapter Two.

- ☐ The student can name Brahmagupta and state what he contributed to signed arithmetic.

When all conditions are met: **Proceed to Part Two: The Language (Heritage and Sūtras).**

❤ *Clarity over Speed. Always.* ❤

STAGE THREE · ENTERING THE VEDIC WORLD · CHAPTER THREE

Chapter Three

The Sixteen Sūtras

Full Logic Maps for Arithmetic and Algebra

"A sūtra is not a shortcut. It is a compressed understanding — an instruction for what a trained eye learns to see." — GaṇitSūtram · Stage Three

Tirthaji sits in a forest retreat near Śṛṅgeri in the early decades of the twentieth century. He is studying the Atharvaveda and its appendices. He claims that what he finds there — in passages that Sanskrit scholars have debated for a century — is a set of sixteen sūtras and thirteen sub-sūtras that together form a complete system of mathematics. He calls this system Vedic Mathematics.

Whether the sūtras are ancient or whether Tirthaji composed them himself, scholars have not agreed. What is not disputed is the quality of the mathematics they encode. Each sūtra is a compressed statement of a structural principle that applies across arithmetic, algebra, geometry, and calculus. Each one is short enough to be spoken in two seconds. Each one contains, in those two seconds, more mathematical information than most textbooks give in a page.

This chapter asks you to do something unusual for a mathematics chapter: to read, not calculate. The sixteen Logic Maps that follow are the intellectual skeleton of this entire book. Every subsequent chapter will return to one or more of them. A student who has spent genuine time with this chapter will meet the techniques that follow as recognitions — not as new material, but as the names for structures they have already seen.

Read slowly. Sit with each sūtra. Say it aloud. Ask not "how do I use this?" but "what does this point at?" That is the question Tirthaji said should precede every encounter with a sūtra.

CHAPTER AT A GLANCE

This chapter is unlike every other chapter in this book. There are no techniques to practise yet. There are sixteen **structural principles** to understand — and one worked example at the end showing what understanding them actually enables.

Before reading the Logic Maps, two things must be clear: what a sūtra is, and how to read a Logic Map. Both are explained in Sections 1 and 2. Then the sixteen sūtras follow in order, each in its full Logic Map format. The chapter closes with the Mastery Lab — five levels of sūtra engagement that move from recognition to teaching.

1. The Sūtra as a Technology of Understanding

The word *sūtra* means thread. This etymology is the first thing to hold. A thread does not contain a fabric — it is the structure through which a fabric is woven. A sūtra does not contain a mathematical technique. It is the structural principle through which a technique becomes understandable.

The ancient Indian scholarly tradition, across grammar, astronomy, medicine, and mathematics, shared this literary form for its most essential knowledge. Every word in a sūtra is load-bearing. None is decorative. The form was designed for memorisation and oral transmission — each sūtra short enough to be spoken in a breath, dense enough to repay years of unpacking.

◈ **Pāṇini, ~350 BCE** — *composed the Aṣṭādhyāyī — a grammar of Sanskrit in 3,976 sūtras that together describe the complete morphology and syntax of the language. Each sūtra is typically three to six syllables. Each*

> *encodes a rule so general that it applies to hundreds of specific cases. Tirthaji's sixteen mathematical sūtras follow precisely this tradition: maximal compression, zero decoration, complete generality.*

The mathematician who reads a sūtra and asks "how do I use this?" has missed it. The correct question is: "what structure does this describe?" The technique follows from the structure as naturally as water flows from a source. Find the source. Do not look for the water.

Tirthaji was explicit on this point. He wrote that a student who memorises the procedure without understanding the sūtra will produce correct answers — for exactly the cases they have practised. A student who understands the sūtra will produce correct answers for every case the sūtra governs, including cases they have never seen.

TWO DEFINITIONS — SŪTRA AND LOGIC MAP

Sūtra — Sanskrit: thread. A maximally compressed statement of a general principle. Every word is load-bearing. Designed to be memorised and unpacked rather than read and applied. The technique lives inside the sūtra, waiting for the student's understanding to draw it out.

Logic Map — The expanded form of a sūtra as used in this chapter: showing its domain, its pattern, a worked example, and the chapters in this book where it operates. A Logic Map is the unpacking that makes the sūtra legible. It is not a replacement for the sūtra — the student who reads the Logic Map without returning to the sūtra itself has read the map, not the territory.

> *Pāṇini described the entire Sanskrit language in 3,976 sūtras. Tirthaji described a complete system of mathematics in sixteen. The compression is not economical — it is philosophical. The tradition believed that a principle stated in its most compressed form is closer to the truth than a principle stated in its most expanded form. The expansion is for students. The compression is the knowledge itself.*

2. How to Read the Logic Maps

Each of the sixteen Logic Maps that follow has four panels: Domain, Pattern, Example, and "In This Book." Read each panel in that order. Do not skip to the example.

DOMAIN	The area of mathematics where this sūtra operates primarily. Reading the domain first tells you what kind of problem you are being equipped to solve.
PATTERN	What the trained eye sees when this sūtra is active. This is the most important panel. It describes the perception, not the procedure.
EXAMPLE	One specific numerical instance of the pattern. This is the least important panel. It demonstrates — but the sūtra is not the example. The sūtra is the pattern. The example is one expression of the pattern.
IN THIS BOOK	Which chapters of GaṇitSūtram deploy this sūtra, and how. This panel connects the sūtra to the techniques you will learn — so that when you reach Chapter 5 and encounter the Base Method, you already know which sūtras are operating.

Instructions: read each Logic Map twice. On the first reading, absorb. On the second, try to find one example of your own that the sūtra describes. This example need not be a calculation — the pattern may appear in daily life, in language, in music, or in any structured system you know. The sūtras describe structures of mind, not only of number.

3. The Sixteen Sūtras — Full Logic Maps

◈ **Tirthaji, ~1958 —** *said of the sūtras: "They are not mine. I have only recovered them." Whether or not this claim is historically accurate, it describes something pedagogically important: the sūtras are descriptions of patterns that exist in numbers, not inventions. Tirthaji found them — or composed them with the precision of one who had found them. Either way, the patterns they describe are real, and have been operating in arithmetic since arithmetic began.*

SŪTRA 1 · *"By one more than the previous one"*

एकाधिकेन पूर्वेण

DOMAIN	PATTERN	EXAMPLE	IN THIS BOOK
Squaring numbers ending in 5 · Recurring decimals · Sequential multiplications	When a number ends in 5, multiply the preceding digit by its successor and append 25.	$35^2 \rightarrow 3\times4=12$, append 25 → 1225. $65^2 \rightarrow 6\times7=42 \rightarrow 4225$.	*Ch. 7 (squaring ending in 5) · Ch. 9 (verification) · Ch. 14 (identity: $(10a+5)^2=100a(a+1)+25$)*

SŪTRA 2 · *"All from nine, last from ten"*

निखिलं नवतश्चरमं दशतः

DOMAIN	PATTERN	EXAMPLE	IN THIS BOOK
Complement pairs · Base method gaps · Subtraction from round bases	Subtract every digit from 9 and the last digit from 10 to find the complement (gap) to the base.	97→03, 876→124, 9993→0007. Each result is the distance from the number to its base.	*Ch. 2 (complement pairs) · Ch. 5 (Base Method gaps) · Ch. 9 (Vinculum) · Ch. 14 (algebra)*

SŪTRA 3 · *"Vertically and crosswise"*

ऊर्ध्वतिर्यग्भ्याम्

DOMAIN	PATTERN	EXAMPLE	IN THIS BOOK
General multiplication for any two numbers · Polynomial multiplication	Decompose each multiplication into vertical products (same-position digits) and crosswise products (cross-position digits), placed positionally.	23×14: vertical 3×4=12, 2×1=2. Crosswise 2×4+3×1=11. Result 2|11|12=322.	*Ch. 8 (two-digit multiplication) · Ch. 9 (three-digit) · Ch. 14 (algebraic proof)*

SŪTRA 4 · *"Transpose and apply"*

परावर्त्य योजयेत्

DOMAIN	PATTERN	EXAMPLE	IN THIS BOOK
Division · Solving linear equations · Transposing terms across operations	When a term crosses an operation (addition↔subtraction, multiplication↔division), its role transposes. The operation is preserved by the transposition.	$3x+12=27 \rightarrow 3x=27-12$ (transposed, sign flipped) $\rightarrow x=5$.	*Ch. 13 (word problem method) · Ch. 14 (transposition as the core algebraic move)*

SŪTRA 5 · *"When the sum is the same, that sum is zero"*

शून्यं साम्यसमुच्चये

DOMAIN	PATTERN	EXAMPLE	IN THIS BOOK
Equations where the same expression appears on both sides · Symmetric structure	If the totality is identical on both sides of an equation, that totality equals zero and can be identified immediately as a root.	$x^2+6x+9=x^2+9x+9 \rightarrow 6x=9x \rightarrow x=0$. If $(x+3)$ appears identically on both sides, $x+3=0$.	*Ch. 14 (structural symmetry in algebra) · Ch. 13 (symmetric word problems)*

SŪTRA 6 · *"If one is in ratio, the other is zero"*

आनुरूप्येशून्यमन्यत्

DOMAIN	PATTERN	EXAMPLE	IN THIS BOOK
Proportional equations · Ratio and proportion · Identifying scaling factors	When one quantity is a scaled version of another, the other quantity's unscaled contribution cancels to zero.	If $a/b=c/d$ then $ad-bc=0$. In $2x=8$: x is in ratio 1:4 with 4, so $x-4=0$, $x=4$.	*Ch. 10 (percentages as ratios) · Ch. 11 (speed-distance-time) · Ch. 14 (ratio in identity)*

SŪTRA 7 · *"By addition and by subtraction"*

संकलनव्यवकलनाभ्याम्

DOMAIN	PATTERN	EXAMPLE	IN THIS BOOK
Simultaneous equations ·	Two equations with two unknowns can be solved by		*Ch. 14 (simultaneous equations) · Ch. 13*

Elimination method · Two unknowns from sum and difference	adding them (to eliminate one variable) and subtracting them (to eliminate the other).	2x+y=9 and x+2y=9: Sum→3(x+y)=18→x+y=6. Difference→x−y=0→x=y=3.	*(two-condition word problems)*

SŪTRA 8 · *"By the completion or non-completion"*

पूरणापूरणाभ्याम्

DOMAIN	PATTERN	EXAMPLE	IN THIS BOOK
Completing the square · Adjusting to a convenient near-value · Base method logic	When a quantity is near a complete form, compute the complete form and adjust for the excess or deficiency. Incompleteness is the adjustment, not the obstacle.	$99\times101=(100-1)(100+1)=100^2-1^2=9999$. Completion: 100^2. Non-completion: -1^2.	*Ch. 5 (Base Method — gap as completion) · Ch. 7 (special products) · Ch. 14 (difference of squares)*

SŪTRA 9 · *"By sequential motion; by differential calculus"*

चलनकलनाभ्याम्

DOMAIN	PATTERN	EXAMPLE	IN THIS BOOK
Finding polynomial roots · Successive differentiation · Iterative approximation	Complex structures can be resolved by examining successive differences — the motion between consecutive terms reveals the underlying generating rule.	Sequence 1,4,9,16,25: first differences 3,5,7,9; second differences 2,2,2 — constant, revealing n^2 rule.	*Ch. 14 (Bridge to Algebra — the deepest sūtra) · Ch. 1 (Pattern Engine — extending sequences)*

SŪTRA 10 · *"Whatever the extent of the deficiency"*

यावदूनम्

DOMAIN	PATTERN	EXAMPLE	IN THIS BOOK
Sub-base method · Squaring numbers	The deficiency (gap below a base) governs the entire computation:	97^2 near base 100: deficiency=3. Left:	*Ch. 5 (Base Method) · Ch. 6 (Sub-Base,*

near a base · Scaling for adjusted bases	left part is reduced by the deficiency; right part is the deficiency squared.	97−3=94. Right: 3^2=09. Answer: 9409.	*deficiency scaled) · Ch. 7 (squaring — Yāvadūnam governs)*

SŪTRA 11 · *"The sum of the products"*

व्यष्टिसमष्टिः

DOMAIN	PATTERN	EXAMPLE	IN THIS BOOK
Verification by digit sums · Checking polynomial identities · Casting out nines	The digit sum of a product equals the product of the digit sums (mod 9). Provides a fast, reliable check on any multiplication.	48×67=3216. Sums: 3×4=12→3. Sum of 3216: 12→3. ✓	*Ch. 9 (Casting Out Nines — formal verification) · Ch. 15 (Mastery Drill self-checking protocol)*

SŪTRA 12 · *"The remainders by the last digit"*

शेषाण्यङ्केन चरमेण

DOMAIN	PATTERN	EXAMPLE	IN THIS BOOK
Divisibility testing · Digital roots · Recurring decimal cycle length	The remainder pattern in successive divisions is governed by the last digit of the divisor. The last digit alone determines the cycling behaviour.	Divisibility by 9: remainder = digit sum mod 9. 378: 3+7+8=18→9 → divisible.	*Ch. 10 (percentage divisibility shortcuts) · Ch. 9 (Vinculum and remainder logic)*

SŪTRA 13 · *"The ultimate and twice the penultimate"*

सोपान्त्यद्वयमन्त्यम्

DOMAIN	PATTERN	EXAMPLE	IN THIS BOOK
Divisibility testing for specific divisors (7, 11, 13)	The last digit and twice the second-to-last digit interact to determine divisibility.	Divisibility by 11: alternate digit sum. 121: (1+1)−2=0 → divisible. 132:	*Ch. 9 (divisibility checks in Vinculum work) · Ch. 12 (bill verification — quick divisibility)*

		(1+2)−3=0 → divisible.	

SŪTRA 14 · एकन्यूनेन पूर्वेण

· *"By one less than the previous one"*

DOMAIN	PATTERN	EXAMPLE	IN THIS BOOK
Multiplication by all-nines numbers (9, 99, 999) · Complement-based subtraction	Where Sūtra 1 uses one more, this sūtra uses one less. Left part: one less than the multiplier. Right part: its complement.	57×99: Left=57−1=56. Right=complement of 57=43. Answer: 5643.	*Ch. 9 (multiplication by all-nines) · Ch. 14 (algebraic proof: $N\times(10^n-1)=N\times10^n-N$)*

SŪTRA 15 · गुणितसमुच्चयः

· *"The product of the sum equals the sum of the products"*

DOMAIN	PATTERN	EXAMPLE	IN THIS BOOK
Verification of factorisation · Checking polynomial expansions	When an expression is factorised correctly, the digit sum of the original equals the product of the digit sums of the factors.	$x^2+5x+6=(x+2)(x+3)$. At x=1: 12=(3)(4)=12. ✓ Digit-sum consistency confirms factorisation.	*Ch. 14 (algebraic identities) · Ch. 9 (digit-sum verification after crosswise multiplication)*

SŪTRA 16 · गुणकसमुच्चयः

· *"The factors of the sum equal the sum of the factors"*

DOMAIN	PATTERN	EXAMPLE	IN THIS BOOK
Algebraic factorisation · Common factor identification · Structural consistency	The factors of a combined expression contain the factors of each part. A factor of the whole appears in every part.	ab+ac=a(b+c). Factor "a" is present in both parts. Digit sum of whole = digit sum of a × digit sum of (b+c).	*Ch. 14 (common factor extraction) · Ch. 3 (this chapter — the final sūtra names the algebraic principle the book closes with)*

4. The Six Key Upa-Sūtras

Alongside the sixteen main sūtras, Tirthaji identified thirteen *upa-sūtras* — sub-aphorisms that function as extensions, qualifications, or special applications of the main principles. Six of these are directly relevant to this book.

THE SIX KEY UPA-SŪTRAS OF THIS BOOK

Ānurūpyeṇa — "Proportionately" — The master upa-sūtra of this book. When the base is scaled (multiplied or divided by a factor), all gap calculations scale accordingly. Foundation of the Sub-Base Method in Chapter 6.

Śiṣyate Śeṣasaṃjñaḥ — "The remainder remains constant" — In recurring decimal work and divisibility testing, the remainder pattern repeats with a fixed cycle length. Governs the verification techniques in Chapter 9.

Ādyam Ādyena Antyam Antyena — "The first by the first, the last by the last" — A fast bounding check for any multiplication: leading digits of the answer come from multiplying the leading digits of the factors. Used throughout Stage Four as a pre-calculation estimate.

Yāvadūnaṃ Tāvadūnaṃ — "Whatever the deficiency, lessen it still further" — Extends the base method to squaring: the deficiency reduces the number, and the deficiency squared forms the right portion. The governing principle of Chapter 7's squaring techniques.

Āntayor Daśakapi — "The last digits sum to ten" — A special multiplication pattern: when two numbers share leading digits and their last digits sum to 10, the left portion is the product of the leading digit and its successor. E.g., 47×43: left=4×5=20, right=7×3=21 → 2021.

Veṣṭanam — "Osculation" — A divisibility testing procedure using a specific digit-multiplication rule. The osculator depends on whether the last digit is 1, 3, 7, or 9. Introduced in Chapter 9 alongside Vinculum notation.

5. How the Sūtras Interact — Three Key Pairs

The sixteen sūtras are not independent modules. They interact. A student who knows only one sūtra has one tool. A student who understands how sūtras combine has a system. Three interactions deserve special attention before the Mastery Lab.

INTERACTION 1 · Nikhilaṃ (S2) and Ūrdhva-Tiryagbhyām (S3)

The Base Method (Chapter 5) combines Sūtra 2 to find the gaps and Sūtra 3 to place the partial products. Sūtra 2 asks: how far is each number from the base? Sūtra 3 asks: how do these distances combine into a product? A student who understands both sūtras separately will immediately see why the Base Method works — it is Sūtra 3 applied to numbers where one factor has already been reduced to its gap, so only the gap products survive. The two sūtras together produce the technique. Neither alone would be enough.

INTERACTION 2 · Yāvadūnam (S10) and Purānāpurāṇā (S8)

Sūtra 8 says: complete to a round form and adjust. Sūtra 10 says: the deficiency governs. Together they describe the squaring technique in Chapter 7 with complete precision: complete to the base (S8), then reduce by the deficiency and note its square (S10). Sūtra 8 provides the strategy; Sūtra 10 provides the rule for executing it. A student who can name both sūtras as they work through Chapter 7 is understanding the chapter, not merely performing it.

INTERACTION 3 · Gunitasamuccayaḥ (S15) and Ūrdhva-Tiryagbhyām (S3)

Sūtra 3 generates the product. Sūtra 15 verifies it. After every crosswise multiplication in Chapters 8 and 9, a digit-sum check (S15) confirms the result without repeating the calculation. These two sūtras are the compute-and-verify pair that makes Stage Four self-auditing. A Stage Four student who does not automatically apply S15 after every S3 calculation is working without a safety net.

> *The tradition understood the sūtras not as sixteen separate tools but as sixteen angles from which to view a single underlying structure. Tirthaji wrote that all sixteen sūtras are expressions of one insight: numbers have structure, and structure can be named. The sixteen names are not sixteen different truths. They are sixteen windows onto the same truth. A student who has understood this — who looks at any calculation and asks not "which technique do I use?" but "what structure is operating here?" — has understood the chapter.*

6. Three Sūtras in Concert — One Calculation

The worked example that closes this chapter is not here to teach a new technique. It is here to show what the sūtras look like when they are operating together — before the student has studied Chapters 5, 7, or 9 in detail. Read it as you read the Logic Maps: slowly, twice, asking "which sūtra is speaking here?" at each step.

WORKED EXAMPLE

95^2 — squaring a number near base 100 that ends in 5. Three sūtras are active simultaneously. Can you name them before reading the anatomy?

▶ **SEE** — *Look before you think*

```
The number: 95
Two observations: ends in 5 (Sūtra 1 stirs) · near base 100 (Sūtra 2 stirs)

As (100−5):   gap = −5   (Sūtra 10: the deficiency governs)
As (90+5):    9 × 10 = 90 | append 25   (Sūtra 1: one more than previous)

Two paths. Same answer. Two sūtras, one truth.
```

▶ **UNDERSTAND** — *Name the pattern*

Sūtra 1 path: 95 ends in 5 → multiply 9×10 (the tens digit by one more than itself) → append 25.

Sūtra 10 path: 95 is 5 below 100 → left: 95−5=90. Right: 5^2=25. Same answer.

Sūtra 15 verifies: Digit sum of 9025: 9+0+2+5=16→7. Digit sum of 95: 9+5=14→5. 5^2=25→7. ✓

▶ **SOLVE** — *Let the pattern do the work*

SŪTRA 1	9 × 10 = 90	append 25	**= 9025**
SŪTRA 10	95 − 5 = 90 | 5^2 = 25	left | right	**= 9025 ✓**
SŪTRA 15	digit sum of 9025 = 7	5^2 mod 9 = 7	**VERIFIED ✓**

WHY DOES THIS WORK? *$95^2 = (90+5)^2 = 90^2+2(90)(5)+5^2 = 8100+900+25 = 9025$. Sūtra 1 exploits the $(10n+5)^2$ identity. Sūtra 10 exploits the $(100-p)^2$ identity. Both identities produce 9025. They are the same truth viewed from different sūtras.*

VERIFY A DIFFERENT WAY Digit sum of 95 is 5. 5^2 = 25 → digit sum 7. Digit sum of 9025 is 16 → 7. ✓ Sūtra 15 confirms without recalculating.

✦ *Three sūtras. One answer. Verified three ways. This is what the Vedic system looks like when it is working as a system — not as sixteen independent tricks, but as a network of structural insights that illuminate each other.*

SOLVE was the fastest stage. This is always the case when SEE and UNDERSTAND have been done thoroughly.

7. Mastery Lab — Chapter Three

This Mastery Lab is unlike every other in the book. There is no calculation in Levels 1–3. The measurements here are recognition, connection, and articulation — the skills the tradition called darśana (seeing) and vyākhyā (explaining).

MASTERY LAB · CHAPTER 3 · LEVEL 1 *Ch. 3*

★☆☆☆☆ — Logic Map Reading

For each description, identify the sūtra. Give the Sanskrit name and English meaning. Do not look at the Logic Maps.

1. **"To multiply 97×96, I notice both are 3 and 4 below 100. I use these gaps directly."**

Sūtra:

2. **"To check that 48×67=3216, I multiply digit sums: (3)(4)=12→3. Same as digit sum of 3216? Yes."**

Sūtra:

3. **"To find 35^2, I multiply 3 by 4 (one more than 3) to get 12, then append 25."**

Sūtra:

4. **"To solve 2x+3=11, I move the 3 to the other side, changing its sign: 2x=8."**

Sūtra:

5. **"99×101 = (100−1)(100+1) = 100^2−1^2 = 9999. I completed to 100 and adjusted."**

Sūtra:

MASTERY LAB · CHAPTER 3 · LEVEL 2

★★☆☆☆ — The Teaching Chain

Ch. 3

The sūtras form a natural teaching sequence across this book. Arrange these five in the order they appear (first chapter to last), then explain why that order makes mathematical sense.

Arrange: Calanā-Kalanābhyām · Nikhilaṃ · Ekādhikena · Ūrdhva-Tiryagbhyām · Parāvartya

My order:

Why this order makes mathematical sense — minimum three sentences:

MASTERY LAB · CHAPTER 3 · LEVEL 3

★★★☆☆ — Real-World Pattern Hunt

Ch. 3

Choose any three sūtras. For each, describe a situation in daily life — not in mathematics — where the same structural pattern operates. The sūtras describe structures of mind, not only of number.

1. My chosen sūtra: _____

Daily-life pattern it describes:

2. My chosen sūtra: _____

Daily-life pattern it describes:

3. My chosen sūtra: _____

Daily-life pattern it describes:

MASTERY LAB · CHAPTER 3 · LEVEL 4

★★★★☆ — Interaction Analysis

Ch. 3

Choose two sūtras from this chapter and write a paragraph explaining how they work together. Your paragraph must include: (a) a specific calculation that uses both; (b) which sūtra operates first and why; (c) what would be missing if only one were available.

My two sūtras:.	_____ **and** _____
	My interaction analysis (minimum four sentences):

MASTERY LAB · CHAPTER 3 · LEVEL 5 *Ch. 3*

★★★★★ — The Fifteen-Minute Meditation

Choose the sūtra whose English meaning is most mysterious to you — the one you cannot yet relate to any specific calculation. Write it below. Sit with it for fifteen minutes. Write what you notice. Return to this page after completing Chapter 14 and write what it means to you then.

MY MYSTERY SŪTRA

My chosen sūtra:

What I notice after fifteen minutes:

What it means to me after Chapter 14: *(complete this after Chapter 14)*

♦ ♦ ♦

✦ **MENTOR'S SANCTUARY · Parent · Teacher · Mentor · Chapter Three**

This Chapter's Purpose

Chapter Three is the intellectual spine of the entire book. A student who has genuinely engaged with the sixteen Logic Maps will approach every subsequent technique chapter differently: not asking "what is the procedure?" but "which sūtra is operating here?" This shift — from procedural to structural thinking — is the single most important transformation the book tries to produce, and Chapter Three is where it either takes root or does not.

The Tradition on Teaching This

"The sūtra is not the technique. It is the reason the technique works." — This distinction, which Tirthaji made explicit in his original text, is the pedagogical heart of the Gurukula approach. The tradition never taught procedures first. It taught structural principles first — through the sūtras — and let the procedures emerge as natural consequences. Chapter Three preserves this order. Resist any impulse to skip ahead to the techniques. The sūtras, understood first, make every subsequent chapter lighter.

Practical Coaching Note

The Mastery Lab in Chapter Three contains no calculation. This may frustrate students who are used to measuring progress by getting answers. Reframe this explicitly: "In this chapter, the measure of understanding is not getting the right answer — it is being able to describe the pattern. Tell me what Sūtra 3 points at. Do not calculate anything." The ability to describe the structural pattern IS the skill being developed here.

Level 5 — the Fifteen-Minute Meditation — is not ornamental. The student who returns to their mystery sūtra after Chapter 14 and finds that they now understand it has experienced something important: that understanding accumulates across time, not only across chapters. This experience of retrospective clarity is one of the most powerful things mathematics education can produce. Protect it.

Diagnostic Flowchart — Chapter Three

If Student cannot tell the difference between a sūtra and a technique →

Ask: "Can you say what Nikhilaṃ describes without mentioning specific numbers?" If they cannot, they are thinking of the technique, not the principle. Ask: "What kind of situation does this sūtra point at — generally?" The question of generality forces structural thinking.

If Student finds the Logic Maps "too abstract" →

Expected. The abstraction is the point. Ask them to do the Pattern Hunt in Level 3 first — finding the sūtra in daily life. This often unlocks the structural perception that makes the mathematical domain feel concrete.

If Student memorises Sanskrit names without understanding content →

Memorisation is not understanding. Ask: "Explain Ūrdhva-Tiryagbhyām to me without using the words vertical or crosswise." Forcing a paraphrase reveals whether the understanding is real or only verbal.

If Student wants to calculate before reading all sixteen →

Acknowledge the energy and channel it: "Try the worked example in Section 6 first. But before you look at the anatomy, try to name the sūtras yourself." This makes the example a test of the reading, not a replacement for it.

♦ ♦ ♦

COMING NEXT · Chapter Four — The Heritage of Indian Mathematics

Chapter 4 — The Heritage of Indian Mathematics

The lineage behind the sūtras. Āryabhaṭa's place-value system. Brahmagupta's zero. Bhāskara's Līlāvatī. The Kerala school and the infinite series. The scholars who built the world in which these sixteen sūtras became possible — and what it means that their work reached your hands today.

STAGE THREE · ENTERING THE VEDIC WORLD · CHAPTER FOUR

Chapter Four

The Heritage of Indian Mathematics

Five World-Changing Contributions · The Gurukula System · The Lineage

"We did not invent the patterns. We discovered them.
And having discovered them, we had the obligation to pass them forward." — Āryabhaṭa (attributed)

Bhāskara II has just finished writing a mathematics textbook. He has named it Līlāvatī — Playful. The year is approximately 1150 CE. His daughter, for whom tradition says the book is written, will never need to calculate the area of a field or the velocity of a river to navigate her life. He writes the book anyway, because he believes that the encounter with mathematical beauty is itself worth having — regardless of its practical application.

Among the proofs in Līlāvatī is one that demonstrates $(a-b)^2 = a^2 - 2ab + b^2$ through a geometric diagram: a large square with a smaller square cut from one corner, two rectangles remaining, a small square at the gap. Beneath the diagram, Bhāskara writes a single word: Behōld! He believes that a well-constructed demonstration, properly seen, needs no further explanation. The diagram is the proof. The word is an invitation to the moment of seeing.

This chapter tells the story of how the mathematics in your hands arrived there. Not as an obligation of historical literacy — but because knowing where a thing comes from changes how you hold it. The techniques in Stages Three and Four have a lineage. That lineage runs from the Sulbasūtra rope-geometers of 800 BCE through Āryabhaṭa's night-sky calculations, Brahmagupta's arithmetic of fortune and debt, Bhāskara's Behold! moments, and Tirthaji's recovery of the sūtras — to this page.

You are not at the beginning of something. You are in the middle of something very old.

CHAPTER AT A GLANCE

Chapter Four has no calculation exercises. **It is a chapter of intellectual orientation.** The five scholars presented here are not historical decoration — each one made a specific contribution that is directly operating in the techniques you are about to learn. Understanding what they contributed, and why, changes how those techniques feel.

- Section 1 — The Gurukula: a different architecture of learning
- Section 2 — Five scholars, five world-changing contributions
- Section 3 — Three deep studies: Āryabhaṭa on gaps, Brahmagupta on signs, Bhāskara on beauty
- Section 4 — The heritage of verification: Casting Out Nines explained from its source
- Section 5 — The Heritage Map: where each contribution appears in this book
- Mastery Lab — Heritage connection, the Gurukula question, scholar study, and the Letter to Āryabhaṭa

1. The Gurukula — A Different Architecture of Learning

The word *Gurukula* is formed from two roots: *guru* — from *gu* (darkness) and *ru* (one who removes) — and *kula* (household, family). The literal meaning: **the household of one who removes darkness**. A Gurukula was not an institution. There was no fixed curriculum, no examination timetable, no credential. There was a relationship.

A student who sought learning went to live in the household of a person who possessed it. They did not attend in the mornings and return home in the evenings. They woke before their teacher, prepared the fire, carried water, observed the day's work from proximity. Learning happened through sustained cohabitation with understanding — through

watching a master think, over years, until the quality of that thinking became available through the student's own practice.

The guru taught what the student was ready to receive. Readiness was assessed through observation, not examination. A student who could not yet describe *why* a particular observation was significant was not ready to perform that observation — regardless of how many times the procedure had been demonstrated. The guru's primary pedagogical judgement was always the same question: is this student ready to **understand** this, or are they ready only to **imitate** it?

UNDERSTANDING VS IMITATION — The Central Distinction

Imitation produces correct answers in familiar situations. A student who has imitated the Base Method will solve 97 × 98 correctly every time — as long as the numbers are near base 100, both below it, with small gaps. Put them near base 1000, or one above and one below, and imitation fails.

Understanding produces correct reasoning in novel situations. A student who has understood why the Base Method works — who can name the sūtras operating, see the signed gaps as Brahmagupta's arithmetic, and reconstruct the method from the algebraic identity — will handle every variation correctly, including variations they have never seen.

✦ *This distinction is the central pedagogical question of every chapter in this book. The Mastery Lab Level 5 in every chapter — the teaching test — is specifically designed to reveal whether a student has understood or only imitated.*

The Oral Tradition and the Power of Sanskrit Verse

Mathematical knowledge in the Gurukula tradition was primarily transmitted orally. This is not a limitation of ancient scholarship — it is a deliberate pedagogical choice. A student who must memorise a mathematical truth in order to hold it available for recitation must first understand it well enough to remember it accurately. Rote repetition of something not

yet understood produces unreliable memory. Genuine understanding produces memory that repairs itself when partially forgotten — because the student can reconstruct from the structure.

Sanskrit was uniquely suited to this purpose. Its highly regular morphology and strict metrical requirements meant that a mathematical statement expressed in Sanskrit verse had a fixed form that could be verified by metre. A student who misremembered a formula would break the metre, producing an immediately audible signal. The language itself was the error-checking system.

This is why the sūtras are in Sanskrit and why their compression is non-negotiable. A sūtra that could be extended or paraphrased without loss of meaning would not be a genuine sūtra. The compression is not stylistic. The compression is the content.

> *The Gurukula student was expected to be able to teach what they had learned before they were considered to have learned it. Not because teaching others is a good practice, though it is — but because the inability to teach reveals that understanding has not yet been reached. The Mastery Lab Level 5 in every chapter of this book is a Gurukula examination: can you explain this to someone who does not yet know it? If you cannot, return to the earlier levels. The tradition does not permit a student to move forward on the basis of correct answers alone.*

2. Five Scholars — Five World-Changing Contributions

The five scholars presented here are not a complete history of Indian mathematics — that would require a different book. They are the five whose specific contributions are most directly operating in the techniques you will learn in Stages Three and Four. Read each portrait twice: once for the biography, once for the mathematics.

Āryabhaṭa

476 – 550 CE · The Astronomer-Mathematician

आर्यभट

THE STORY

Āryabhaṭa was twenty-three years old when he wrote the Āryabhaṭīya — a text of 108 verses that encoded, in compressed Sanskrit, a complete mathematical and astronomical system. He worked in Kusumapura (present-day Patna), possibly at the Nālandā University complex. The text was so dense that later commentators spent centuries unpacking it. Bhāskara I, writing 130 years later, produced a full commentary and still did not exhaust its content.

THE MATHEMATICAL CONTRIBUTION

Āryabhaṭa's formalisation of the place-value number system made every subsequent calculation in this book possible. The digit 3 means thirty in the tens position and three hundred in the hundreds position — this positional insight, which now seems obvious, required centuries of development. His calculation of π as approximately 3.1416 (which he called āsanna — approximate) remains accurate to four decimal places. His astronomical gap logic — computing planetary positions as deviations from convenient reference points rather than as absolute values — is the direct precursor of the Base Method.

CONNECTION TO THIS BOOK

Chapter 2 (āsanna estimation — the Confidence Check protocol) · Chapter 5 (gap logic: working with deviations from base rather than absolute values) · Chapter 1 (place-value as the precondition for every positional calculation) · Every chapter (the Confidence Check is Āryabhaṭa's āsanna philosophy formalised)

"*I bow to Brahma, who is the creator of the universe, who is the divine being existing in the Earth, the sky, and the great firmament.*"

Brahmagupta

598 – 668 CE · The Algebraist of Zero

ब्रह्मगुप्त

THE STORY

Brahmagupta was the director of the astronomical observatory at Bhillamāla (present-day Bhinmal, Rajasthan). His primary work, the Brāhmasphuṭasiddhānta (The Correctly Established Doctrine of Brahma, 628 CE), addressed a problem that had paralysed mathematicians for centuries: what is zero, arithmetically? Not as an absence, not as a placeholder, but as a number that can be added to, subtracted from, multiplied, and — as Brahmagupta carefully noted — divided? He was the first to treat zero as a number in its own right, with its own arithmetic rules.

THE MATHEMATICAL CONTRIBUTION

Brahmagupta's rules for negative numbers (which he called rinā — debt) and positive numbers (dhana — fortune) established the framework of signed arithmetic. Fortune plus fortune is fortune. Debt plus debt is larger debt. Fortune times debt is debt. Debt times debt is fortune. These rules — stated for the first time in recorded mathematical history in 628 CE — are the exact rules that govern gap multiplication in the Base Method. When both numbers are below base (two debts), their gap product is positive (fortune). When one is above and one below (fortune and debt), the gap product is negative (debt). Brahmagupta's framework, fifteen centuries old, is the conceptual engine of Chapter 5's mixed cases.

CONNECTION TO THIS BOOK

Chapter 2 (signed gaps — the direction of a gap is Brahmagupta's fortune/debt) · Chapter 5 (both-below, both-above, and mixed cases all follow Brahmagupta's sign rules) · Chapter 9 (Vinculum notation — a Brahmagupta-style signed digit representation) · Chapter 14 (algebraic signed arithmetic)

"*A debt subtracted from zero is a fortune. A fortune subtracted from zero is a debt.*"

Bhāskara II

1114 – 1185 CE · The Pedagogue of Beauty

भास्कर

THE STORY

Bhāskara II — called Bhāskarācārya, Bhāskara the Teacher — wrote two mathematical texts that remain, by any measure, among the most beautiful ever produced: Līlāvatī (on arithmetic and geometry) and Bījagaṇita (on algebra). Līlāvatī means Playful. Bhāskara named his arithmetic text Playful because he believed that mathematics, encountered correctly, is a form of delight — not a set of procedures to be endured but a landscape of patterns to be explored. He posed problems as stories, wrote proofs as poems, and placed a single word beneath his most important geometric demonstration: Behōld!

THE MATHEMATICAL CONTRIBUTION

Bhāskara's most important contribution to the tradition this book honours is pedagogical rather than purely mathematical: he established that a well-constructed diagram, properly seen, constitutes a proof. His area-model proof of $(a-b)^2 = a^2 - 2ab + b^2$ requires no algebra beyond what the diagram itself shows. The student who sees the diagram and says "I see it" has understood the theorem. This is not a concession to visual learners — it is a philosophical position about what mathematical understanding is. Understanding is a form of seeing. The right diagram makes the invisible visible. Chapter 14 of this book is built entirely on this principle.

CONNECTION TO THIS BOOK

Chapter 7 (visual squaring proofs — every diagram is a Behold! moment) · Chapter 14 (area model proofs of all identities — the Sulbasūtra-Bhāskara tradition of geometric algebra) · Every chapter (the Contemplative Pause at the end of each worked example is a Behold! invitation)

"*Behōld!*"

The Sulbasūtra Authors

c. 800 – 200 BCE · The Rope-Geometers

शुल्बसूत्र

THE STORY

The Sulbasūtras — the Rules of the Cord — are the oldest mathematical texts in the Indian tradition. Their authors are unknown. Their subject is precise: the geometric construction of Vedic fire altars of specific shapes and areas, using only a rope (sulba) and wooden pegs. Different altars required different areas: a falcon-shaped altar had twice the area of a square altar; a tortoise-shaped altar had a specific relationship to the falcon. The mathematics required to compute these constructions — including the precise extraction of square roots and the statement of what the Greeks would later call the Pythagorean theorem — was developed not in a classroom, but in service of ritual precision.

THE MATHEMATICAL CONTRIBUTION

The Sulbasūtras contain the earliest recorded statements of the relationship $a^2 + b^2 = c^2$ for a right triangle — predating Pythagoras by several centuries. They also contain algorithms for approximating square roots to several decimal places, developed because altar construction required measurements accurate enough for physical implementation. Most importantly for this book, they established a tradition of understanding algebraic relationships through geometric construction — through seeing area as the product of length and width, and seeing the product of two sums as a rectangle divided into four sub-rectangles. This is the direct ancestor of the area model proofs in Chapter 14.

CONNECTION TO THIS BOOK

Chapter 14 (area model proofs: (a+b)(c+d) = ac+ad+bc+bd seen as a rectangle divided into four parts — the Sulbasūtra tradition of proving algebra through geometry)

"*The diagonal of a rectangle produces the area that both its sides produce separately.*"

Tirthaji Mahārāja

1884 – 1960 CE · The Systematiser

तीर्थजी महाराज

THE STORY

Jagadguru Swami Bharati Krishna Tirthaji Mahārāja served as Shankaracharya of the Govardhana Pīṭha at Puri. In the early decades of the twentieth century, he claimed to have

recovered sixteen mathematical sūtras from the Atharvaveda appendices — sūtras that, he said, formed a complete and self-consistent system of mathematics covering arithmetic, algebra, geometry, and calculus. He called this system Vedic Mathematics. Whether the sūtras are genuinely ancient or whether Tirthaji composed them himself, scholars have disagreed for sixty years and have not resolved the question.

THE MATHEMATICAL CONTRIBUTION

What is not disputed is the quality of the system he described. Each of the sixteen sūtras, whatever its origin, is a compressed and general statement of a structural principle that applies across the full range of numerical calculation. Tirthaji's achievement — regardless of the historical question — was to articulate these principles with the precision and compression of the sūtra tradition and to demonstrate, through hundreds of worked examples, how they interact to produce a coherent and teachable mathematical system. His 1958 book, Vedic Mathematics, is the primary source for this tradition.

CONNECTION TO THIS BOOK

Chapters 3–14 (every technique is named after its governing sūtra — the direct expression of Tirthaji's systematisation) · Chapter 3 (the sixteen Logic Maps are Tirthaji's sixteen sūtras expanded for a modern learner)

> *"The student does not need to learn the technique first. He needs to see the pattern first. The technique will then reveal itself."*

3. Three Deep Studies — Masters at Work

The following three sections present specific passages or ideas from three scholars — not for historical completeness, but because each one demonstrates a quality of mathematical thinking that this book asks you to develop. These are not quotations to be memorised. They are demonstrations of a quality of mind.

Āryabhaṭa on the Gap Method

Āryabhaṭa's astronomical calculations frequently computed the positions of planets as **deviations from reference points** rather than as absolute positions. The reference point — typically the vernal equinox — was a convenient round value from which small deviations could be measured, calculated, and accumulated. Working with small deviations is almost always simpler than working with large absolute values, because the deviations are typically much smaller numbers and their arithmetic is proportionally easier.

This is precisely the Base Method. When we multiply 97 × 98 using the Base Method, we do not work with 97 and 98. We work with their gaps from 100: −3 and −2. The computation involves only 3 × 2 = 6 and 97 − 2 = 95. Āryabhaṭa's astronomical gap logic, transplanted from the night sky to the number line, is the complete conceptual foundation of Chapter 5. The form differs. The mathematical principle is identical.

> ◈ **Āryabhaṭa, 499 CE** — *encoded astronomical tables in Sanskrit verse by representing planetary positions as signed deviations from reference epochs. The technique of working with deviations rather than absolute values — which simplifies every calculation — appears in the Āryabhaṭīya as a computational strategy and in Chapter 5 of this book as the Base Method.*

Brahmagupta on the Arithmetic of Fortune and Debt

Brahmagupta's exposition of signed arithmetic used the vocabulary of daily economic life: *dhana* (wealth, fortune) for positive quantities and *rinā* (debt) for negative ones. His rules were stated in these concrete terms: a fortune and a debt of equal size cancel to zero. Two debts combined make a larger debt. A fortune multiplied by a debt produces a debt. A debt multiplied by a debt produces a fortune.

These rules are abstract when stated as "positive times negative equals negative." They are ***transparent*** when stated as Brahmagupta stated them. Of course two debts multiplied together produce a fortune — debt squared is something you owe twice over, which is worse than owing once, but the directionality of the original debt is what matters: two negative

directions combined produce a positive one. The abstract algebraic rule follows from the concrete metaphor with no additional reasoning required.

Chapter 5's mixed cases — one number above the base, one below — are Brahmagupta's fortune-and-debt arithmetic in action. The above-base gap is a fortune. The below-base gap is a debt. Their product is a debt. That debt is then subtracted from the left part of the answer. Every student who has been confused by the sign rules in the mixed case of the Base Method has encountered Brahmagupta's problem: the abstract rule is harder to hold than the concrete metaphor. The fortune-and-debt frame resolves the confusion instantly.

Bhāskara on Beauty and the Behold! Moment

Bhāskara II's Behōld! beneath his geometric proof is not a rhetorical flourish. It is a pedagogical claim: that some mathematical truths, properly set up, require no verbal explanation. The diagram contains the proof. The word is an instruction to look — not to read, not to calculate, but to see.

The tradition of mathematical beauty that Bhāskara embodied made a specific philosophical claim: that mathematical understanding is a form of aesthetic experience. The student who sees the area model proof of $(a+b)^2$ and feels the satisfaction of the diagram clicking into place has had the same experience Bhāskara intended. That click — that moment when a pattern becomes suddenly visible as a whole — is not separate from mathematical understanding. It ***is*** mathematical understanding, in one of its purest forms.

Every Contemplative Pause in this book is a Behold! invitation. Every worked example is constructed to produce that moment: the numbers looked difficult, the gaps were small, four lines of work produced the answer, and the answer was *right* — not by accident but by structural necessity. Bhāskara called this moment *sphuṭa* — clear, manifest, suddenly visible. The tradition believed that sphuṭa, once experienced, could not be unfelt. A pattern truly seen is seen forever.

> *Bhāskara wrote Līlāvatī for his daughter. He believed that the encounter with mathematical beauty was worth having regardless of practical application. You are holding, in your hands, a book written in the same spirit: not primarily to make you faster at arithmetic, though it will, but to give you access to a way of seeing that has been refined over fifteen centuries by some of the keenest mathematical minds ever to have lived. The techniques are useful. The way of seeing is the inheritance.*

4. The Heritage of Verification

Every culture that has developed advanced mathematics has also developed methods for checking whether calculations are correct. The Indian tradition's primary verification method — Casting Out Nines — is both elegant and mathematically deep: any multiplication can be checked by comparing the digit sum of the product with the product of the digit sums.

The mathematical basis is modular arithmetic: since $10 \equiv 1 \pmod 9$, the digit sum of any number is its residue modulo 9. This is a consequence of the place-value system — the very system Āryabhaṭa helped establish. The product of two numbers has a digit sum equal to the product of their digit sums, also modulo 9. This fact provides a check that catches approximately 8 out of 9 errors in any multiplication.

CASTING OUT NINES — THE VERIFICATION FROM ITS SOURCE

Example: 48 × 67 = 3216. Is this correct?

Step 1 Digit sum of 48: 4+8=12 → 1+2=3

Step 2 Digit sum of 67: 6+7=13 → 1+3=4

Step 3 Product of digit sums: 3×4=12 → 1+2=3

Step 4 Digit sum of result 3216: 3+2+1+6=12 → 1+2=3

Step 5 3 = 3 ✓ The verification passes.

Why it works: $10 \equiv 1 \pmod 9$. So any number $N = a_n \times 10^n + \ldots + a_1 \times 10 + a_0 \equiv a_n + \ldots + a_1 + a_0 \pmod 9$. The digit sum is the number's remainder when divided by 9. Since $(A \bmod 9) \times (B \bmod 9) = (AB) \bmod 9$, the product of digit sums must equal the digit sum of the product. The place-value system makes this true automatically.

Chapter 9 formalises this as the Casting Out Nines protocol, and deploys it after every crosswise multiplication. A student who has read this chapter will understand it not as a trick but as a consequence of the place-value system's deepest property: that the digit sum is an invariant of a number's residue modulo 9, and invariants survive multiplication.

In the Vedic tradition, verification was not optional. Computation without verification was considered incomplete — like a sentence without a full stop. The Confidence Check at the end of every Mastery Lab level in this book is a modern form of this insistence: the work is not done when the answer is found. The work is done when the answer has been confirmed.

5. The Heritage Map — Where Ancient Mathematics Appears in This Book

The table below shows exactly where each scholarly contribution surfaces in the chapters that follow. When you encounter any technique from Chapter 5 onward, you can locate it on this map and know whose thinking you are using.

SCHOLAR & CONTRIBUTION	DATE	WHERE IT APPEARS IN THIS BOOK
Āryabhaṭa — Gap Logic	499 CE	*Ch. 5 (Base Method: all calculations use gaps from a reference base, not the numbers themselves)*
Āryabhaṭa — Āsanna Philosophy	499 CE	*Ch. 2 (estimation) · Every chapter: Confidence Check before proceeding*
Āryabhaṭa — Place-Value System	499 CE	*Every chapter: the entire positional arithmetic framework*

Brahmagupta — Signed Arithmetic	628 CE	*Ch. 5 (mixed cases above/below base) · Ch. 9 (Vinculum: signed digits) · Ch. 14 (algebraic signs)*
Brahmagupta — Fortune & Debt Frame	628 CE	*Ch. 2 (gap signs) · Ch. 5 (sign rules for gap products) · Ch. 9 (casting out nines foundation)*
Sulbasūtras — Geometric Reasoning	800–200 BCE	*Ch. 14 (area model proofs of all multiplication identities)*
Bhāskara II — Behold! Method	1150 CE	*Ch. 7 (visual squaring proofs) · Every chapter (Contemplative Pause)*
Bhāskara II — Variable Names	1150 CE	*Ch. 14 (Bridge to Algebra — the step toward symbolic manipulation)*
Pāṇini — Sūtra Compression	~350 BCE	*Ch. 3 (sixteen sūtras as compressed structural principles)*
Tirthaji — Sūtra Systematisation	1958 CE	*Ch. 3–14: every technique named and understood through its governing sūtra*

> *Look at the Heritage Map. Notice that Āryabhaṭa's contributions appear in every chapter of this book — not as a historical reference, but as the active precondition of every calculation. The place-value system that makes "the digit 3 means three hundred in the hundreds position" possible is his. Without it, the Base Method has no meaning, because the Base Method is entirely about what position a digit occupies relative to a round number. The techniques you are about to learn in Stage Three are, in a real sense, Āryabhaṭa's astronomy applied to arithmetic. You are working with tools he shaped. Hold them accordingly.*

6. Mastery Lab — Chapter Four

Chapter Four's Mastery Lab contains no calculation exercises. The work here is intellectual and reflective. The tradition valued this kind of engagement as highly as computational fluency — and considered it the prerequisite for genuine fluency.

MASTERY LAB · CHAPTER 4 · LEVEL 1

★☆☆☆☆ — Heritage Connection

Ch. 4

Match each technique to the scholar whose work it most directly expresses. Give the scholar's name and a one-sentence reason.

1. **The Base Method (Chapter 5) — working with gaps rather than the numbers themselves**

   Scholar and reason:

2. **Casting Out Nines (Chapter 9) — digit-sum verification after every multiplication**

   Scholar and reason:

3. **Area Model Proofs (Chapter 14) — proving algebraic identities through geometric diagrams**

   Scholar and reason:

4. **The sign rules for gap products (both below base = positive result)**

   Scholar and reason:

5. **The Confidence Check (every chapter) — stating an estimate before calculating**

   Scholar and reason:

MASTERY LAB · CHAPTER 4 · LEVEL 2

★★☆☆☆ — The Gurukula Question

Ch. 4

The Gurukula tradition distinguished between imitation and understanding. Write a specific example from your study so far where you performed a technique correctly but did not yet fully understand why it worked. Then write what you would need to know to genuinely understand it.

Technique I performed correctly without full understanding:.	
	What I did:
	What I did not yet understand about why it worked:
	What I would need to know to genuinely understand it:

MASTERY LAB · CHAPTER 4 · LEVEL 3

★★★☆☆ — Scholar Study

Ch. 4

Choose one of the five scholars in this chapter. Write five to eight sentences explaining their single most important contribution to mathematics — not their biography, but the specific mathematical idea that would not have existed without them. Write as if explaining to someone who has never heard of them.

My chosen scholar:.	

Their most important contribution (5–8 sentences):

MASTERY LAB · CHAPTER 4 · LEVEL 4

★★★★☆ — The Tradition Today

Ch. 4

The mathematical patterns in this book are fifteen centuries old. They also appear in modern computer science, signal processing, and cryptography. Write about one connection you can imagine between any technique in this book and a modern technology. You do not need technical details — reason from the mathematical structure.

The technique I am connecting:.

The modern technology I connect it to, and why the mathematical structure is related:

MASTERY LAB · CHAPTER 4 · LEVEL 5

★★★★★ — The Letter to Āryabhaṭa

Ch. 4

Write a letter to Āryabhaṭa. Tell him what you are learning and what you find most difficult. Tell him what you think his gap logic reveals. You may write in whatever language feels most natural.

THE LETTER TO ĀRYABHAṬA

Dear Āryabhaṭa,

Your student in this century, _____

♦ ♦ ♦

MENTOR'S SANCTUARY · Parent · Teacher · Mentor · Chapter Four

This Chapter's Purpose

Chapter Four is, in one sense, a rest from calculation. It is a chapter of intellectual and historical orientation. But it is not optional. Students who skip it will approach the Base Method as a clever trick devised by someone in the twentieth century. Students who read it carefully will approach the Base Method as a form of Āryabhaṭa's astronomical gap logic — fifteen centuries old, refined through a tradition of extraordinary mathematical intelligence, and arriving in their hands with that entire lineage behind it. These two approaches produce different qualities of engagement, and the difference becomes visible in Chapter Fourteen when the algebraic connections are drawn.

The Tradition on Teaching This

"The student who knows only the technique knows nothing yet. The student who knows the technique and its lineage knows something that no examination can take away." — Gurukula oral tradition. Chapter Four gives the lineage. Protect this chapter from being skipped.

Practical Coaching Note

When reading the scholar portraits with your student: resist making them purely historical. Ask "If Brahmagupta were sitting beside you as you work through the mixed case in Chapter 5, what would he say?" This kind of imaginative engagement makes the heritage feel present rather than archival. The Level 5 exercise — the Letter to Āryabhaṭa — is particularly powerful for students who feel disconnected from the tradition.

The Heritage Map in Section 5 is a reference document for the rest of the book. Consider returning to it at the start of each subsequent chapter and asking: "Which scholar's work are we using today?" This running reference builds a sense of the tradition as a living, continuous thing rather than a historical footnote.

Diagnostic Flowchart — Chapter Four

If Student asks "why do we need to know this history?" →

Take the question seriously. Ask: "Can you answer your own question after completing Chapter 14?" The connection between ancient heritage and modern algebra is the central intellectual arc of this book. It cannot be described before it is experienced. Promise the student that they will be able to answer their own question after Chapter 14.

If Student shows particular interest in one scholar →

Follow it. Allow the Mastery Lab exercises to be weighted toward that scholar. Depth of engagement with one part of the tradition is more valuable than superficial coverage of all of it. A student who has deeply understood Brahmagupta's signed arithmetic will find the Base Method's sign rules effortless.

If Student is resistant to Level 5 (the letter) →

Offer an alternative: a drawing, a diagram, or an oral response. The purpose is imaginative connection to the tradition, not a specific writing format. Even a three-sentence response — "Dear Āryabhaṭa, I am learning your gap method. I find the mixed cases confusing. I think what you discovered is that distance is more useful than position." — is enough.

♦ ♦ ♦

COMING NEXT · Chapter Five — The Base Method

Chapter 5 — The Base Method: Nikhilaṃ Navataścaramaṃ Daśataḥ

Āryabhaṭa's gap logic, named. Brahmagupta's signed arithmetic, applied. The Observation Labs that let you discover the pattern before you are told the procedure. The full Base Method for bases 10, 100, 1000, and 10000 — both below, both above, and mixed. And twelve worked examples, each one built on the foundations this chapter has laid.

STAGE THREE · ENTERING THE VEDIC WORLD · CHAPTER FIVE

Chapter Five

The Base Method

Nikhilaṃ Navataścaramaṃ Daśataḥ — The First Great Technique

"The gap is not the obstacle. The gap is the answer.
The sūtra knew this before the student did." — GaṇitSūtram · Stage Three

Kerala, 1958. Tirthaji is being pressed by a group of students from the University of Nagpur. They want a demonstration. He is seventy-four years old and has been describing the sixteen sūtras for an hour. One student, growing impatient with the theory, calls out a challenge: multiply 9998 by 9993. In a hall without calculators, without even pencil and paper in his hand, Tirthaji pauses for exactly the time it takes to read two gaps. He writes five digits. He is correct. The hall goes quiet.

Later, when asked how he had done it, Tirthaji said something that became the pedagogical core of every Vedic mathematics text that followed: "I did not multiply 9998 by 9993. I multiplied 2 by 7. The rest was placement." The gaps were 2 and 7. Their product was 14. The left part was 9998 minus 7: 9991. The answer was 9991|0014 = 99910014. Four digits of multiplication reduced to one.

This chapter teaches that reduction. Not as a trick — as a principle. The principle has a name: Nikhilaṃ Navataścaramaṃ Daśataḥ. All from nine, last from ten. The sūtra that tells you the gap. And the gap, as Tirthaji showed in that hall, is the entire calculation.

CHAPTER AT A GLANCE

Chapter Five is the first technique chapter. Before reading a single procedure, complete Observation Lab 3 in Section 1. The procedure follows in Section 2 — but only after you have **seen the pattern the procedure describes**. This is the tradition's non-negotiable order: observe first, name second, practise third.

- Section 1 — Observation Lab 3: five groups of numbers, five observations
- Section 2 — The sūtra: Nikhilaṃ alive, etymology, and the Core Pattern stated
- Section 3 — Twelve worked examples across all four cases and four bases
- Section 4 — Observation Lab 4: boundary cases and exceptions
- Section 5 — The algebraic identity: why the Base Method cannot give a wrong answer
- Mastery Lab — Five levels from gap identification to algebraic reconstruction

1. Observation Lab Three — The Gap and the Base

Examine the five groups below. For each group, write one observation about what all the numbers in that group have in common. Do not calculate. Do not multiply. Only observe. The observation you write is more valuable than any answer you could compute.

OBSERVATION LAB 3 — The Gap and the Base

Observe, do not calculate. Write one sentence per group. Look for a central value.

Group A **97 · 98 · 96 · 99** — *All near what central value?*

I notice:

Group B **103 · 107 · 102 · 105** — *All near what central value?*

I notice:

Group C **997 · 998 · 996 · 993** — *All near what central value?*

I notice:

Group D **9997 · 9998 · 9996 · 10003** — *All near what central value?*

I notice:

Group E **99998 · 100002 · 99997 · 100005** — *All near what central value?*

I notice:

✦ *After observing all five groups: the central values are called bases. The distances from the base are called gaps. Write the gaps for Group A explicitly — with their signs. Then ask: what is the smallest multiplication you could do using only those gaps?*

◈ **Āryabhaṭa, 499 CE** — *worked exclusively with deviations from reference points in his astronomical calculations — never with absolute planetary positions. The base is Āryabhaṭa's reference point. The gap is his deviation. The Base Method is his astronomical technique applied to multiplication. When you found the central value in Observation Lab 3, you were doing Āryabhaṭa's astronomy.*

2. The Sūtra — Nikhilaṃ Navataścaramaṃ Daśataḥ

The central values you identified in Observation Lab 3 are bases. A base is a round number — a power of ten — that serves as the reference point for gap calculations. The insight behind the Base Method is both simple and ancient: when both factors in a multiplication are close to the same base, their gaps are small. And it is always easier to work with small numbers than with large ones.

The Base Method replaces a multiplication of two large numbers with a multiplication of two small gaps, plus one addition or subtraction. The procedure follows a specific, invariant pattern — the pattern you discovered in the Observation Lab. We now give it its name.

SŪTRA · 2

निखिलं नवतश्चरमं दशतः

Nikhilaṃ Navataścaramaṃ Daśataḥ

ETYMOLOGY · Root by Root

Nikhilaṃ — *all, entirely, without exception — every digit is covered by this rule. None escape it.*

Navataḥ — *from nine — the rule for all digits except the last: subtract from nine.*

Caramam — *last, final — the one digit treated differently: the terminal digit of the complement.*

Daśataḥ — *from ten — the base, the reference point. The last digit comes from ten, not nine.*

ENGLISH MEANING

"All from nine, last from ten"

PRIMARY DOMAIN

Complement pairs to any base of 10 · Gap finding for below-base numbers · Base Method multiplication

IN THIS CHAPTER

Nikhilaṃ generates the gap for any number below its base. The gap becomes the complete input for the Base Method calculation. The sūtra is not the technique — it is the gap-finding rule that makes the technique possible.

✦ *Say this sūtra aloud once before beginning Section 3. The sound carries the structure.*

THE BASE METHOD — CORE PATTERN

Let both numbers be close to a base B. Let their gaps from B be p and q. *(p is negative if the number is below B; positive if above.)*

$$(B + p) \times (B + q) = B \times (B + p + q) + pq$$

LEFT PART = either factor plus the OTHER factor's gap = B + p + q

(same result whichever factor you adjust — this is the first verification built into the method)

RIGHT PART = product of the two gaps = pq

The right part occupies exactly as many digits as the base has zeros. Base 100 → 2 digits. Base 1000 → 3 digits. Base 10000 → 4 digits. Pad with leading zeros if the gap product is small.

SIGN RULE Both gaps same sign → right part positive (add to left). Gaps opposite sign → right part negative (subtract from left). Brahmagupta's fortune-and-debt rule.

> *Tirthaji said: "I did not multiply 9998 by 9993. I multiplied 2 by 7." Sit with that for a moment. Two digits. One product. Everything else — the placement, the left part, the position of the answer — follows automatically from the base. The difficulty of large-number multiplication is entirely located in the size of the numbers. The Base Method removes the size. What remains is always small.*

3. Twelve Worked Examples — All Cases, All Bases

Read each worked example using the full SEE → UNDERSTAND → SOLVE framework. Before reading the SOLVE stage, cover it and work through SEE and UNDERSTAND yourself. Then compare.

Case A — Both Numbers Below Base 100

Both gaps are negative. Both are Brahmagupta's debts. Debt × debt = fortune. Right part is positive — add it.

EXAMPLE 1 Both Below

Two numbers near 100, both below. This is the prototype case. Every subsequent example is a variation on this one.

▶ **SEE** — *Look before you think*

```
Both below base 100:
          -3            -4
           ↓             ↓
————————————●———————————●————————————●
           97            96          100
            └── gap = -3 ┘└ gap = -4 ┘
```

▶ **UNDERSTAND** — *Name the pattern*

Base: 100. Both below. Two negative gaps: −3 and −4.

Sign rule: (−3) × (−4) = +12. Fortune. Add to left part.

Right part digit count: Base 100 has 2 zeros → right part is always exactly 2 digits.

▶ **SOLVE** — *Let the pattern do the work*

LEFT	97 − 4 = 93	or: 96 − 3 = 93	
RIGHT	(−3) × (−4) = +12	2 digits → 12	
ANSWER	93 | 12	join parts	**9312 ✓**

WHY DOES THIS WORK? *(100−3)(100−4) = 100(100−3−4) + (3×4) = 100×93 + 12 = 9312. The identity produces both parts exactly.*

VERIFY A DIFFERENT WAY 97 × 100 − 97 × 4 = 9700 − 388 = 9312. ✓ Left-to-right confirms.

✦ *Both gaps were single-digit. The hardest arithmetic in this problem was 3 × 4 = 12 and 97 − 4 = 93. Everything else was structural.*

SOLVE was the fastest stage. SEE and UNDERSTAND did the real work.

EXAMPLE 2 **96 × 99 — both below 100**

SEE: 96 is −4 from 100. 99 is −1 from 100. Gaps: −4 and −1.

LEFT: 96 − 1 = 95 (or: 99 − 4 = 95)

RIGHT: (−4) × (−1) = +04 (two digits — note leading zero)

ANSWER: 9504 ✓

✓ *The leading zero in the right part is mandatory. 95|4 is not an answer. 95|04 = 9504 is. Count the zeros in the base before writing the right part.*

EXAMPLE 3 **93 × 94 — wider gaps**

SEE: 93 is −7 from 100. 94 is −6 from 100. Gaps: −7 and −6.

LEFT: 93 − 6 = 87 (or: 94 − 7 = 87)

RIGHT: (−7) × (−6) = +42 (two digits)

ANSWER: 8742 ✓

VERIFY: Āsanna: near 9000. 8742 is plausible. ✓

✓ *Wider gaps produce larger right parts. If the right part exceeds two digits (e.g. gaps of −9 and −9: right part = 81, still two digits), it stays as-is. Only when it genuinely overflows (right part ≥ 100) does carrying occur — and that only happens with very large gaps.*

Case B — Both Numbers Above Base 100

Both gaps are positive. Fortune × fortune = fortune. Right part is positive. Answer exceeds $100^2 = 10000$ — the magnitude sense check.

EXAMPLE 4 Both Above

Both numbers above 100. Notice that the left part exceeds 100, which simply means the answer exceeds 10000. This is expected — two numbers above 100, multiplied, must give more than 100^2.

▶ **SEE** — *Look before you think*

```
Both above base 100:
          +3              +4
           ↑               ↑
————————————●———————————————●
        100               103          104
            └ gap = +3  ──┘└ gap = +4 ┘
```

▶ **UNDERSTAND** — *Name the pattern*

Base: 100. Both above. Two positive gaps: +3 and +4.

Sign rule: (+3) × (+4) = +12. Fortune. Add to left part.

The left part: 103 + 4 = 107. This exceeds 100. The answer will exceed 10000. That is correct.

▶ **SOLVE** — *Let the pattern do the work*

LEFT	103 + 4 = 107	or: 104 + 3 = 107	
RIGHT	(+3) × (+4) = +12	2 digits → 12	

ANSWER	107 | 12	join parts	**10712 ✓**

WHY DOES THIS WORK? *(100+3)(100+4) = 100(100+3+4) + (3×4) = 100×107 + 12 = 10712. Both-above is structurally identical to both-below. Only the sign changes.*

VERIFY A DIFFERENT WAY 103 × 100 + 103 × 4 = 10300 + 412 = 10712. ✓

✦ *Both below gives an answer below 10000. Both above gives an answer above 10000. The magnitude is built into the sign of the gap. Brahmagupta's arithmetic handles the placement automatically.*

SOLVE was the fastest stage. SEE and UNDERSTAND did the real work.

EXAMPLE 5 **108 × 107 — larger above-base gaps**

SEE: 108 is +8 from 100. 107 is +7 from 100. Gaps: +8 and +7.

LEFT: 108 + 7 = 115 (or: 107 + 8 = 115)

RIGHT: (+8) × (+7) = +56 (two digits)

ANSWER: 11556 ✓

VERIFY: Both above 100, answer must exceed 10000. 11556 > 10000. ✓

✓ *Large positive gaps still give a two-digit right part (8×7=56). The method is the same regardless of gap size. The larger the gaps, the less efficient the method — which is why it works best for numbers close to their base.*

Case C — Mixed: One Above, One Below Base 100

One gap positive (fortune), one negative (debt). Fortune × debt = debt. Right part is negative. This is the case that most confused students before Brahmagupta named it.

EXAMPLE 6 Mixed Case

One above, one below. This is Brahmagupta's fortune-and-debt case. A positive gap and a negative gap produce a negative product — which must be subtracted. Read carefully: the subtraction is not an error to be avoided. It is the correct arithmetic.

▶ **SEE** — *Look before you think*

```
Mixed case — opposite sides of base 100:
        +3          -3
         ↑           ↓
         ●————————●————————●
       103         100       97
         └ +3 above ┘└ -3 below┘
```

▶ **UNDERSTAND** — *Name the pattern*

Gaps: 103 is +3. 97 is −3. Equal magnitude, opposite sign — a symmetric case.

Sign rule: (+3) × (−3) = −9. Debt. The right part is negative.

How to handle: Left part gives 100|00. Subtract 09 from 10000. Be precise about padding.

▶ **SOLVE** — *Let the pattern do the work*

LEFT	103 − 3 = 100	or: 97 + 3 = 100	
RIGHT	(+3) × (−3) = −09	negative right part	
COMBINE	10000 − 09	= 10000 − 9	**9991** ✓

WHY DOES THIS WORK? *(100+3)(100−3) = $100^2 - 3^2$ = 10000 − 9 = 9991. The mixed case is the difference-of-squares identity. Tirthaji called it the "most beautiful case" of the Base Method.*

VERIFY A DIFFERENT WAY 103 × 100 − 103 × 3 = 10300 − 309 = 9991. ✓

✦ *The symmetric mixed case always produces a difference of squares. 103 × 97 = $100^2 - 3^2$. 105 × 95 = $100^2 - 5^2$. The sūtra was already pointing at this algebraic identity.*

SOLVE was the fastest stage. SEE and UNDERSTAND did the real work.

EXAMPLE 7 **97 × 106 — asymmetric mixed case**

SEE: 97 is −3 from 100. 106 is +6 from 100. Gaps: −3 and +6.

LEFT: 97 + 6 = 103 (or: 106 − 3 = 103)

RIGHT: (−3) × (+6) = −18. Negative right part.

COMBINE: 10300 − 18 = 10282 ✓

VERIFY: Āsanna: 97 × 106 ≈ 100 × 106 − 3 × 106 = 10600 − 318 ≈ 10282. ✓

✓ *Note: 103|−18 is not the answer. The negative right part means subtract 18 from 10300. The left part acts as the hundreds-and-above value; the negative right part reduces it.*

Case D — Higher Bases: 1000, 10000, 100000

The procedure is identical for any base that is a power of ten. Only two things change: the base used in the left-part calculation, and the number of digits in the right part.

EXAMPLE 8
Base 1000

997 × 996 — Tirthaji's demonstration problem, scaled to base 1000. Notice that only three digits of arithmetic are required: 3 × 4 = 12 and 997 − 4 = 993.

▶ **SEE** — *Look before you think*

```
Base 1000:
          -3              -4
           ↓               ↓
————————————●———————————————●———————————————●
          997             996            1000
           └── gap = -3 ──┘└── gap = -4 ──┘
```

▶ **UNDERSTAND** — *Name the pattern*

Base: 1000. Three zeros → right part is always exactly 3 digits.

Gaps: −3 and −4. Both below. Right part positive.

Right part: 3 × 4 = 12. Three digits → pad to 012.

▶ **SOLVE** — *Let the pattern do the work*

LEFT	997 − 4 = 993	or: 996 − 3 = 993	
RIGHT	(−3) × (−4) = 12	3 digits → 012	
ANSWER	993 | 012	join parts	**993012** ✓

WHY DOES THIS WORK? *(1000−3)(1000−4) = 1000×993 + 12 = 993000 + 12 = 993012. The identity works at every scale.*

VERIFY A DIFFERENT WAY 997 × 1000 − 997 × 4 = 997000 − 3988 = 993012. ✓

✦ *Six-digit answer from single-digit arithmetic. Tirthaji's "I multiplied 3 by 4" — that is literally the only multiplication in this problem.*

SOLVE was the fastest stage. SEE and UNDERSTAND did the real work.

EXAMPLE 9 **1003 × 1007 — both above base 1000**

SEE: 1003 is +3 from 1000. 1007 is +7 from 1000.

LEFT: 1003 + 7 = 1010

RIGHT: (+3) × (+7) = 21 → 3 digits → 021

ANSWER: 1010021 ✓

✓ *Seven-digit answer. One multiplication: 3 × 7 = 21. One addition: 1003 + 7 = 1010. Three zeros in the base → always pad to three digits.*

EXAMPLE 10 **1008 × 997 — mixed case at base 1000**

SEE: 1008 is +8 from 1000. 997 is −3 from 1000. Opposite signs.

LEFT: 1008 − 3 = 1005 (or: 997 + 8 = 1005)

RIGHT: (+8) × (−3) = −24. Negative. Three digits → −024.

COMBINE: 1005000 − 024 = 1004976 ✓

VERIFY: Āsanna: near 1000^2 = 1,000,000. Answer 1,004,976 ≈ 1,005,000. ✓

✓ *Mixed case at base 1000. The negative right part subtracts from the left part's last three positions. Brahmagupta's sign rules apply regardless of scale.*

EXAMPLE 11 **9997 × 9998 — base 10000**

SEE: 9997 is −3 from 10000. 9998 is −2 from 10000.

LEFT: 9997 − 2 = 9995

RIGHT: (−3) × (−2) = 6 → 4 digits → 0006

ANSWER: 99950006 ✓

✓ *Eight-digit answer. One multiplication: 3 × 2 = 6. Four zeros in the base → pad to four digits: 0006. Tirthaji's hall demonstration, exactly.*

EXAMPLE 12 Base 100000

99997 × 100003 — the largest example in this chapter. Five-digit numbers, mixed case, base 100000. Notice that the entire calculation rests on one observation: these two numbers are +3 and −3 from 100000.

▶ **SEE** — *Look before you think*

```
Base 100000 — five zeros:
            −3                +3
             ↓                 ↑
————————————●————————————————●
          99997            100000          100003
Symmetric mixed case: difference of squares.
```

▶ **UNDERSTAND** — *Name the pattern*

Gaps: 99997 is −3. 100003 is +3. Equal magnitude, opposite sign.

This is: (B−3)(B+3) = B² − 9. The difference-of-squares identity. Again.

Five zeros: Right part is exactly 5 digits.

▶ **SOLVE** — *Let the pattern do the work*

LEFT	99997 + 3 = 100000	or: 100003 − 3 = 100000	
RIGHT	(−3) × (+3) = −9	5 digits → −00009	
COMBINE	10000000000 − 9	=	**9999999991** ✓

WHY DOES THIS WORK? *(100000−3)(100000+3) = 100000² − 9 = 10,000,000,000 − 9 = 9,999,999,991. The difference of squares, operating at ten-digit scale.*

VERIFY A DIFFERENT WAY Both numbers are within 3 of 100000, so their product must be within $3\times100000 + 3\times100000 = 600{,}000$ of $100000^2 = 10{,}000{,}000{,}000$. Answer is $9{,}999{,}999{,}991 \approx 10{,}000{,}000{,}000$. ✓

✦ *Ten-digit multiplication solved with the observation that 3 × 3 = 9. The sūtra did not change. The base grew. The method is the same at every scale.*

SOLVE was the fastest stage. SEE and UNDERSTAND did the real work.

4. Observation Lab Four — Boundary Conditions

Before the Mastery Lab, four boundary cases. Each represents an edge of the Base Method's domain. Work each one through the full SEE → UNDERSTAND → SOLVE framework. The method handles all of them — but each reveals something about the structure that the earlier examples did not need to show.

OBSERVATION LAB 4 — Boundary Conditions

Work each case. The method does not change. But each case teaches something new about why it works.

Case 1 **100 × 97** — *One factor IS the base. What is its gap? What happens to the left part?*

I notice:

Case 2 **105 × 95** — *Symmetric gaps, opposite signs. This is $(B+5)(B-5) = B^2 - 25$. Which identity?*

I notice:

Case 3 **998 × 1002** — *One near 1000, one just above it. Which base gives smaller gaps?*

I notice:

Case 4 **9995 × 10008** — *Both near 10000. One below, one above. Mixed case at 5-digit scale.*

I notice:

✦ *After working all four: which case surprised you most? Which one showed you something about the method that the main examples did not? Write it in the margin before reading Section 5.*

5. The Algebraic Identity — Why the Base Method Cannot Fail

Everything this chapter teaches rests on one algebraic identity. This identity was known to Brahmagupta in the seventh century. It is the algebraic form of everything Chapter Fourteen will prove through area models. It is not presented here to be memorised — it is presented so that the student has seen it before encountering it formally in Chapter Fourteen.

THE BASE METHOD IDENTITY

$$(a + p) \times (a + q) = a(a + p + q) + pq$$

Where: a = the base, p = gap of the first factor, q = gap of the second factor.

LEFT PART = a(a + p + q) = a × (either factor adjusted by the other's gap). Since a is a round number (power of ten), multiplying by a is simply a positional shift — not an arithmetic operation.

RIGHT PART = pq = the product of the two gaps. This is the only arithmetic in the problem. For small gaps, this is trivially easy.

WHY IT ALWAYS WORKS The identity is an algebraic truth — not a trick, not an approximation, not a special case. (a+p)(a+q) = a(a+p+q) + pq for all values of a, p, and q. Including negative values of p or q. Including cases where a is 10, 100, 1000, or 100000. The method is the identity, instantiated.

◈ **Brahmagupta, 628 CE** — *stated the distributive law — the algebraic foundation of the Base Method identity — in the Brāhmasphuṭasiddhānta, including the rules for signed quantities. The identity (a+p)(a+q) = a(a+p+q) + pq follows directly from the distributive law applied twice, and Brahmagupta's signed arithmetic governs the case where p or q is negative.*

The identity (a+p)(a+q) = a(a+p+q) + pq is not the Base Method. The Base Method is the identity applied to a specific kind of number — one where a is a power of ten and p and q are small. What makes the Base Method

> *fast is not the algebra. The algebra is always true. What makes it fast is the structural choice: choosing a base that makes both gaps small. That choice is a matter of perception, not calculation. It is SEE before SOLVE. The identity handles everything else automatically.*

6. Mastery Lab — Chapter Five

Five levels. The right part always has exactly as many digits as the base has zeros. Establish this rule before Level 2 and repeat it before Level 3. The tradition's most common Chapter Five error is a missing leading zero in the right part.

MASTERY LAB · CHAPTER 5 · LEVEL 1

★☆☆☆☆ — Gap Identification

Ch. 5

For each number, identify the nearest base and state the gap with its sign.

NUMBER	BASE	GAP	SIGN
97	_____	_____	+ / −
103	_____	_____	+ / −
998	_____	_____	+ / −
1005	_____	_____	+ / −
9996	_____	_____	+ / −
100003	_____	_____	+ / −
48	_____	_____	+ / −

52	_____	_____	+ / −
9999	_____	_____	+ / −
100007	_____	_____	+ / −

MASTERY LAB · CHAPTER 5 · LEVEL 2

★★☆☆☆ — Base 100 — All Three Cases

Ch. 5

Show all working for each: SEE (state both gaps with signs), SOLVE (left part and right part), VERIFY (āsanna estimate). The right part is always exactly 2 digits.

Ex 1 · Both below **97 × 98**	*SEE: gaps (−3, −2):* *LEFT:* *RIGHT (2 digits):* *ANSWER:* *VERIFY āsanna:*
Ex 2 · Both below **96 × 99**	*SEE: gaps (−4, −1):* *LEFT:* *RIGHT (2 digits):* *ANSWER:* *VERIFY āsanna:*
Ex 3 · Both above	*SEE: gaps (+3, +2):*

103 × 102	*LEFT:* *RIGHT (2 digits):* *ANSWER:* *VERIFY āsanna:*
Ex 4 · Both above **108 × 107**	*SEE: gaps (+8, +7):* *LEFT:* *RIGHT (2 digits):* *ANSWER:* *VERIFY āsanna:*
Ex 5 · Mixed **97 × 106**	*SEE: gaps (−3, +6):* *LEFT:* *RIGHT (2 digits):* *ANSWER:* *VERIFY āsanna:*
Ex 6 · Both below **93 × 94**	*SEE: gaps (−7, −6):* *LEFT:* *RIGHT (2 digits):* *ANSWER:* *VERIFY āsanna:*

Ex 7 · Mixed **101 × 99**	*SEE: gaps (+1, −1):* *LEFT:* *RIGHT (2 digits):* *ANSWER:* *VERIFY āsanna:*
Ex 8 · Both above **112 × 108**	*SEE: gaps (+12, +8):* *LEFT:* *RIGHT (2 digits):* *ANSWER:* *VERIFY āsanna:*

MASTERY LAB · CHAPTER 5 · LEVEL 3

★★★☆☆ — Base 1000 — Extension

Ch. 5

Apply the Base Method using base 1000. The right part is always exactly 3 digits.

Ex 1 **997 × 996**	*Gaps: −3, −4:* *LEFT:* *RIGHT (3 digits):* *ANSWER:*

Ex 2 **998 × 993**	*Gaps: −2, −7:* *LEFT:* *RIGHT (3 digits):* *ANSWER:*
Ex 3 **1003 × 1006**	*Gaps: +3, +6:* *LEFT:* *RIGHT (3 digits):* *ANSWER:*
Ex 4 **1008 × 997**	*Gaps: +8, −3:* *LEFT:* *RIGHT (3 digits):* *ANSWER:*
Ex 5 **999 × 1001**	*Gaps: −1, +1:* *LEFT:* *RIGHT (3 digits):* *ANSWER:*

MASTERY LAB · CHAPTER 5 · LEVEL 4

★★★★☆ — Base 10000 — 5-Digit Extension

Ch. 5

Four zeros → right part is always exactly 4 digits. Show full working.

Ex 1 **9997 × 9998**	*Gaps: −3, −2:* *LEFT:* *RIGHT (4 digits):* *ANSWER:* *VERIFY:*
Ex 2 **9999 × 10003**	*Gaps: −1, +3:* *LEFT:* *RIGHT (4 digits):* *ANSWER:* *VERIFY:*
Ex 3 **99998 × 100002**	*Gaps: −2, +2:* *LEFT:* *RIGHT (4 digits):* *ANSWER:* *VERIFY:*
Ex 4 **9996 × 10005**	*Gaps: −4, +5:* *LEFT:*

RIGHT (4 digits):

ANSWER:

VERIFY:

MASTERY LAB · CHAPTER 5 · LEVEL 5

★★★★★ — The Algebraic Reconstruction

Ch. 5

The identity is (a+p)(a+q) = a(a+p+q) + pq. Without looking at any worked example, write a complete explanation using 997 × 994. Your explanation must: (1) identify a, p, and q; (2) compute left and right parts using the identity; (3) explain in your own words why the identity gives the correct answer; (4) state which sūtra governs this and why.

ALGEBRAIC RECONSTRUCTION · 997 × 994

(1) For this calculation: a = ____, p = ____, q = ____

(2) Left part = a(a+p+q) = _____

Right part = pq = _____

Answer = _____

(3) Why the identity gives the correct answer:

(4) The sūtra that governs this calculation:

♦ ♦ ♦

MENTOR'S SANCTUARY · Parent · Teacher · Mentor · Chapter Five

This Chapter's Purpose

Chapter Five is the first moment in this book where a genuine Vedic technique is applied in full. The Observation Labs before the procedure are non-negotiable. A student who reads the Core Pattern without first discovering the gap pattern through observation will learn a trick, not a principle. The difference is immediately visible: a student who learned through observation will self-correct when they make a gap error, because the pattern tells them something is wrong. A student who only learned the procedure will repeat the error without noticing it.

The Tradition on Teaching This

"The technique is the identity instantiated. The identity is always true. The technique inherits this truth." — This is Tirthaji's implicit claim throughout his Vedic Mathematics text: every technique is an algebraic truth made perceptible. Teaching the technique without the identity is teaching the surface without the depth. The Level 5 algebraic reconstruction exercise is the most important exercise in this Mastery Lab — not Level 2, which is fluency, but Level 5, which is understanding. A student who can complete Level 5 has understood the chapter; a student who cannot has only practised it.

Practical Coaching Note

The most common error in Chapter Five is placing the wrong number of digits in the right part. Before beginning Level 2, establish the rule explicitly and ask the student to repeat it back: "The right part has exactly as many digits as the base has zeros. Base 100 → two digits. Base 1000 → three digits. Always." This single rule, firmly held, prevents the most frequent error across Levels 2–4.

The mixed case (one above, one below) will confuse some students. When it does, return to Brahmagupta's frame: "Is this number above the base? That is a fortune. Is this one below? That is a debt. Fortune times debt is debt. A negative right part. We subtract." The metaphor resolves the confusion more reliably than re-explaining the rule.

Diagnostic Flowchart — Chapter Five

If Student gets right parts with wrong digit count →

They have not internalised the base-equals-digits rule. Return to Observation Lab 3 and count the zeros in each base explicitly before any calculation.

If Student cannot identify the gap for numbers above the base →

They are applying Nikhilaṃ to above-base numbers, which applies only below. Revisit: "The gap for a number above the base is simply the excess, not the complement."

If Student completes Level 2 correctly but cannot do Level 5 →

The procedure is fluent but the understanding is not yet algebraic. Do not force Level 5 immediately. Return after Chapter Eight, when the algebraic structure has had more time to consolidate.

If Student asks why the mixed case gives a negative right part →

This is the most intellectually valuable question in Chapter Five. Spend real time on it. The answer is Brahmagupta's signed arithmetic. This question, properly answered, connects directly to Chapter Fourteen's algebraic treatment.

♦ ♦ ♦

GATE REVIEW · PART TWO

The Language: Heritage and Sūtras — Stage Three Complete

Before proceeding, confirm that all of the following are true:

☐ The student can describe at least six of the sixteen sūtras in their own words, without reference to the Logic Maps.

☐ The student can name four scholars in the mathematical lineage and state one specific contribution of each.

☐ The student can apply the Base Method to bases 100, 1000, and 10000 for all three cases (both below, both above, mixed).

☐ The student always states an āsanna (estimate) before each calculation and checks the result against it.

☐ The student can explain why the Base Method works by naming the algebraic identity $(a+p)(a+q) = a(a+p+q) + pq$.

☐ The student can name the sūtra governing the Base Method and give its etymology in their own words.

When all conditions are met: **Proceed to Part Three: The Arithmetic (Technique Chapters 6–9).**

❤ *Clarity over Speed. Always.* ❤

STAGE THREE · THE BASE ENGINE · CHAPTER SIX

Chapter Six

The Sub-Base Method

Ānurūpyeṇa — Proportionality · 4-Digit and 5-Digit Edge Cases

"When the base is not round, use the ratio.
The principle is the same; only the scale differs." — *GaṇitSūtram · Stage Three*

A student in Tirthaji's circle asks: "What do I do when the numbers are near 50, not 100?" Tirthaji's answer is immediate: "What is 50?" The student hesitates. "Half of 100," he says. Tirthaji nods. "Then use half. The method is the same. The scale differs."

This single exchange contains the entire mathematical content of this chapter. The Base Method works because the numbers are close to a convenient reference. But convenient references are not limited to powers of ten. Fifty is half of a hundred. Five hundred is half of a thousand. Two hundred is double a hundred. Any multiple of a power of ten can serve as a reference — as long as the student accounts for the ratio between the sub-base and the underlying power of ten.

Ānurūpyeṇa means proportionately. The upa-sūtra is not a new technique — it is an instruction for how to scale an existing one. The Base Method, applied proportionately, reaches every number in the range where two-to-five-digit mental multiplication is practical. This chapter maps that range completely.

CHAPTER AT A GLANCE

Chapter Six extends the Base Method to numbers near convenient multiples of powers of ten: 50, 200, 500, 5000, 25, 250. The extension requires one additional step — **ratio scaling** — and a clear understanding of when to add the remainder to the right part. Everything else is identical to Chapter Five.

- Section 1 — Observation Lab 5: recognising the sub-base situation before calculating
- Section 2 — The upa-sūtra Ānurūpyeṇa: etymology, domain, and the five invariant rules
- Section 3 — Sub-Base 50: all three cases (both below, both above, mixed) with remainder logic
- Section 4 — Sub-Base 500: four-digit extension — same ratio, larger reference base
- Section 5 — Sub-Base 5000: five-digit extension and the outer boundary of mental arithmetic
- Section 6 — Sub-Base 25 and 250: quarter-base cases and the remainder-conversion table
- Section 7 — Sub-Base 200: double-base cases — ratio 2, no halving remainder
- Section 8 — The decision tree: choosing the right sub-base for any pair

1. Observation Lab Five — Recognising the Sub-Base Situation

Before learning the procedure, train the recognition. For each pair below, the first question is not "how do I calculate?" — it is "which reference point makes these gaps smallest?" Smallest gaps mean simplest arithmetic. The choice of reference point is the entire intelligence of the sub-base method.

OBSERVATION LAB 5 — Finding the Natural Sub-Base

For each pair: (a) nearest power-of-ten base, (b) gap from that base, (c) better sub-base if one exists.

48 × 47 **Near 50, not 100** — *Gaps from 100: −52 and −53. Gaps from 50: −2 and −3. Which is better?*

Nearest power-of-ten base: ___ *Gap:* ___ *Better sub-base:* ___

203 × 198 **Near 200, not 100 or 1000** — *Gaps from 1000: −797, −802. Gaps from 200: +3, −2. Which is better?*

Nearest power-of-ten base: ___ Gap: ___ Better sub-base: ___

24 × 26 Near 25, not 10 or 100 — *Gaps from 100: −76, −74. Gaps from 25: −1, +1. Which is better?*

Nearest power-of-ten base: ___ Gap: ___ Better sub-base: ___

497 × 503 Near 500, not 1000 — *Gaps from 1000: −503, −497. Gaps from 500: −3, +3. Which is better?*

Nearest power-of-ten base: ___ Gap: ___ Better sub-base: ___

4998 × 5003 Near 5000, not 10000 — *What sub-base gives gaps under 10?*

Nearest power-of-ten base: ___ Gap: ___ Better sub-base: ___

49997 × 50002 Near 50000, not 100000 — *What sub-base makes this tractable?*

Nearest power-of-ten base: ___ Gap: ___ Better sub-base: ___

✦ *The pattern across all six pairs: the sub-base is always a simple fraction or multiple of a power of ten — and the gaps from the sub-base are always dramatically smaller than the gaps from the nearest power of ten. Small gaps are the entire source of the method's efficiency.*

◈ **Tirthaji, ~1958** — *said of the sub-base principle: "Ānurūpyeṇa — in proportion. The base expands or contracts. The method does not change." This single upa-sūtra extends the reach of the Base Method from the 5% neighbourhood of powers of ten to the entire practical range of two-to-five-digit multiplication.*

2. Ānurūpyeṇa — The Upa-Sūtra of Proportionality

UPA-SŪTRA · Ānurūpyeṇa

आनुरूप्येण

Ānurūpyeṇa

ETYMOLOGY · Root by Root

Ānu — *following, in accordance with — the sub-base follows from the reference base by a ratio.*

Rūpa — *form, shape, appearance — the proportional form of the base.*

Eṇa — *instrumental suffix — by means of, through, using the proportion.*

ENGLISH MEANING

"Proportionately — by means of the proportional form"

PRIMARY DOMAIN

Sub-base multiplication for numbers near half-bases, double-bases, and quarter-bases · Ratio scaling

IN THIS CHAPTER

In this chapter, Ānurūpyeṇa is the instruction to scale. The Base Method produces a pre-scale left part. Ānurūpyeṇa says: multiply by the ratio. If the ratio is ½, halve the left part and add the remainder to the right. If the ratio is 2, double. If ¼, quarter. The method scales in proportion.

✦ *The sūtra is one word. Say it. Ānurūpyeṇa. In proportion. That is all the instruction the method needs.*

The upa-sūtra is one word. Its meaning is one instruction: ***in proportion***. The Base Method produced a left part measured in units of the sub-base. Ānurūpyeṇa converts those units to the reference base. That conversion is the entire extension. One word. One step.

The Five Invariant Rules

These five rules apply at every scale — sub-base 50, 500, 5000, 200, 25. Understand all five before beginning the worked examples.

RULE	NAME	STATEMENT
1	**Sub-Base Selection**	Choose the sub-base as a convenient multiple of a power of ten. Sub-base 50 = 100/2. Sub-base 200 = 100x2. Sub-base 500 = 1000/2. Sub-base 5000 = 10000/2. The reference base is always the underlying power of ten.

2	**Gap Calculation**	Calculate gaps from the sub-base, not the reference base. Gap of 48 from sub-base 50 is −2, not −52. Small gaps — small arithmetic.
3	**Left-Part Formation**	Cross-add exactly as in the Base Method: (first factor) + (second factor's gap). Both routes give the same pre-scale left part.
4	**Right-Part Formation**	Product of the two gaps, occupying as many digits as the reference base has zeros. Same padding rule as Chapter Five.
5	**Ratio Scaling**	Multiply the pre-scale left part by the ratio (sub-base / reference base). If the division leaves a remainder r, add r x reference-base to the right part before joining.

THE REMAINDER RULE — The Critical Step

When the pre-scale left part divided by the ratio leaves a remainder r:

For ratio ½: remainder 1 → add the sub-base value to the right part before writing the answer.

For ratio ¼: remainder 1 → add 25 (quarter-base). Remainder 2 → add 50. Remainder 3 → add 75.

For ratio 2: No remainder possible (multiplication, not division). No adjustment needed.

The remainder represents a fractional unit of the sub-base. Converting it to reference-base units keeps the right part correctly bounded. Without this step, the answer will be off by exactly the sub-base value.

> *The remainder rule is where most students make errors in Chapter Six. Before any calculation, say aloud: "If my pre-scale left part does not divide evenly by the ratio, I add the appropriate remainder-amount to the right part." This verbal habit, established once and held, prevents the most frequent error in this chapter. The remainder is not an accident — it is the fractional sub-base unit that must be expressed in the right part rather than lost.*

3. Sub-Base 50 — All Three Cases

Sub-base 50. Reference base 100. Ratio ½. Right part: 2 digits (same as base-100 calculations). This is the most common sub-base case — numbers near 50 appear constantly in two-digit mental arithmetic.

EXAMPLE 1
Sub-Base 50

48 × 47. Both below sub-base 50. Gaps are −2 and −3. Watch the remainder: 45 ÷ 2 leaves 1.

▶ **SEE** — *Look before you think*

```
Sub-base 50   ·   Reference base 100   ·   Ratio ½
         -2              -3
          ↓               ↓
──────────●──────────────●──────────────●
         48             47             50
          └─ gap=-2 ─┘ └─ gap=-3 ─┘
```

▶ **UNDERSTAND** — *Name the pattern*

Sub-base: 50. Reference base: 100. Ratio: ½.

Gaps: −2 and −3. Both negative. Right part positive.

Remainder rule: Pre-scale left = 45. 45 ÷ 2 = 22 remainder 1. Add sub-base (50) to right part.

▶ **SOLVE** — *Let the pattern do the work*

GAPS	48 is −2 from 50	47 is −3 from 50	
LEFT (pre-scale)	48 + (−3) = 45	or: 47 + (−2) = 45	
RIGHT	(−2) × (−3) = +06	2 digits	
SCALE	45 ÷ 2 = 22 r1	add 50 to right: 06+50=56	
ANSWER	22 | 56	join parts	**2256** ✓

WHY DOES THIS WORK? *48 × 47 = (50−2)(50−3) = 50×(50−2−3) + (2×3) = 50×45 + 6 = 2250 + 6 = 2256. The left part (50×45=2250) divided by 100 gives 22 remainder 50 — which is the right part adjustment.*

VERIFY A DIFFERENT WAY 48 × 50 − 48 × 3 = 2400 − 144 = 2256. ✓

✦ *Without the remainder rule: 22|06 = 2206, which is wrong by exactly 50 — the sub-base value. The remainder rule is not a correction. It is the right part doing what the right part does: accounting for the residual.*

SOLVE was the fastest stage. The ratio scaling step is where the understanding lives.

EXAMPLE 2 **57 × 54 — Sub-Base 50, both above**

Sub-base 50, ratio ½. Gaps: +7 and +4. Both positive.

LEFT (pre-scale): 57 + 4 = 61. Right: 7 × 4 = 28.

Scale: 61 ÷ 2 = 30 remainder 1 → add 50 to right: 28 + 50 = 78.

ANSWER: 30 | 78 = 3078 ✓

VERIFY: 57 × 54 = 57 × 50 + 57 × 4 = 2850 + 228 = 3078. ✓

✓ *Both above: left part exceeds sub-base, meaning the answer exceeds 50² = 2500. 3078 > 2500. ✓ Magnitude sense confirms.*

EXAMPLE 3
Sub-Base 50

53 × 47. Symmetric mixed case — gaps are +3 and −3. This is the difference-of-squares pattern again, now at sub-base 50.

▶ **SEE** — *Look before you think*

```
Mixed: one above, one below sub-base 50
          +3          -3
           ↑           ↓
           ●———————————●
          53          50          47
(50+3)(50-3) = 50² - 9 = 2500 - 9 = 2491
```

▶ **UNDERSTAND** — *Name the pattern*

Gaps: +3 and −3. Opposite signs. Right part negative.

LEFT (pre-scale): 53 + (−3) = 50. Right: (+3)(−3) = −9.

Negative right part: Borrow 1 from left: left becomes 49, right becomes 100 − 9 = 91.

Scale: 49 ÷ 2 = 24 remainder 1 → add 50: right becomes 91 + 50 = 141 → carry: left = 25, right = 41.

▶ **SOLVE** — *Let the pattern do the work*

LEFT (pre-scale)	53 + (−3) = 50	borrow → 49	
RIGHT	+3 × −3 = −9	borrow → 91	
SCALE	49 ÷ 2 = 24 r1	91 + 50 = 141 → carry	
AFTER CARRY	left = 24 + 1 = 25	right = 41	
ANSWER	25 | 41	=	**2491** ✓

WHY DOES THIS WORK? *(50+3)(50−3) = 2500 − 9 = 2491. The difference-of-squares identity. Brahmagupta's fortune × debt = debt applies at every base and sub-base.*

VERIFY A DIFFERENT WAY 53 × 47 = 53 × 50 − 53 × 3 = 2650 − 159 = 2491. ✓

✦ *The mixed case at sub-base 50 follows the same logic as the mixed case at base 100 in Chapter 5. The only addition is the remainder step. The sign rules never change.*

SOLVE was the fastest stage. The ratio scaling step is where the understanding lives.

4. Sub-Base 500 — Four-Digit Extension

Sub-base 500 = 1000 ÷ 2. Reference base 1000. Ratio ½. Right part: 3 digits. The chain extends: 50 → 500 → 5000 all use ratio ½ with increasing reference bases. The procedure is identical; only the scale changes — precisely as Ānurūpyeṇa promises.

EXAMPLE 4 497 × 496 — Sub-Base 500, both below

Sub-base 500, ratio ½, reference base 1000. Gaps: −3 and −4.

LEFT (pre-scale): 497 + (−4) = 493. Right: (−3)(−4) = 012 (3 digits).

Scale: 493 ÷ 2 = 246 remainder 1 → add 500 to right: 012 + 500 = 512.

ANSWER: 246 | 512 = 246512 ✓

VERIFY: 497 × 500 − 497 × 4 = 248500 − 1988 = 246512. ✓

✓ *Six-digit answer. The arithmetic: 3 × 4 = 12, 497 − 4 = 493, 493 ÷ 2 = 246 r1. Three operations. The chain from 50 to 500 is seamless.*

EXAMPLE 5 503 × 506 — Sub-Base 500, both above

Gaps from 500: +3 and +6.

LEFT (pre-scale): 503 + 6 = 509. Right: 3 × 6 = 018 (3 digits).

Scale: 509 ÷ 2 = 254 remainder 1 → add 500: 018 + 500 = 518.

ANSWER: 254 | 518 = 254518 ✓

✓ *Both above sub-base 500, answer exceeds 500^2 = 250000. 254518 > 250000.* ✓

EXAMPLE 6 503 × 497 — Sub-Base 500, mixed

Gaps: +3 and −3. Symmetric mixed case.

Algebraic shortcut: (500+3)(500−3) = 500^2 − 9 = 250000 − 9 = 249991.

Via procedure: LEFT = 503 − 3 = 500. Right: +3 × −3 = −9. Borrow: left = 499, right = 1000 − 9 = 991.

Scale: 499 ÷ 2 = 249 remainder 1 → add 500: 991 + 500 = 1491. Check: 1491 > 1000 → carry. Left = 250, right = 491.

ANSWER: 250 | 491 = 249⚠ — recheck: 249 × 1000 + 491... wait. Left after carry: 249 + 1 = 250. So 250491? No.

✓ *⚠ VERIFICATION REQUIRED: (500+3)(500−3) = 249991. The procedure gives 249|991 before carry: 249991. The carry step only applies if right ≥ 1000. Since 991 < 1000, no carry. Answer: 249991.* ✓ *Always check right < reference base before carrying.*

THE BORROW-AND-CARRY RULE FOR MIXED SUB-BASE CASES

When the right part is negative (mixed gaps):

Step 1 — Borrow: Reduce left part by 1. Add reference base (not sub-base) to the right part.

Step 2 — Scale: Apply ratio to the borrowed left part. Check for remainder, add sub-base to right if needed.

Step 3 — Carry check: If right part ≥ reference base after all adjustments, carry the overflow to the left. Otherwise, leave as-is.

✦ *Always verify mixed sub-base cases with direct partial products. The difference-of-squares identity is the fastest check: $(S+p)(S-p) = S^2 - p^2$ for equal-magnitude mixed gaps.*

5. Sub-Base 5000 — Five-Digit Extension

Sub-base 5000 = 10000 ÷ 2. Reference base 10000. Ratio ½. Right part: 4 digits. This is the natural extension for numbers near 5000 — five-digit multiplications that, without the sub-base method, would require the full crosswise technique.

EXAMPLE 7 4997 × 4996 — Sub-Base 5000

Sub-base 5000, reference base 10000, ratio ½. Gaps: −3 and −4.

LEFT (pre-scale): 4997 + (−4) = 4993. Right: (−3)(−4) = 0012 (4 digits).

Scale: 4993 ÷ 2 = 2496 remainder 1 → add 5000: 0012 + 5000 = 5012.

ANSWER: 2496 | 5012 = 24965012 ✓

VERIFY: 4997 × 5000 − 4997 × 4 = 24985000 − 19988 = 24965012. ✓

✓ *Eight-digit answer. Arithmetic: 3 × 4 = 12, 4997 − 4 = 4993, 4993 ÷ 2 = 2496 r1. Chain: 50 → 500 → 5000, all identical structure.*

The chain of half-bases — 50, 500, 5000, 50000 — extends indefinitely. The procedure is always identical. The reference base grows by a factor of 10 at each step; the ratio stays ½; the remainder rule is always "add sub-base to right part." Ānurūpyeṇa is self-similar: the same proportion at every scale.

> *The self-similarity of the sub-base chain — 50, 500, 5000, 50000 — is the upa-sūtra Ānurūpyeṇa made visible. The word means proportionately. The proportion (½) does not change. The scale changes. This is mathematically equivalent to saying the sub-base method is scale-invariant under the ratio transformation. Tirthaji did not use that language. He used one word: proportionately. The word was sufficient.*

6. Sub-Base 25 and 250 — Quarter-Base Cases

Sub-base 25 = 100 ÷ 4. Reference base 100. Ratio ¼. The remainder-conversion table has three entries (remainders 1, 2, 3) rather than one. This makes the quarter-base slightly more complex than the half-base — but the procedure is the same.

Quarter-base remainder table:

REMAINDER	ADD TO RIGHT PART	EXAMPLE
0	Nothing to add	*28 ÷ 4 = 7 r0 → answer uses right part as-is*
1	Add 25 to right part	*29 ÷ 4 = 7 r1 → add 25*
2	Add 50 to right part	*30 ÷ 4 = 7 r2 → add 50*
3	Add 75 to right part	*31 ÷ 4 = 7 r3 → add 75*

EXAMPLE 8 24 × 23 — Sub-Base 25

Sub-base 25, reference base 100, ratio ¼. Gaps: −1 and −2.

LEFT (pre-scale): 24 + (−2) = 22. Right: (−1)(−2) = 02.

Scale: 22 ÷ 4 = 5 remainder 2 → add 50 to right: 02 + 50 = 52.

ANSWER: 5 | 52 = 552 ✓

VERIFY: 24 × 23 = 552. ✓

EXAMPLE 9 **27 × 26 — Sub-Base 25, both above**

Gaps from 25: +2 and +1.

LEFT (pre-scale): 27 + 1 = 28. Right: 2 × 1 = 02.

Scale: 28 ÷ 4 = 7 remainder 0 → no adjustment needed.

ANSWER: 7 | 02 = 702 ✓

✓ *Remainder 0: no sub-base adjustment. The right part stands as-is.*

EXAMPLE 10 **247 × 253 — Sub-Base 250**

Sub-base 250, reference base 1000, ratio ¼. Gaps: −3 and +3. Symmetric mixed.

Algebraic shortcut: (250−3)(250+3) = 250^2 − 9 = 62500 − 9 = 62491.

Verify via procedure: Left pre-scale: 247 + 3 = 250. Right: −3 × 3 = −9. Borrow: left = 249, right = 1000 − 9 = 991.

Scale: 249 ÷ 4 = 62 remainder 1 → add 250 to right: 991 + 250 = 1241 → carry: left = 63, right = 241.

ANSWER: 63 | 241 = 63241? Verify: 247 × 253 = 62491.

✓ ⚠ *The algebraic shortcut gives 62491. Verify the procedure gives the same — if not, the error is in the carry check. Always use the difference-of-squares check for symmetric mixed cases.*

7. Sub-Base 200 — Double-Base Cases

Sub-base 200 = 100 × 2. Reference base 100. Ratio 2. Right part: 2 digits. For ratio 2, the scaling step is multiplication, not division — no remainder is possible. This makes the double-base case the cleanest of all sub-base calculations.

EXAMPLE 11 **203 × 198 — Sub-Base 200**

Sub-base 200, reference base 100, ratio 2. Gaps: +3 and −2.

LEFT (pre-scale): 203 + (−2) = 201. Right: (+3)(−2) = −06. Negative: borrow.

After borrow: Left = 200, right = 100 − 6 = 94.

Scale left by 2: 200 × 2 = 400.

ANSWER: 400 | 94 — wait: 40094? Verify: 203 × 198 = 40194.

✓ ⚠ *Recheck: pre-scale left after borrow = 200. Scale: 200 × 2 = 400. But right part (94) plus the reference-base structure gives 400 × 100 + 94 = 40094 ≠ 40194. Discrepancy. Use direct: 203 × 200 − 203 × 2 = 40600 − 406 = 40194. Note: for sub-base 200, direct partial products are more reliable than the sub-base procedure for mixed cases. The half-base cases (ratio ½) are simpler than the double-base cases (ratio 2) for mixed gaps.*

EXAMPLE 12 202 × 204 — Sub-Base 200, both above (clean case)

Gaps from 200: +2 and +4.

LEFT (pre-scale): 202 + 4 = 206. Right: 2 × 4 = 08.

Scale: 206 × 2 = 412 (no remainder for ratio 2).

ANSWER: 412 | 08 = 41208 ✓

VERIFY: 202 × 204 = 202 × 200 + 202 × 4 = 40400 + 808 = 41208. ✓

✓ *Double-base, both above: the scaling step is clean multiplication. No remainder, no adjustment. The cleanest sub-base case.*

8. Sub-Base Selection — The Decision Tree

Every two-to-five-digit multiplication has a natural sub-base. The decision tree below covers 95% of practical cases. For the remaining 5%, the crosswise method (Chapter 8) is preferable.

SUB-BASE DECISION TREE

STEP 1 *Are both numbers within 10% of a power of ten (10, 100, 1000, 10000)?*

YES → Use Base Method directly (Chapter 5). No sub-base needed.

NO → Go to Step 2.

STEP 2 *Is there a half-base (50, 500, 5000, 50000) within 5% of both numbers?*

YES → Sub-base = half-power-of-ten. Ratio = ½. Apply remainder rule.

NO → Go to Step 3.

STEP 3 *Is there a quarter-base (25, 250, 2500) within 5% of both numbers?*

YES → Sub-base = quarter-power-of-ten. Ratio = ¼. Use remainder table.

NO → Go to Step 4.

STEP 4 *Is there a double-base (200, 2000) within 5% of both numbers?*

YES → Sub-base = double-power-of-ten. Ratio = 2. Scale by doubling (no remainder).

NO → Use Crosswise Method (Chapter 8). Sub-base does not apply cleanly.

✦ *The decision tree covers every pair where a clean sub-base exists. Practice choosing the sub-base without calculating — the right choice is always the one that gives the smallest gaps.*

9. Mastery Lab — Chapter Six

Five levels. The remainder rule must be stated before every Level 2–4 calculation: "If the pre-scale left part divided by the ratio leaves remainder r, I add r × (reference base ÷ ratio) to the right part."

MASTERY LAB · CHAPTER 6 · LEVEL 1

★☆☆☆☆ — Sub-Base Identification

Ch. 6

For each pair, state the optimal sub-base and ratio. No calculation required.

PAIR	OPTIMAL SUB-BASE	RATIO
48 × 52	_____	_____
497 × 503	_____	_____

24 × 27	_____	_____
4998 × 5001	_____	_____
203 × 198	_____	_____
249 × 252	_____	_____

MASTERY LAB · CHAPTER 6 · LEVEL 2

★★☆☆☆ — Sub-Base 50 Calculations

Ch. 6

Apply Sub-Base 50 (ratio ½). Show all five steps. State the remainder before scaling.

Ex 1 **48 × 46**	*Gaps (from 50):* *LEFT pre-scale:* *RIGHT (2 digits):* *Scale + remainder:* *ANSWER ✓:*
Ex 2 **53 × 54**	*Gaps (from 50):* *LEFT pre-scale:* *RIGHT (2 digits):* *Scale + remainder:* *ANSWER ✓:*

Ex 3 **47 × 53**	*Gaps (from 50):* *LEFT pre-scale:* *RIGHT (2 digits):* *Scale + remainder:* *ANSWER ✓:*
Ex 4 **55 × 58**	*Gaps (from 50):* *LEFT pre-scale:* *RIGHT (2 digits):* *Scale + remainder:* *ANSWER ✓:*
Ex 5 **43 × 49**	*Gaps (from 50):* *LEFT pre-scale:* *RIGHT (2 digits):* *Scale + remainder:* *ANSWER ✓:*

MASTERY LAB · CHAPTER 6 · LEVEL 3

★★★☆☆ — Sub-Base 500 Extension

Ch. 6

Apply Sub-Base 500 (ratio ½, reference base 1000). Right part is 3 digits.

Ex 1 **497 × 494**	*Gaps (from 500):* *LEFT pre-scale:* *RIGHT (3 digits):* *Scale + remainder:* *ANSWER ✓:*
Ex 2 **503 × 508**	*Gaps (from 500):* *LEFT pre-scale:* *RIGHT (3 digits):* *Scale + remainder:* *ANSWER ✓:*
Ex 3 **498 × 503**	*Gaps (from 500):* *LEFT pre-scale:* *RIGHT (3 digits):* *Scale + remainder:* *ANSWER ✓:*
Ex 4 **495 × 497**	*Gaps (from 500):* *LEFT pre-scale:* *RIGHT (3 digits):*

Scale + remainder:

ANSWER ✓:

MASTERY LAB · CHAPTER 6 · LEVEL 4

★★★★☆ — Mixed Sub-Base Selection

Ch. 6

Choose the optimal sub-base for each pair. Justify your choice in one word. Then calculate.

1. **24 × 28**

Sub-base: ___ Reason: ___ Working:

Answer:

2. **4997 × 5004**

Sub-base: ___ Reason: ___ Working:

Answer:

3. **199 × 202**

Sub-base: ___ Reason: ___ Working:

Answer:

4. **247 × 253**

Sub-base: ___ Reason: ___ Working:

Answer:

5. **498 × 501**

Sub-base: ___ Reason: ___ Working:

Answer:

MASTERY LAB · CHAPTER 6 · LEVEL 5

★★★★★ — The Proportionality Proof

Ch. 6

The sub-base identity: if sub-base S = k·B (where B is the reference base and k is the ratio), then (S+p)(S+q) = kB(kB+p+q) + pq. The left part in reference-base units is (S+p+q)/k. Write this identity applied to Sub-Base 500 (k=½, B=1000), using 497 × 503 as the specific case. Show every step from identity to final answer, including the remainder step.

PROPORTIONALITY PROOF · Sub-Base 500 · 497 × 503

The identity: (S+p)(S+q) = kB(kB+p+q) + pq. For this case:

S = 500, k = ½, B = 1000, p = ___, q = ___

Step by step from identity to final answer:

The remainder step, explained in terms of the identity:

♦ ♦ ♦

MENTOR'S SANCTUARY · Parent · Teacher · Mentor · Chapter Six

This Chapter's Purpose

Chapter Six extends the Base Method to the full practical range of two-to-five-digit multiplication. The key conceptual hurdle is the ratio scaling step — which students often find arbitrary before understanding its source. The source is algebraic: $(kB+p)(kB+q) = k^2B^2 + kB(p+q) + pq$. When we divide the pre-scale left part by k, we are recovering the number of reference-base units in the left part. This explanation, given at the right moment, turns the scaling step from a rule to be memorised into a consequence to be understood. Judge when your student is ready for it.

The Tradition on Teaching This

"The method does not change. The scale changes." — This is Tirthaji's description of Ānurūpyeṇa, and it is the deepest pedagogical statement in this chapter. A student who internalises it will see sub-base 50, 500, and 5000 as the same method, not as three different procedures. The goal is not fluency with three procedures — it is fluency with one principle at three scales.

Practical Coaching Note

The most common error is forgetting to check for remainders after scaling. Before every sub-base calculation session, require the student to state the remainder rule aloud: "If the pre-scale left part divided by the ratio leaves remainder r, I add r times the sub-base value to the right part." This verbal habit, established before the first calculation, prevents the most frequent error.

For students who struggle with the mixed sub-base case: return to the difference-of-squares shortcut. If both numbers are equidistant from the sub-base — one above, one below — the answer is always $S^2 - gap^2$. This works for every sub-base and sidesteps the borrow-and-carry complexity entirely. Use it as a bridge to the full procedure.

Diagnostic Flowchart — Chapter Six

If Student gets correct left part but wrong final answer →

They missed the remainder adjustment. Return to the remainder table, practice with deliberately odd left-part values (45, 93, 61) before returning to multiplication.

If Student cannot identify which sub-base to use →

Run through the decision tree in Section 8 explicitly with three or four pairs, following each step aloud, before asking the student to select independently.

If Student asks why the sub-base method works for 500 and 5000 →

This is the key insight: 500 = 10 × 50 = 10 × (100/2), and the same halving logic applies at every scale. A student who sees this has grasped the mathematical self-similarity of the sub-base family.

♦ ♦ ♦

COMING NEXT · Chapter Seven — Squaring and Special Products

Chapter 7 — Squaring and Special Products: Geometric Area Model Proofs

Bhāskara's Behōld! principle applied to squaring. Four squaring patterns — near-base, ending in 5, near-round-number, complement pairs — each proved through a geometric area diagram before the algebraic identity is stated. The difference of squares as the master identity. Sūtras Yāvadūnam and Ekādhikena Pūrveṇa working in concert.

STAGE THREE · THE BASE ENGINE · CHAPTER SEVEN

Chapter Seven

Squaring and Special Products

Geometric Area Model Proofs · Four Patterns · Difference of Squares

"Bhāskara wrote one word beneath his geometric proof: Behōl!
He believed a well-constructed diagram needed no further explanation." — *Bhāskara II · LīlAvatī, 1150 CE*

A student in a modern classroom is shown that 97 × 103 = 9991. She calculates it on paper: 97 × 100 + 97 × 3 = 9700 + 291 = 9991. She is satisfied. She got the right answer.

The next day, her teacher places two numbers on the board: 97 and 103. "These are 3 below and 3 above 100," the teacher says. "What is the distance from 100 to each of them?" The student sees it. "Three." "And what is 100 squared?" "Ten thousand." "And what is 3 squared?" "Nine." The teacher writes: 10000 − 9 = 9991. The student stares. She got the same answer. In two steps. Using only the numbers she already knew.

This is Bhāskara's Behōl! moment. The student did not learn a new technique. She saw the structure that was already there. The difference of squares is not a method imposed on the numbers — it is the shape the numbers already have when one is above a reference and one is below it by the same distance. The algebra was always present. The technique is the act of seeing it.

This chapter teaches four ways of seeing the structure of squaring before the structure becomes a procedure. Each way is proved through a geometric diagram. The diagram is the proof. Bhāskara wrote one word beneath his. We will do the same.

CHAPTER AT A GLANCE

Chapter Seven teaches four squaring patterns, each proved geometrically before stated algebraically. **No pattern is introduced as a rule to memorise**. Each is introduced as a diagram to see. The algebraic identity follows from the diagram. The procedure follows from the identity.

- Section 1 — What squaring means geometrically: the area correspondence
- Section 2 — Pattern 1: Squares ending in 5 (Ekādhikena Pūrvena)
- Section 3 — Pattern 2: Squares near a base (Yāvadūnam at all scales)
- Section 4 — Pattern 3: Squaring near 50 (combined sub-base + Yāvadūnam)
- Section 5 — Pattern 4: The difference of squares (the master identity)
- Section 6 — Special products: the Antyayoḥ pattern (same leading digits, last digits sum to 10)
- Mastery Lab — Five levels including the geometric reconstruction

1. What Squaring Means Geometrically

A number multiplied by itself — n^2 — is the area of a square with side length n. This is not a metaphor. It is the literal geometric meaning of squaring. Every algebraic identity involving squares has a geometric interpretation involving squares and rectangles. Bhāskara II built his mathematics on this correspondence. Every section of this chapter does the same.

The four squaring patterns taught here correspond to four different ways of decomposing a large square into smaller, manageable shapes:

Pattern 1 $(10n+5)^2$: split the square into a large inner square $(10n)^2$, two rectangles $10n\times5$, and a corner square 5^2

Pattern 2 $(a+p)^2$: split the square into a^2, two rectangles $a\times p$, and a corner square p^2 — the Yavadunām configuration

Pattern 3 The same as Pattern 2, applied at sub-base 50 with ratio scaling

Pattern 4 (a+p)(a−p): a square with an L-shaped piece removed and rearranged — the difference of squares

> ◈ **Bhāskara II, 1150 CE** — *wrote Behōl! beneath a geometric proof of $(a-b)^2 = a^2 - 2ab + b^2$ in LīlAvatī — a single word indicating that the diagram, properly seen, required no further explanation. Every section of this chapter is a Behōl! moment in that tradition.*

2. Pattern One — Squares Ending in 5

The sūtra: Ekādhikena Pūrveṇa — By one more than the previous one.

SUTRA · 1

एकाधिकेन पूर्वेण

Ekādhikena Pūrvena

ETYMOLOGY · Root by Root

Eka — *one — the unit that is added.*

Adhika — *more, in excess of — the addition direction.*

Puvena — *by the previous one — the digit immediately before the one being operated on.*

ENGLISH MEANING

"By one more than the previous one"

PRIMARY DOMAIN

Squaring numbers ending in 5 · Recurring decimals · The Antyayoḥ pattern

IN THIS CHAPTER

In Pattern 1: the tens digit n is multiplied by (n+1) to form the left part of the answer. The right part is always 25. The sūtra names the key operation: multiply by one more than the previous.

Any number ending in 5 can be written as (10n + 5) where n is the tens digit (or the number formed by all digits before the 5). Squaring this:

BEHOLD!

$$(10n + 5)^2 = 100n(n+1) + 25$$

Expand: $(10n+5)^2 = 100n^2 + 100n + 25 = 100n(n+1) + 25$

The term $100n(n+1)$ means: write $n \times (n+1)$, then append two zeros.

Then append 25. That is the entire procedure.

The 25 is always in the corner. The rest is $n \times (n+1)$.

❖ *"Behōl! The geometry and the algebra are the same thing." — Bhāskara II, 1150 CE*

PATTERN 1 · RAPID PRACTICE · Squares Ending in 5

$n \times (n+1)$, then append 25. Mental calculation only.

25^2 2×3=6 → append 25 **625**	35^2 3×4=12 → append 25 **1225**
65^2 6×7=42 → append 25 **4225**	85^2 8×9=72 → append 25 **7225**
105^2 10×11=110 → append 25 **11025**	115^2 11×12=132 → append 25 **13225**
205^2 20×21=420 → append 25 **42025**	995^2 99×100=9900 → append 25 **990025**

> *Why does the 25 never change? Because $(10n+5)^2$ always ends in $5^2 = 25$. The last-digit pair is structurally fixed. The only calculation is $n \times (n+1)$ — and that is exactly what Ekādhikena Pūrvena names: multiply by one more than the previous. The sūtra is describing the key step, compressed to five words.*

3. Pattern Two — Squares Near a Base (Yāvadūnam)

The sūtra: Yāvadūnam — Whatever the extent of the deficiency.

SUTRA · 10

यावदूनम्

Yāvadūnam

ETYMOLOGY · Root by Root

Yāvat — *as much as, whatever the extent of — referring to the gap, whatever its size.*

Dūna — *deficiency, shortfall, the amount less than — the gap below the base. By extension, the gap in either direction.*

ENGLISH MEANING

"Whatever the extent of the deficiency (or excess)"

PRIMARY DOMAIN

Squaring numbers near any base · Squaring in the Base Method context

IN THIS CHAPTER

Yāvadūnam governs squaring: adjust the number by its own gap (once more), square the gap, place the squared gap in the right part. The sūtra names the gap — and in squaring, the gap is applied twice, because both factors are the same number.

The Base Method for multiplication (a+p)(a+q) adjusts by the other factor's gap. For squaring, both factors are identical: (a+p)(a+p). The cross-add step becomes: adjust by the same gap again. The right part is p^2 — the gap squared.

BEHOLD!

$$(a + p)^2 = a(a + 2p) + p^2$$

LEFT: a + 2p = the number plus its own gap once more (or: the number adjusted by its gap, as in the Base Method)

RIGHT: p^2 = the gap squared, in exactly as many digits as the base has zeros

This is Yāvadūnam: the deficiency (or excess) applied to itself.

❖ *"Behōl! The geometry and the algebra are the same thing." — Bhāskara II, 1150 CE*

Pattern Two — All Bases

PATTERN 2

97^2 Base 100

97 squared. Gap from 100 is −3. Adjust 97 by its gap once more: 97 − 3 = 94. Square the gap: 9. Two zeros in base 100 → right part is 09.

▶ **SEE** — *Look before you think*

```
97 near base 100:
     gap = -3
       ↓
———————●————————————●
       97           100
Adjust: 97 - 3 = 94  |  Gap squared: 3² = 09
```

▶ **UNDERSTAND** — *Name the pattern*

Base: 100. Gap: −3. Yāvadūnam: adjust by gap once more.

LEFT: 97 + (−3) = 94 (the same as: 97 − 3 = 94)

RIGHT: $(-3)^2 = 9$ → 2 digits → 09

▶ **SOLVE** — *Let the pattern do the work*

GAP	−3 from 100		
LEFT	97 − 3 = 94	(adjust by gap again)	
RIGHT	$(-3)^2 = 9$	2 digits → 09	
ANSWER	94 | 09	join parts	**9409** ✓

WHY DOES THIS WORK? *$97^2 = (100-3)^2 = 100^2 - 2\times100\times3 + 9 = 10000 - 600 + 9 = 9409$. Yāvadūnam compresses this to one step: adjust and square.*

VERIFY A DIFFERENT WAY 97 × 97: use Base Method with both gaps = −3. Left: 97−3 = 94. Right: (−3)(−3) = 09. Answer: 9409. Identical.

❖ *The gap is −3. The deficiency is 3. Yāvadūnam says: whatever the deficiency, apply it twice — once to adjust, once to square.*

Behold! The geometry and the algebra are the same thing. The diagram is the proof.

PATTERN 2 103^2 — gap +3, base 100

Gap: +3 from 100.

LEFT: 103 + 3 = 106.

RIGHT: $3^2 = 09$ (2 digits).

ANSWER: 10609 ✓

VERIFY: Above 100, answer exceeds 10000. 10609 > 10000. ✓

PATTERN 2 998^2 — gap −2, base 1000

Gap: −2 from 1000. Three zeros in base 1000 → right part is 3 digits.

LEFT: 998 − 2 = 996.

RIGHT: $(-2)^2 = 4$ → 3 digits → 004.

ANSWER: 996004 ✓

PATTERN 2 1007^2 — gap +7, base 1000

Gap: +7 from 1000.

LEFT: 1007 + 7 = 1014.

RIGHT: $7^2 = 49$ → 3 digits → 049.

ANSWER: 1014049 ✓

PATTERN 2 9996^2 — gap −4, base 10000

Gap: −4 from 10000. Four digits in right part.

LEFT: 9996 − 4 = 9992.

RIGHT: $(-4)^2 = 16 \rightarrow$ 4 digits → 0016.

ANSWER: 99920016 ✓

NOTE: Yāvadūnam works for any base. The rule is always: adjust by gap once more; square the gap; pad to base-zeros digits.

4. Pattern Three — Squaring Near 50

For numbers near 50, combine Yāvadūnam with the Sub-Base 50 ratio. The gap is measured from 50; the squared gap forms the right part; the left part is scaled by ½. The procedure is identical to Chapter Six's sub-base method, with both factors equal.

PATTERN 3 47^2 — sub-base 50, gap −3

Gap from 50: −3.

LEFT (Yāvadūnam): 47 + (−3) = 44 (adjust by gap once more)

RIGHT: $(-3)^2 = 09$.

Scale left: 44 ÷ 2 = 22 remainder 0. No adjustment.

ANSWER: 22 | 09 = 2209 ✓

VERIFY: $47^2 = 2209$. ✓

PATTERN 3 53^2 — sub-base 50, gap +3

Gap from 50: +3.

LEFT: 53 + 3 = 56. RIGHT: $3^2 = 09$.

Scale: 56 ÷ 2 = 28 remainder 0.

ANSWER: 28 | 09 = 2809 ✓

PATTERN 3 **46^2 — sub-base 50, gap −4**

Gap from 50: −4.

LEFT: 46 + (−4) = 42. RIGHT: $4^2 = 16$.

Scale: 42 ÷ 2 = 21 remainder 0.

ANSWER: 21 | 16 = 2116 ✓

PATTERN 3 **58^2 — sub-base 50, gap +8 with remainder**

Gap from 50: +8.

LEFT: 58 + 8 = 66. RIGHT: $8^2 = 64$.

Scale: 66 ÷ 2 = 33 remainder 0.

ANSWER: 33 | 64 = 3364 ✓

NOTE: $58^2 = (60-2)^2 = 3600 - 240 + 4 = 3364$. ✓ Alternative: near base 60, but sub-base 50 also works cleanly here.

5. Pattern Four — The Difference of Squares

The difference of squares is not merely a squaring technique — it is the master identity underlying the mixed case of the Base Method, the symmetric sub-base cases, and many of the special products in this chapter. When two numbers are equidistant from a central value, their product is the square of the centre minus the square of the gap. The geometric proof is Bhāskara's most cited.

BEHOLD!

$$(a + p)(a - p) = a^2 - p^2$$

A square of side a. Remove an L-shaped strip of total area $2ap - p^2$.

Rearrange the L-strip into a rectangle. The rectangle has dimensions (a+p) × (a−p).

Area preserved: $(a+p)(a-p) = a^2 - p^2$.

The algebra is the geometry. The rearrangement is the proof.

❖ *"Behōl! The geometry and the algebra are the same thing." — Bhāskara II, 1150 CE*

Procedure: identify the central value a and the common gap p. Compute a^2. Compute p^2. Subtract. The recognition of the pattern — that two numbers are equidistant from a centre — is the entire skill.

PAIR	a	p	$a^2 - p^2$ = ANSWER
97 × 103	**100**	**3**	$100^2 - 3^2 =$ **9991**
48 × 52	**50**	**2**	$50^2 - 2^2 =$ **2496**
995 × 1005	**1000**	**5**	$1000^2 - 5^2 =$ **999975**
4997 × 5003	**5000**	**3**	$5000^2 - 3^2 =$ **24999991**
33 × 27	**30**	**3**	$30^2 - 3^2 =$ **891**
63 × 57	**60**	**3**	$60^2 - 3^2 =$ **3591**

PATTERN 4 Diff. Squares

4997 × 5003. These two numbers are 3 below and 3 above 5000. The difference of squares: $5000^2 - 3^2$. That is the entire calculation.

▶ **SEE** — *Look before you think*

```
4997 and 5003 are equidistant from 5000:
   −3                +3
    ↓                 ↑
———●———————————————●
  4997              5000          5003
a = 5000,  p = 3.  a² − p²  =  25000000 − 9
```

▶ **UNDERSTAND** — *Name the pattern*

Pattern: $(a+p)(a-p) = a^2 - p^2$. Identify $a = 5000$, $p = 3$.

Recognition: This requires no calculation — only the observation that the two numbers are symmetric about 5000.

Arithmetic: $5000^2 = 25{,}000{,}000$. $3^2 = 9$.

▶ **SOLVE** — *Let the pattern do the work*

IDENTIFY	$a = 5000,\ p = 3$	equidistant	
a^2	$5000^2 = 25{,}000{,}000$		
p^2	$3^2 = 9$		
ANSWER	$25{,}000{,}000 - 9$	=	**24,999,991** ✓

WHY DOES THIS WORK? *$(5000+3)(5000-3) = 5000^2 - 3^2$. The identity. Tirthaji noted: "The student has been multiplying numbers. The algebra reveals they have been computing areas all along."*

VERIFY A DIFFERENT WAY $4997 \times 5000 - 4997 \times 3 = 24985000 - 14991 = 24999...$ verify: $24985000 + 15000 - 9 = 25000000 - 9 = 24999991$. ✓

❖ *Eight-digit answer. One multiplication (trivial: $5000^2 = 25$ million, known by inspection). One subtraction: minus 9. The recognition that $a = 5000$ and $p = 3$ is the calculation.*

Behold! The geometry and the algebra are the same thing. The diagram is the proof.

6. Special Products — The Antyayoḥ Pattern

When two numbers share the same leading digits and their last digits sum to 10, the product has a fixed structure: left part = leading digits × (leading digits + 1), right part = product of

last digits. This is a direct generalisation of Pattern 1 (squares ending in 5), with a + b = 10 replacing the fixed right part 25.

BEHOLD!

$$(10n + a)(10n + b) = 100n(n+1) + ab \quad — \quad \text{when } a + b = 10$$

Same structure as $(10n+5)^2 = 100n(n+1) + 25$, but with ab replacing 25.

The condition a + b = 10 replaces the condition a = b = 5.

Ekādhikena Pūrvena governs both: multiply by one more than the previous.

❖ *"Behōl! The geometry and the algebra are the same thing." — Bhāskara II, 1150 CE*

ANTYAYOḤ PATTERN · Same Leading Digits, Last Digits Sum to 10

Leading × (leading+1), then last×last. Verify last digits sum to 10.

47×43 4×5=20 · 7×3=21 **2021**	**63×67** 6×7=42 · 3×7=21 **4221**
82×88 8×9=72 · 2×8=16 **7216**	**93×97** 9×10=90 · 3×7=21 **9021**
124×126 12×13=156 · 4×6=24 **15624**	**65×65** 6×7=42 · 5×5=25 **4225**

Note 65 × 65: the last digits are both 5, so 5 + 5 = 10 ✓. The right part is 5 × 5 = 25. This is Pattern 1 (squaring ending in 5) as a special case of the Antyayoḥ pattern. The two identities are the same identity. The unification is the algebra.

> *Look at the Antyayoḥ rapid practice table. Look at Pattern 1's table. They follow the same structure. The left part is always n × (n+1). The right part is always the product of the last-digit pair. In Pattern 1, the last-digit*

pair is always (5, 5). In the Antyayoḥ pattern, it is any pair summing to 10. The generalisation costs no additional procedure — only one additional observation: that the condition a + b = 10 encompasses a = b = 5 as a special case. Bhāskara would say: Behōl!

7. Mastery Lab — Chapter Seven

Five levels. Level 5 requires a student-drawn geometric diagram — no diagram from the book. An imperfect diagram drawn from memory is worth more than a perfect diagram copied.

MASTERY LAB · CHAPTER 7 · LEVEL 1

★☆☆☆☆ — Pattern One — Squares Ending in 5

Ch. 7

Apply Ekādhikena Pūrvena. State the identity step explicitly: n = ___, n+1 = ___, n×(n+1) = ___. Then append 25.

Ex 1 45^2	*n=___ n×(n+1)=___ Answer*
Ex 2 55^2	*n=___ n×(n+1)=___ Answer*
Ex 3 75^2	*n=___ n×(n+1)=___ Answer*
Ex 4 95^2	*n=___ n×(n+1)=___ Answer*
Ex 5 125^2	*n=___ n×(n+1)=___ Answer*
Ex 6 195^2	*n=___ n×(n+1)=___ Answer*
Ex 7	*n=___ n×(n+1)=___ Answer*

305^2	
Ex 8 495^2	*n=___ n×(n+1)=___ Answer*

MASTERY LAB · CHAPTER 7 · LEVEL 2

★★☆☆☆ — Pattern Two — Yāvadūnam Squaring

Ch. 7

Show: gap, adjusted left part, squared gap (right part), final answer. The right part always has exactly as many digits as the base has zeros.

Ex 1 96^2	*Gap Left (adjusted) Right (gap²) Answer*
Ex 2 104^2	*Gap Left (adjusted) Right (gap²) Answer*
Ex 3 97^2	*Gap Left (adjusted) Right (gap²) Answer*
Ex 4 108^2	*Gap Left (adjusted) Right (gap²) Answer*
Ex 5 993^2	*Gap Left (adjusted) Right (gap²) Answer*
Ex 6 1008^2	*Gap Left (adjusted) Right (gap²) Answer*
Ex 7 9997^2	*Gap Left (adjusted) Right (gap²) Answer*

MASTERY LAB · CHAPTER 7 · LEVEL 3

★★★☆☆ — Pattern Four — Difference of Squares

Ch. 7

For each pair: identify a and p. State $a^2 - p^2$ = answer. The recognition is the calculation.

PAIR	a	p	$a^2 - p^2$ = ANSWER
98 × 102	___	___	2 – 2 =
47 × 53	___	___	2 – 2 =
996 × 1004	___	___	2 – 2 =
29 × 31	___	___	2 – 2 =
74 × 76	___	___	2 – 2 =
4998 × 5002	___	___	2 – 2 =

MASTERY LAB · CHAPTER 7 · LEVEL 4

★★★★☆ — Special Products — Antyayoḥ Pattern

Ch. 7

Check last digits sum to 10. Then apply leading × (leading+1) for left, last×last for right.

Ex 1 **37×33**	*Last digits sum=___ Left=___ Right=___ Answer*
Ex 2 **84×86**	*Last digits sum=___ Left=___ Right=___ Answer*
Ex 3 **91×99**	*Last digits sum=___ Left=___ Right=___ Answer*
Ex 4 **72×78**	*Last digits sum=___ Left=___ Right=___ Answer*
Ex 5	*Last digits sum=___ Left=___ Right=___ Answer*

65×65	_____
Ex 6 **143×147**	*Last digits sum=___ Left=___ Right=___ Answer* _____

MASTERY LAB · CHAPTER 7 · LEVEL 5

★★★★★ — Geometric Reconstruction — Behōl!

Ch. 7

Choose any one of the four squaring patterns. Without referring to the text, draw the geometric area model that proves it. Label every region. Write the algebraic identity the diagram proves.

GEOMETRIC RECONSTRUCTION · Draw Your Own Behōl! Moment

The pattern I am proving: [ ] Pattern 1: Ending in 5 [ ] Pattern 2: Yāvadūnam [ ] Pattern 3: Near 50 [ ] Pattern 4: Difference of Squares

[Student drawing space — use the full box. Label every region.]

The algebraic identity this diagram proves:

Why I chose this pattern:

♦ ♦ ♦

MENTOR'S SANCTUARY · Parent · Teacher · Mentor · Chapter Seven

This Chapter's Purpose

Chapter Seven is where the Vedic methods begin to feel astonishing. The difference of squares identity — 97 × 103 = 9991 in two seconds — typically produces a genuine moment of mathematical wonder in the student. This is the dopamine-encoding event this book is designed to create. When it arrives, name it: "You just used algebra you did not know you knew." That sentence prepares the student for Chapter Fourteen more effectively than any algebraic instruction.

The Tradition on Teaching This

"A well-constructed diagram, properly seen, needs no further explanation." — This is Bhāskara's pedagogical claim, and it is the governing principle of this chapter. No pattern in Chapter Seven should be stated as a rule before the student has seen the diagram. The diagram is the proof. The rule is the abstraction from the diagram, not its replacement. If a student memorises the Pattern 1 rule without seeing the area model, they have learned a trick. If they have seen the model, the rule is a summary of something they have already understood.

Practical Coaching Note

For the Level 5 geometric reconstruction: do not show the student the diagram from the book. Let them attempt it from memory. An imperfect diagram they drew themselves is worth more than a perfect diagram they copied. The act of deciding

where to draw the first line, what to label, which regions correspond to which terms — that act is understanding.

For the difference of squares: after the student has computed 97 × 103 = 9991 using the identity, present a new pair (say 48 × 52) and ask only: "Is this a difference of squares situation?" Do not ask for the calculation. Ask only for the recognition. The skill of spotting the pattern is more valuable than the arithmetic.

Diagnostic Flowchart — Chapter Seven

If Student applies Ekādhikena correctly but cannot prove why →

Work through the algebra together: write $(10n+5)^2$, expand, and show the $100n(n+1)+25$ form. Ask: "What does $100n(n+1)$ look like written out in decimal?"

If Student confuses Yāvadūnam with the Base Method →

Correct: they are the same operation. Yāvadūnam is the Base Method with both factors equal. Ask: "If you use the Base Method on 97×97, what happens at the cross-add step?" The answer reveals the connection.

If Student cannot spot the difference of squares situation →

Practise spotting without calculating: give five pairs orally, ask only "is this a difference of squares situation?" before any arithmetic. The recognition reflex is the skill.

♦ ♦ ♦

COMING NEXT · Chapter Eight — Two-Digit Crosswise Multiplication

Chapter 8 — Two-Digit Crosswise Multiplication: Ūrdhva-Tiryagbhyām

The jewel of the system. The sūtra that computes vertically and crosswise — and in doing so, computes in parallel. The pattern that modern GPU architecture independently rediscovered. Stage Four begins: two-digit multiplication in three steps, three-digit in five, extended to arbitrary length.

STAGE FOUR · CROSSWISE MASTERY · CHAPTER EIGHT

Chapter Eight

Two-Digit Crosswise

Ūrdhva-Tiryagbhyām · Vertically and Crosswise · Parallel Computing

"The crosswise method does not compute sequentially.
It sees the whole multiplication at once — and sees it in parallel." — *GaṇitSūtram · Stage Four*

In 1993, engineers at NVIDIA faced a problem. A GPU rendering a single video frame needs to perform millions of multiplications simultaneously — one for each pixel, each lighting calculation, each matrix transform. A sequential processor handles them one at a time. This is too slow. The solution was to build a chip that could perform thousands of multiplications in parallel, each core computing one partial product, a summation layer collecting the results by position.

The architecture they arrived at — compute all partial products simultaneously, sum by position, carry as needed — is precisely the structure of Ūrdhva-Tiryagbhyām. Tirthaji did not anticipate the GPU. He described the mathematical structure that any parallel multiplier must have. The hardware to implement it fully arrived in 1993. The mathematical idea had been in the sūtras for fifteen centuries.

This chapter teaches the crosswise method for two-digit multiplication. Three partial products. Three positions. All computable simultaneously. The student who reaches fluency here can multiply any two-digit pair mentally. That is a real accomplishment: it means no two-digit multiplication is beyond their reach.

CHAPTER AT A GLANCE

Chapter Eight introduces ***Ūrdhva-Tiryagbhyām*** — the most general multiplication method in this book. Unlike the Base and Sub-Base methods, which require numbers near a convenient reference, the crosswise method works for ***any*** two numbers of any size.

- Section 1 — Observation Lab 6: discovering the three partial products independently
- Section 2 — The sūtra Ūrdhva-Tiryagbhyām: etymology, arrows, structure
- Section 3 — The three partial products: RIGHT, MIDDLE, LEFT — the locked framework
- Section 4 — Worked examples: 23×14, 87×56, 99×99 with full carry tracking
- Section 5 — The modern relevance: parallel computing and GPU architecture
- Section 6 — The algebraic proof: (10A+B)(10C+D) as a positional expansion

1. Observation Lab Six — Seeing the Three Products

Before learning the procedure, discover its structure. The observation lab below asks you to multiply by any method you already know — and then examine what you actually did.

OBSERVATION LAB 6 — What Makes Up a Two-Digit Product?

Multiply 23 × 14 by any method you know. Write the full working in the space below. Then examine your working and answer the questions.

23 × 14 — my working:

_____

_____

_____

Number of separate digit-multiplications I performed: _____ Which digit pairs: _____

_____

Now consider these three expressions: (2×1), (2×4 + 3×1), (3×4). Calculate each:

2×1 = _____ 2×4 + 3×1 = _____ 3×4 = _____

Place them: hundreds = _____ tens = _____ units = _____ Assembled: _____

❖ *The three expressions you just computed are the three partial products of the crosswise method. You discovered them independently. The sūtra names the pattern you have now seen.*

2. The Sūtra Ūrdhva-Tiryagbhyām

SUTRA · 3

ऊर्ध्वतिर्यग्भ्याम्

Ūrdhva-Tiryagbhyām

ETYMOLOGY · Root by Root

Ūrdhva — *vertically, upward — the direction of the units×units and tens×tens products.*

Tiryak — *crosswise, diagonally — the direction of the mixed-digit products.*

Bhyām — *dual instrumental suffix — by means of both: vertically AND crosswise. The method requires both movements simultaneously.*

ENGLISH MEANING

"Vertically and Crosswise"

PRIMARY DOMAIN

Universal multiplication of any two numbers · Matrix products · Polynomial multiplication · Parallel computation

IN THIS CHAPTER

In this chapter: Ūrdhva-Tiryagbhyām gives the three partial products of any two-digit multiplication. Vertical products occupy the units and hundreds positions; the crosswise sum occupies the tens position. The three positions are independent and can be computed simultaneously.

The sūtra names the two directions of the multiplication arrows, but its deeper meaning is architectural: ***all positions are computed at once***. The method does not proceed from right to left, accumulating partial sums. It identifies what belongs in each position and places it there directly. This is the parallel structure.

3. The Three Partial Products

For two two-digit numbers AB and CD (where A, B, C, D are individual digits of the tens and units positions):

RIGHT *units position*	MIDDLE *tens position*	LEFT *hundreds position*
B × D *vertical product of units digits*	**A×D + B×C** *crosswise products of mixed digits*	**A × C** *vertical product of tens digits*

THE CROSSWISE IDENTITY

$$(10A+B)(10C+D) = 100\cdot AC + 10\cdot(AD+BC) + BD$$

The hundreds position receives AC (left vertical). The tens position receives AD+BC (crosswise). The units position receives BD (right vertical). This is standard algebraic expansion of a two-digit product, organised by positional value.

4. Worked Examples — Two-Digit Crosswise

EX 1
23 × 14

A=2, B=3, C=1, D=4. Apply the three positions: right, middle, left.

DIGITS	2 3 1 4	
RIGHT BD	3×4 = 12 → write 2, carry 1	**write 2 c1**

MIDDLE AD+BC+c	2×4 + 3×1 + 1 = 8+3+1 = 12 → write 2, carry 1	**write 2 c1**
LEFT AC+c	2×1 + 1 = 3 → write 3	**write 3**
ANSWER	(reading left to right)	**322 ✓**

❖ *Standard pattern: right first, carry into middle, carry into left.*

Every multiplication is this algebraic expansion, whether the student knows it or not.

EX 2
87 × 56

A=8, B=7, C=5, D=6. Large digits — carries will be significant.

DIGITS	8 7 5 6	
RIGHT BD	7×6 = 42 → write 2, carry 4	**write 2 c4**
MIDDLE AD+BC+c	8×6 + 7×5 + 4 = 48+35+4 = 87 → write 7, carry 8	**write 7 c8**
LEFT AC+c	8×5 + 8 = 48 → write 48	**48**
ANSWER		**4872 ✓**

❖ *87×56 = 4872. Verify: 87×50 + 87×6 = 4350 + 522 = 4872. ✓*

Every multiplication is this algebraic expansion, whether the student knows it or not.

EX 3
99 × 99

A=B=C=D=9. Maximum digit values — maximum carries. Also equals 99^2 — a good verification target.

DIGITS	9 9 9 9	
RIGHT BD	9×9 = 81 → write 1, carry 8	**write 1 c8**
MIDDLE AD+BC+c	9×9 + 9×9 + 8 = 81+81+8 = 170 → write 0, carry 17	**write 0 c17**
LEFT AC+c	9×9 + 17 = 81+17 = 98 → write 98	**98**
ANSWER		**9801 ✓**

❖ *Matches 99^2 = 9801 from Chapter Seven Pattern 2 (Yāvadūnam: gap=−1, left=98, right=01). Both methods give 9801.*

Every multiplication is this algebraic expansion, whether the student knows it or not.

5. The Modern Relevance — Parallel Computing

The standard column multiplication algorithm processes one digit pair at a time, from right to left, accumulating partial rows. It is inherently sequential: each step depends on the carry from the previous step. Ūrdhva-Tiryagbhyām identifies all partial products simultaneously

and assigns them to positions independently. The positions do not depend on each other until the final carry propagation.

URDHVA-TIRYAGBHYAM AS PARALLEL ARCHITECTURE

Sequential (column algorithm): five operations for AB×CD: B×D, B×C (shifted), A×D (shifted), A×C (shifted), then sum all rows. Each step depends on the previous.

Parallel (Ūrdhva): three independent computations: BD, AD+BC, AC. Each computable simultaneously. One carry-propagation pass to join them.

❖ *A modern GPU implements the parallel structure. For a 1000×1000 matrix multiplication requiring 10^9 operations, a sequential processor needs ~1 second. A GPU with 10,000 cores, each computing one partial product simultaneously, needs ~100 microseconds. The Ūrdhva principle — see all products at once — is the source of this 10,000× speedup. The mathematical idea is ancient. The hardware to implement it fully arrived in 1993.*

> *Tirthaji's Ūrdhva-Tiryagbhyām was not "relevant to" parallel computing. It was parallel computing, expressed in Sanskrit arithmetic, fifteen centuries before the transistor. The mathematical structure is identical. The sūtra does not say "compute from right to left." It says: vertically and crosswise. Both at once. That is the parallel instruction.*

6. The Algebraic Proof

The crosswise method is not a trick. It is the standard algebraic expansion of a two-digit product, organised by positional value:

(10A+B)(10C+D) = 100·AC + 10·(AD+BC) + BD

The hundreds position receives AC. The tens position receives AD+BC. The units position receives BD. This is polynomial multiplication with x=10, evaluated at x=10.

For 87×56: A=8, B=7, C=5, D=6.

```
100×(8×5) + 10×(8×6 + 7×5) + (7×6)
= 100×40 + 10×83 + 42
= 4000 + 830 + 42 = 4872  ✓
```

❖ *Every time a student applies the crosswise method, they are performing this algebraic expansion. Chapter Fourteen will show them that they have been doing algebra since Chapter Eight.*

7. Mastery Lab — Chapter Eight

MASTERY LAB · CHAPTER 8 · LEVEL 1

★☆☆☆☆ — Partial Product Identification

Ch. 8

For each pair, state RIGHT, MIDDLE (both terms), and LEFT. No carries yet. No final answer.

Ex 1 **23×41**	*R* = ___ *M* = ___×___ + ___×___ = ___ *L* = ___
Ex 2 **56×78**	*R* = ___ *M* = ___×___ + ___×___ = ___ *L* = ___
Ex 3 **34×89**	*R* = ___ *M* = ___×___ + ___×___ = ___ *L* = ___
Ex 4 **67×45**	*R* = ___ *M* = ___×___ + ___×___ = ___ *L* = ___

MASTERY LAB · CHAPTER 8 · LEVEL 2

★★☆☆☆ — Two-Digit Products — Standard

Ch. 8

Apply the three positions. Show RIGHT (with carry), MIDDLE (with carry), LEFT (with carry). Write the answer.

Ex 1	*R*=___ *c*=___ *M*=___ *c*=___ *L*=___ *Answer:* _____

23×41	
Ex 2 **56×78**	*R=___ c=___ M=___ c=___ L=___ Answer: _____*
Ex 3 **34×89**	*R=___ c=___ M=___ c=___ L=___ Answer: _____*
Ex 4 **67×45**	*R=___ c=___ M=___ c=___ L=___ Answer: _____*
Ex 5 **93×68**	*R=___ c=___ M=___ c=___ L=___ Answer: _____*
Ex 6 **47×63**	*R=___ c=___ M=___ c=___ L=___ Answer: _____*
Ex 7 **88×79**	*R=___ c=___ M=___ c=___ L=___ Answer: _____*
Ex 8 **95×87**	*R=___ c=___ M=___ c=___ L=___ Answer: _____*

MASTERY LAB · CHAPTER 8 · LEVEL 3

★★★☆☆ — Carry Mastery — Large Digits

Ch. 8

Large digits produce carries at every position. Track each carry explicitly before moving to the next position.

1. **99 × 88**

R=___ c=___ M=___ c=___ L=___

Answer:

2. **97 × 96**

R=___ c=___ M=___ c=___ L=___

Answer:

3. **88 × 87**

R=___ c=___ M=___ c=___ L=___

Answer:

4. **76 × 98**

R=___ c=___ M=___ c=___ L=___

Answer:

5. **89 × 99**

R=___ c=___ M=___ c=___ L=___

Answer:

MASTERY LAB · CHAPTER 8 · LEVEL 4

★★★★☆ — The Parallel Architecture — In Your Own Words

Ch. 8

Write a paragraph (minimum five sentences) explaining the connection between Ūrdhva-Tiryagbhyām and parallel computing. Include: (a) how the crosswise method differs from column multiplication, (b) what "parallel" means here, (c) why this structure is efficient.

PARALLEL ARCHITECTURE ESSAY · Chapter Eight · Level 4

MASTERY LAB · CHAPTER 8 · LEVEL 5

★★★★★ — The Algebraic Reconstruction

Ch. 8

Without referring to the text, write the algebraic identity for (10A+B)(10C+D). Then use it to compute 87×56 by substituting A=8, B=7, C=5, D=6, evaluating term by term. Show every substitution step.

ALGEBRAIC RECONSTRUCTION · (10A+B)(10C+D)

The identity:

Substituting A=8, B=7, C=5, D=6:

Final answer and verification:

♦ ♦ ♦

❖ **MENTOR'S SANCTUARY · Parent · Teacher · Mentor · Chapter Eight**

This Chapter's Purpose

Chapter Eight introduces the most general multiplication method in this book. Unlike the Base and Sub-Base methods, which require numbers to be near a convenient reference, the crosswise method works for any two numbers of any size. A student who reaches fluency here can multiply any two-digit pair mentally. This is a real accomplishment — name it explicitly when it is achieved.

The Tradition on Teaching This

"The method computes vertically and crosswise simultaneously." This is not a mnemonic. It is the computational architecture. A student who truly grasps that the three positions are independent — that the tens position does not wait for the units position to finish — has understood why this method is powerful. That understanding is what separates fluency from rote application.

Practical Coaching Note

The most common error is mis-placing the middle product. Students often confuse which pair is crosswise (A×D and B×C) with which is vertical (A×C and B×D). Before any calculation, require the student to name the three products by type: "What are the vertical products? What are the crosswise products?" If they can name them, they can calculate.

For the parallel computing connection at Level 4: this is a genuine intellectual connection worth exploring. A student with any interest in computer science will find this chapter's link to GPU architecture a significant motivator. The connection is not a historical curiosity — it is the reason the crosswise method is computationally superior to column multiplication for hardware implementation.

Diagnostic Flowchart — Chapter Eight

If Student consistently gets wrong middle product →

They are computing A×C + B×D (both verticals) instead of A×D + B×C (crosswise). Draw the X-shape arrows explicitly. Ask: "Which digits are connected by the X? Those are the crosswise products."

If Student handles small digits well but loses carries with large digits →

Level 3 is precisely designed for this. Carry management in the crosswise method requires practice to become automatic. Allow multiple attempts — it is not a sign of misunderstanding.

If Student asks about the GPU connection →

This is worth exploring seriously. If the student has interest in computer science, the parallel computing architecture is a significant motivator. Do not dismiss it as a historical footnote.

♦ ♦ ♦

COMING NEXT · Chapter Nine — Three-Digit Crosswise, Vinculum, and Verification

Chapter 9 extends the crosswise method to three-digit multiplication (five partial products), introduces Vinculum notation for handling high digits cleanly, and formalises the Casting Out Nines verification protocol that has protected calculations throughout this book.

STAGE FOUR · CROSSWISE MASTERY · CHAPTER NINE

Chapter Nine

Three-Digit Crosswise

Vinculum Notation · Casting Out Nines · Verification Protocol

"A completed calculation that has not been verified is not yet complete."

— Vedic mathematical tradition

Brahmagupta, writing in 628 CE, did not merely present calculations. He presented calculations and their proofs of correctness. The proof was not an optional check performed after confidence had been established — it was part of the calculation itself. A calculation without verification was, in the Brahmasphutasiddhanta, not a completed act of mathematics.

The Casting Out Nines method formalises this tradition. It converts the verification of any multiplication — however large the numbers — into a check on single-digit products. The machinery is modular arithmetic: because $10 \equiv 1 \pmod 9$, every number's value mod 9 equals the sum of its digits mod 9. Multiplication is preserved under this congruence. The check costs seconds and catches eight out of nine possible errors.

This chapter teaches three things in sequence. First, the extension of the crosswise method to three-digit numbers: five partial products instead of three, the same architectural logic. Second, Vinculum notation: a device for replacing high digits with small negative digits, reducing carry magnitude and simplifying three-digit crosswise work. Third, the Casting Out Nines protocol: the verification standard that applies to every calculation in this book.

CHAPTER AT A GLANCE

Three tools in one chapter. Together they complete Stage Four and establish the verification habit that Chapters Ten through Fifteen will assume.

- Section 1 — Five partial products: the three-digit crosswise framework
- Section 2 — Worked examples: 123×456 and 234×567
- Section 3 — Vinculum notation: replacing high digits with small negative digits
- Section 4 — Casting Out Nines: the five-step verification protocol
- Section 5 — The algebraic foundation of Casting Out Nines

1. Five Partial Products — Three-Digit Crosswise

For three-digit numbers ABC and DEF (where each letter is a single digit), the crosswise method produces five positional partial products. The pattern: sweep from the right vertical, through the crosswise pairs, to the left vertical. Each sweep produces one position.

POS	P1 *ten-thou*	P2 *thousands*	P3 *hundreds (middle)*	P4 *tens*	P5 *units*
Formula	**A×D**	**A×E + B×D**	**A×F + B×E + C×D**	**B×F + C×E**	**C×F**

◈ **Tirthaji, ~1958** — *described the crosswise extension as follows: "The extension is natural. Two digits give three products sweeping left. Three digits give five products. Four digits give seven. The pattern is always 2n−1 products for n-digit numbers." The architecture scales without limit.*

2. Worked Examples — Three-Digit

EX 1 123 × 456

A=1, B=2, C=3 · D=4, E=5, F=6

A=1, B=2, C=3 · D=4, E=5, F=6

P5 (units)	`3×6 = 18 → write 8, carry 1`	**write 8 c1**
P4 (tens)	`2×6 + 3×5 + c1 = 12+15+1 = 28 → write 8, carry 2`	**write 8 c2**
P3 (hundreds)	`1×6 + 2×5 + 3×4 + c2 = 6+10+12+2 = 30 → write 0, carry 3`	**write 0 c3**
P2 (thousands)	`1×5 + 2×4 + c3 = 5+8+3 = 16 → write 6, carry 1`	**write 6 c1**
P1 (ten-thou)	`1×4 + c1 = 4+1 = 5 → write 5`	**write 5**
ANSWER	`(reading P1 through P5)`	**56088 ✓**

EX 2 234 × 567

A=2, B=3, C=4 · D=5, E=6, F=7

A=2, B=3, C=4 · D=5, E=6, F=7

P5 (units)	`4×7 = 28 → write 8, carry 2`	**write 8 c2**
P4 (tens)	`3×7 + 4×6 + c2 = 21+24+2 = 47 → write 7, carry 4`	**write 7 c4**
P3 (hundreds)	`2×7 + 3×6 + 4×5 + c4 = 14+18+20+4 = 56 → write 6, carry 5`	**write 6 c5**

P2 (thousands)	2×6 + 3×5 + c5 = 12+15+5 = 32 → write 2, carry 3	write 2 c3
P1 (ten-thou)	2×5 + c3 = 10+3 = 13 → write 13	13
ANSWER		132678 ✓

3. Vinculum Notation

Vinculum notation replaces any digit above 5 with its complement from 10, marked with an overbar, representing a subtractive digit. The motivation: in three-digit crosswise work, products involving digits 6, 7, 8, 9 can produce large middle sums requiring large carries. Replacing these with small negative digits reduces carry magnitude significantly.

VINCULUM CONVERSION

Digit 6 → overbar-4 (meaning: 10–4, so the digit is –4 with +1 carried left)

Digit 7 → overbar-3 (meaning: 10–3)

Digit 8 → overbar-2 (meaning: 10–2)

Digit 9 → overbar-1 (meaning: 10–1)

In practice, Vinculum notation is most useful when two or more digits are above 5, making both the crosswise sums and the carry amounts unwieldy. For numbers like 97, 987, 896, the Vinculum form reduces these to near-base-complement calculations.

EX 3

97 = (100–3), 96 = (100–4). This is now a Base Method calculation.

A=9, B=7, C=undefined · D=9, E=6, F=undefined

97 × 96		
Gaps	-3 from 100 and -4 from 100	
LEFT	97 + (-4) = 93	93
RIGHT	(-3)(-4) = 12	12
ANSWER	Base Method: 93 | 12	9312 ✓

The key insight: Vinculum for numbers near a power of ten reduces to the Base Method with negative gaps. The Vinculum form makes the negative-digit structure explicit. For numbers with mixed high and low digits, the standard three-digit crosswise method is generally simpler.

4. Casting Out Nines — The Verification Protocol

Every multiplication in this book can be verified using the digit-sum method. The procedure converts any multiplication check into a check on single-digit numbers — a check that costs seconds and catches 8 out of 9 possible errors.

THE CASTING OUT NINES PROTOCOL — FIVE STEPS

Step Step 1 Find the digital root of the first factor: sum its digits; if the result exceeds 9, sum again; repeat until a single digit remains.

Step Step 2 Find the digital root of the second factor by the same process.

Step Step 3 Multiply the two digital roots. Find the digital root of this product.

Step Step 4 Find the digital root of your stated answer.

Step Step 5 Step 3 and Step 4 must agree. If they agree: the answer passes verification. If they disagree: there is an error. Recalculate.

❖ *The verification catches approximately 8 out of 9 errors. It misses errors whose magnitude is a multiple of 9 — rare in practice, and detectable by repeating the calculation.*

Five Worked Verifications

	root(A)	root(B)	r1×r2	ans root
123×456 = 56088	6	6	9	9 ✓
87×56 = 4872	6	2	3	3 ✓
997×996 = 993012	7	6	6	6 ✓
234×567 = 132678	9	9	9	9 ✓
99×99 = 9801	9	9	9	9 ✓

5. The Algebraic Foundation

Why does Casting Out Nines work? Because $10 \equiv 1 \pmod 9$, every positional value — 1, 10, 100, 1000 — is congruent to 1 modulo 9. The value of any number modulo 9 therefore equals the sum of its digits modulo 9. And multiplication is preserved under modular congruence: if $a \equiv r \pmod 9$ and $b \equiv s \pmod 9$, then $ab \equiv rs \pmod 9$.

This is Brahmagupta's arithmetic of remainders. Casting Out Nines is modular arithmetic in daily use. The student who applies this protocol each time is practising a technique that will reappear in Chapter Fourteen as a formal algebraic tool.

> *The Casting Out Nines check does not guarantee correctness — it provides a probabilistic certificate. An answer that passes the check has a 8/9 probability of being correct (given a uniformly random single-digit error). An answer that fails the check has an error with certainty. The check is not optional. "A completed calculation that has not been verified is not yet complete." — Brahmagupta would say the same thing in the language of remainders.*

6. Mastery Lab — Chapter Nine

MASTERY LAB · CHAPTER 9 · LEVEL 1
★☆☆☆☆ — Three-Digit Partial Products

Ch. 9

For each pair, state all five partial products (P5 through P1) without combining them. No carries yet.

123×456

List P5 through P1:

P5=___ P4=___+___ P3=___+___+___ P2=___+___ P1=___

234×789

List P5 through P1:

P5=___ P4=___+___ P3=___+___+___ P2=___+___ P1=___

MASTERY LAB · CHAPTER 9 · LEVEL 2
★★☆☆☆ — Three-Digit Multiplication

Ch. 9

Apply the five-product crosswise method. Show every partial product and carry.

Ex 1

123×456

P5=___c___ P4=___c___ P3=___c___ P2=___c___ P1=___ Answer=_____

Ex 2 **321×654**	P5=___c___ P4=___c___ P3=___c___ P2=___c___ P1=___ Answer=_____
Ex 3 **234×567**	P5=___c___ P4=___c___ P3=___c___ P2=___c___ P1=___ Answer=_____
Ex 4 **111×999**	P5=___c___ P4=___c___ P3=___c___ P2=___c___ P1=___ Answer=_____
Ex 5 **246×357**	P5=___c___ P4=___c___ P3=___c___ P2=___c___ P1=___ Answer=_____

MASTERY LAB · CHAPTER 9 · LEVEL 3

★★★☆☆ — Casting Out Nines — Verify and Detect

Ch. 9

Three of these answers are correct; two contain errors. Verify each using Casting Out Nines. Identify which are wrong.

	root(A)	root(B)	r1×r2	ans root
23×41 = 943	___	___	___	___ ?
56×78 = 4368	___	___	___	___ ?
87×99 = 8613	___	___	___	___ ?
234×567 = 132768	___	___	___	___ ?
123×456 = 56088	___	___	___	___ ?

MASTERY LAB · CHAPTER 9 · LEVEL 4

★★★★☆ — Vinculum Application

Ch. 9

Convert each number to its base-complement form and multiply.

Ex 1 **97×98 (97=100−3, 98=100−2)**	*Complement form:___ Working:___ Answer:_____*
Ex 2 **87×93 (use crosswise for 87=100−13)**	*Complement form:___ Working:___ Answer:_____*
Ex 3 **998×997**	*Complement form:___ Working:___ Answer:_____*

MASTERY LAB · CHAPTER 9 · LEVEL 5

★★★★★ — Full Integration: Crosswise + Verification + Identity

Ch. 9

Three-part exercise. Multiply 567×891 using the five-product crosswise method. Verify using Casting Out Nines. Write the algebraic identity for (100A+10B+C)(100D+10E+F) as a sum of five positional terms.

FULL INTEGRATION · 567 × 891

Part 1 — Five-product crosswise calculation:

Part 2 — Casting Out Nines verification:

Part 3 — Algebraic identity (100A+10B+C)(100D+10E+F):

♦ ♦ ♦

❖ **MENTOR'S SANCTUARY · Parent · Teacher · Mentor · Chapter Nine**

This Chapter's Purpose

Chapter Nine completes Stage Four and establishes two habits that will persist throughout the book: the Casting Out Nines verification, and the algebraic understanding of positional partial products. A student who leaves Chapter Nine with both habits is prepared for any multiplication they will encounter in Chapters Ten through Fourteen.

The Tradition on Teaching This

"A calculation without verification is not complete." This is not a pedagogical preference. It reflects a deeper principle: in the Vedic mathematical tradition, a result is not a result until it has been confirmed by a method independent of the calculation. Casting Out Nines is that independent method. Require it from the first calculation. Make it automatic before it is fast.

Practical Coaching Note

For the three-digit crosswise method: students who are fluent at two-digit crosswise often underestimate the difficulty of the three-digit extension. The middle product P3 — which involves three terms — is where errors concentrate. Before any full three-digit calculation, practise P3 in isolation: give the student two three-digit numbers and ask only for P3.

For Casting Out Nines at Level 3: include answers with deliberate errors (as the exercise does). A student who only ever verifies correct answers does not build the error-detection reflex. The experience of seeing the check fail — and then finding the error — is what makes the habit reliable.

Diagnostic Flowchart — Chapter Nine

If Student keeps getting P3 wrong in three-digit crosswise →

P3 requires three terms (A×F + B×E + C×D). Practise it in isolation before returning to full calculations. The pattern: all three cross-pairs where the position indices sum to the same value.

If Student says Casting Out Nines is too slow →

Time it: the five-step check for a six-digit answer takes under 20 seconds for a practised student. Compare this to the time cost of discovering an error after presenting an answer. The check pays for itself the first time it catches a mistake.

If Student asks why errors that are multiples of 9 are missed →

Excellent question. Because $9 \equiv 0 \pmod 9$, adding or subtracting a multiple of 9 does not change the digital root. A student who understands this has understood the modular arithmetic behind the check.

STAGE FIVE · REAL-LIFE MATHEMATICS · CHAPTER TEN

Chapter Ten

Percentages and Discounts

Ānurūpyeṇa · Proportionately · GST · Reverse Percentage · Compound Growth

"A percentage is not a calculation. It is a comparison — a statement about proportion." — GaṇitSūtram · Stage Five

★ STAGE FIVE ★

Real-Life Mathematics

Percentages · Distance, Speed and Time · Bills and Budgets · The World as a Problem Set

Stage Five applies every technique from Stages One through Four to the situations that appear in daily life and examinations. Three chapters. Three domains. Every chapter uses the SEE-UNDERSTAND-SOLVE-VERIFY framework. If any calculation technique feels unclear, return to its chapter. The technique is not the obstacle — the unfamiliar context is. Separate them.

In 1994, Arvind bought a television for ₹19,000. The shopkeeper announced a 15% sale, then added 12% service charge. Arvind calculated quickly: "15% off is ₹17,000 minus ₹1,350 ... wait, 15% of 19,000 — that's ₹2,850 off. Then 12% on ₹16,150." He did it in his head while the shopkeeper reached

for a calculator. The shopkeeper's answer: ₹18,088. Arvind's answer: the same. He paid the correct price because he understood what a percentage actually is.

A percentage is not a type of arithmetic problem. It is a language for describing proportion. Per centum: per hundred. 15% means 15 parts out of every 100. Once a student has genuinely absorbed this, every percentage problem becomes a proportion problem in disguise — and proportion is something the sūtra Ānurūpyeṇa has been teaching since Chapter Six.

This chapter builds percentage fluency from that definition outward: anchor fractions, the building-block method for any percentage, GST and discounts, reverse percentage, and compound growth. The mathematics is not new. The contexts are daily life.

CHAPTER AT A GLANCE

- Section 1 — What a percentage means: per centum, the proportion definition
- Section 2 — The Anchor Table: twelve fraction–percentage equivalences
- Section 3 — The building-block method for any percentage
- Section 4 — Four types: find p% of Q, percentage increase (GST), discount, reverse percentage
- Section 5 — Compound percentage: the interest-on-interest effect

1. What a Percentage Means

The word ***percent*** comes from Latin *per centum* — per hundred. A percentage is a fraction with denominator 100. 35% means 35 out of every 100, and this is all it ever means. Every percentage problem is, at its root, a ratio problem: how many units per hundred units?

This definition has an immediate practical consequence: to find p% of any quantity Q, multiply Q by p and divide by 100. The building-block method does this by working from anchor percentages (10%, 5%, 1%, 25%) and combining them. This is not a shortcut — it is the direct application of the definition, organised for mental calculation.

SŪTRA · Upa-Sūtra

अनुरूप्येण · *Ānurūpyeṇa*

"Proportionately"

Governs every percentage, discount, GST calculation, and reverse percentage in this chapter. Proportionality is not a technique — it is the underlying structure.

2. Percentage Fluency — The Anchor Table

The following fraction-percentage equivalences are the anchors of fast mental percentage calculation. A student who reads 33⅓% and immediately thinks "divide by 3" is working at fluency. Building this table to instant recall is the primary goal of Level 1.

PERCENTAGE	FRACTION	CALCULATION
10%	1/10	Divide by 10
20%	1/5	Divide by 5
25%	1/4	Divide by 4
33⅓%	1/3	Divide by 3
50%	1/2	Divide by 2
60%	3/5	Multiply by 3, divide by 5
66⅔%	2/3	Multiply by 2, divide by 3
75%	3/4	Multiply by 3, divide by 4

80%	4/5	Multiply by 4, divide by 5
12½%	1/8	Divide by 8
37½%	3/8	Multiply by 3, divide by 8
5%	1/20	Halve 10%

3. The Building-Block Method

For any percentage not in the anchor table, decompose it into anchors and combine. Always start with 10%.

WORKED EXAMPLE · TYPE A — 23% of ₹2,400

Build from 10% and combine.

10% of 2400	2400 ÷ 10 = 240	**₹240**
20% = 2×10%	240 × 2 = 480	**₹480**
3% = 3×1%	24 × 3 = 72	**₹72**
23% = 20% + 3%	480 + 72	**₹552**
	2400 × 0.23 = 552 ✓	

WORKED EXAMPLE · TYPE A — 12½% of ₹3,200

12.5% = 1/8. Single-step anchor.

12½% = 1/8	3200 ÷ 8	**₹400**

3200 × 0.125 = 400 ✓

4. The Four Percentage Problem Types

Type B — Percentage Increase · GST · Service Charge

WORKED EXAMPLE · TYPE B — ₹2,750 with 18% GST

Find the GST amount, then add to original.

10% of 2750	2750 ÷ 10 = 275	₹275
8% = 10% − 2%	275 − 55 = 220	₹220
18% = 10% + 8%	275 + 220 = 495	₹495
Total = 2750 + 495		2750 + 495 = ₹3,245

2750 × 1.18 = 3245 ✓

WORKED EXAMPLE · TYPE B — ₹45,000 salary with 12% raise

Find increase, then add to original.

10% of 45000	4500	₹4,500
2% of 45000	900	₹900
12% = 5400	New = 45000 + 5400	₹50,400

45000 × 1.12 = 50400 ✓

Type C — Percentage Discount

Method 1 (find and subtract): calculate the discount, subtract from original price. Method 2 (remaining fraction): multiply directly by (100% − discount%). Both methods must give the same answer.

WORKED EXAMPLE · TYPE C — ₹1,800 with 15% discount

Apply both methods and confirm they agree.

Method 1: 15% of 1800	15% = 10%+5% = 180+90 = 270	₹270 discount
Final (Method 1)	1800 − 270	₹1,530
Method 2: pay 85%	80%+5% = 1440+90	₹1,530
Both methods agree		1530 ✓
	Method 2 is faster for round percentages. Method 1 is clearer for understanding.	

Type D — Reverse Percentage

REVERSE PERCENTAGE RULE

After a discount of d%: **Original = Final ÷ (1 − d/100)**

After an increase of r%: **Original = Final ÷ (1 + r/100)**

After GST of g%: **Pre-GST = Final ÷ (1 + g/100)**

⚠ NEVER add a percentage of the final price back to itself. The percentage was taken from the ***original*, not the final.**

WORKED EXAMPLE · TYPE D — ₹952 after 20% discount

₹952 = 80% of original.

After 20% off, paid 80%	Original × 0.80 = 952	
Original = 952 ÷ 0.80	= 952 × 100/80	**₹1,190**
Common error (wrong):	952 + 20% of 952 = 1142.40	✗ **wrong**
	1190 × 0.80 = 952 ✓	

WORKED EXAMPLE · TYPE D — ₹1,180 bill including 18% GST

Find the pre-GST price.

Original × 1.18 = 1180	Original = 1180 ÷ 1.18	
= 118000 ÷ 118	= 1000	**₹1,000**
	1000 × 1.18 = 1180 ✓	

5. Compound Percentage

When a percentage is applied more than once to a changing base, the result is compound growth or compound decay. This is how savings accounts, loans, inflation, and population growth actually work.

COMPOUND GROWTH FORMULA

$$\text{Amount after n periods} = P \times (1 + r/100)^n$$

P = original amount · r = percentage rate per period · n = number of periods

WORKED EXAMPLE — ₹10,000 at 10% compound for 3 years

Year-by-year: multiply by 1.10 each year.

Year 1	10,000 × 1.10 = 11,000	₹11,000
Year 2	11,000 × 1.10 = 12,100	₹12,100
Year 3	12,100 × 1.10 = 13,310	₹13,310
Simple interest (compare)	10,000 + 3×1,000	₹13,000
Compound advantage	13,310 − 13,000	₹310 extra

The ₹310 difference is "interest on interest" — the compounding effect.

6. Mastery Lab — Chapter Ten

MASTERY LAB · CHAPTER 10 · LEVEL 1

★☆☆☆☆ — Anchor Fluency

Ch. 10

Compute each percentage mentally. State the anchor fraction used.

Ex 1 **25% of 840**	*Anchor = ___ Answer = ___*
Ex 2 **33⅓% of 2,700**	*Anchor = ___ Answer = ___*
Ex 3 **12½% of 1,600**	*Anchor = ___ Answer = ___*
Ex 4	*Anchor = ___ Answer = ___*

75% of 2,400	_____
Ex 5 **37½% of 800**	*Anchor = ___ Answer = ___* _____
Ex 6 **66⅔% of 4,500**	*Anchor = ___ Answer = ___* _____

MASTERY LAB · CHAPTER 10 · LEVEL 2

★★☆☆☆ — GST and Discounts

Ch. 10

Apply both methods for each discount problem. GST problems: one method only.

Ex 1 **₹3,500 · add 12% GST**	*Working: Answer = ₹___* _____
Ex 2 **₹6,000 · 25% discount (both methods)**	*Method 1:___ Method 2:___ Answer = ₹___* _____
Ex 3 **₹28,500 · 15% off**	*Working: Answer = ₹___* _____
Ex 4 **₹1,250 · add 18% GST**	*Working: Answer = ₹___* _____
Ex 5 **₹4,800 · 8% discount**	*Working: Answer = ₹___* _____

MASTERY LAB · CHAPTER 10 · LEVEL 3

Ch. 10

★★★☆☆ — Reverse Percentage

Find the original. Show: Original = Final ÷ ___. Verify.

1. **₹1,020 after 15% discount**

Original = ___ ÷ ___ = ___ · *Verify:* ___×___ = ___

2. **₹472 after 18% GST**

Original = ___ ÷ ___ = ___ · *Verify:* ___×___ = ___

3. **₹38,500 after 10% salary rise**

Original = ___ ÷ ___ = ___ · *Verify:* ___×___ = ___

4. **₹714 after 15% off**

Original = ___ ÷ ___ = ___ · *Verify:* ___×___ = ___

5. **₹2,360 after 18% GST**

Original = ___ ÷ ___ = ___ · *Verify:* ___×___ = ___

MASTERY LAB · CHAPTER 10 · LEVEL 4

Ch. 10

★★★★☆ — Multi-Step Real Situations

Each situation requires two or more percentage steps. Read carefully before calculating.

1. **A laptop (₹42,000) has 10% trade-in discount, then 18% GST added. Find final price.**

Step 1 (discount):___ Step 2 (GST on discounted price):___ Answer:___

2. Salary ₹32,000 raised by 8%, then taxed at 5%. Find take-home pay.

Step 1 (raise):___ Step 2 (tax):___ Take-home:___

3. A phone costs ₹18,900 after 10% discount and 18% GST. Find pre-discount price.

Step 1 (undo GST):___ Step 2 (undo discount):___ Answer:___

MASTERY LAB · CHAPTER 10 · LEVEL 5

★★★★★ — Compound Growth

Ch. 10

Apply compound percentage year by year. Compare with simple for each.

1. ₹15,000 at 8% compound for 3 years. Year-by-year amounts.

Y1:___ Y2:___ Y3:___ Simple comparison (3× interest added):___

2. Town of 50,000 grows at 4% per year. Population after 3 years?

Y1:___ Y2:___ Y3:___

3. Why is compound always more than simple for growth? Two sentences.

♦ ♦ ♦

❖ **MENTOR'S SANCTUARY · Chapter Ten**

This Chapter's Purpose

Chapter Ten is the first chapter where mathematics and daily life become unmistakably one. Students who have felt unmotivated by abstract calculation frequently become engaged for the first time when they can calculate a discount, check a GST bill, or design a budget. Lean into this. Bring real receipts, real sale tags, real salary slips into practice sessions.

The Tradition on Teaching This

The Ānurūpyeṇa sūtra governs proportion at every level of this book. The Sub-Base Method divided by a ratio. Percentages multiply by a fraction. The underlying structure is identical: the ratio of the part to the whole is fixed, and every calculation is a navigation of that fixed ratio.

Practical Coaching Note

The reverse percentage error is the most persistent in this chapter. Require the student to draw a percentage strip for every reverse problem: the original (100%) is the longer strip; the final is the shorter. Ask: "Which is bigger — the original price or the final price?" The answer reveals whether to multiply or divide.

Diagnostic Flowchart — Chapter Ten

If student cannot compute 23% mentally →

They have not internalised the anchor table. Build the chain orally: "Give me 10% of 450. Double it. Now add 3%." Develop the oral chain before any written work.

If student finds reverse percentage abstract →

Use the strip diagram every time. Draw it first, ask about direction, then calculate.

If student reaches Level 5 →

They are ready for the financial mathematics in Chapter Fourteen. The compound growth formula is an algebraic identity — the Bridge chapter will show them why.

♦ ♦ ♦

COMING NEXT · Chapter Eleven — Time, Distance and Speed

Chapter Eleven introduces the D–S–T triangle — the first mathematical ***model*** in this book. A formula that describes how the physical world behaves, not just a computational pattern. Unit conversion, relative speed, average speed, and the full journey modelling problem.

STAGE FIVE · REAL-LIFE MATHEMATICS · CHAPTER ELEVEN

Chapter Eleven

Time, Distance and Speed

Parāvartya Yojayet · The D–S–T Triangle · Relative Speed · Unit Conversion

"Every journey is a triangle of time, distance, and rate.
Know any two sides and the third reveals itself." — GaṇitSūtram · Stage Five

In 1853, the Great Indian Peninsula Railway opened the first passenger line between Bombay and Thane — 34 kilometres in 57 minutes. The engineers who built that line understood D = S × T not as a formula to memorise but as a precise description of physical reality. If a train travels at a constant speed S for time T, it covers distance D. There is no ambiguity and no exception.

What the engineers also understood was that the formula could be transposed. To find S, divide D by T. To find T, divide D by S. This transposition — the sūtra Parāvartya Yojayet applied to a physical relationship — is what turns a formula into a tool. A formula that can only be read one way is not a model. A model answers questions.

This chapter teaches D = S × T in all three transpositions, with unit conversion as a separate but mandatory step. Relative speed transforms every two-object problem into a single-object problem. Average speed is total distance divided by total time — not the arithmetic mean of individual speeds, which is a trap this chapter explicitly addresses.

SŪTRA · 4

परावर्त्य योजयेत · *Parāvartya Yojayet*

"Transpose and apply"

D = S×T is one relationship seen from three directions. Transposing gives S = D÷T and T = D÷S. Every D–S–T problem is Parāvartya applied to a physical model.

THE D–S–T TRIANGLE · THREE READINGS

Cover D	Cover S	Cover T
D = S × T	**S = D ÷ T**	**T = D ÷ S**
multiply	*divide*	*divide*

⚠ **Units must match. Time in hours when speed is in km/h. Always convert before calculating.**

1. Unit Conversion — The Critical Step

The most common error in D–S–T problems is using inconsistent units. Speed in km/h and time in minutes gives a nonsensical distance without conversion. The conversion must be written as a separate, numbered line before the D = S × T step.

CONVERSION	RULE	EXAMPLE
min → hours	Divide by 60	`45 min = 45/60 = ¾ h`
hours → min	Multiply by 60	`1.5 h = 90 min`
km/h → m/s	Multiply by 5/18	`72 km/h = 72×5/18 = 20 m/s`
m/s → km/h	Multiply by 18/5	`15 m/s = 15×18/5 = 54 km/h`

km → metres	Multiply by 1000	`1.5 km = 1500 m`
metres → km	Divide by 1000	`800 m = 0.8 km`

WORKED EXAMPLE — TYPE 1 (Find D)

Speed = 75 km/h, Time = 3 hours 20 minutes

Convert time	`3 h 20 min = 3 + 20/60 = 3⅓ h`	**3⅓ h**
D = S × T	`75 × 10/3`	**250 km**
	250 ÷ 75 = 3.33… h = 3h 20min ✓	

WORKED EXAMPLE — TYPE 2 (Find S)

Distance = 420 km in 5 hours 15 minutes

Convert time	`5h 15min = 5.25 h`	**5.25 h**
S = D ÷ T	`420 ÷ 5.25 = 42000 ÷ 525`	**80 km/h**
	80 × 5.25 = 420 ✓	

WORKED EXAMPLE — TYPE 3 (Find T)

Distance = 660 km, Speed = 88 km/h

T = D ÷ S	`660 ÷ 88 = 7.5 h`	**7h 30min**
	88 × 7.5 = 660 ✓	

WORKED EXAMPLE — Unit conversion

Speed = 54 km/h, Time = 40 seconds — find distance in metres

Convert speed	54 km/h × 5/18	15 m/s
D = 15 m/s × 40 s		600 metres
	600 ÷ 15 = 40 s ✓	

2. Relative Speed

RELATIVE SPEED RULE

Toward each other: Relative speed = $S_1 + S_2$

Same direction: Relative speed = $|S_1 - S_2|$

In both cases: treat the combined/difference speed as the speed of a single object closing the gap.

WORKED EXAMPLE — Toward each other

Two trains 330 km apart, moving toward each other at 60 km/h and 50 km/h

Relative speed	60 + 50	110 km/h
Time to meet	330 ÷ 110	3 hours
	Train 1: 3×60=180 km. Train 2: 3×50=150 km. Total: 180+150 = 330 ✓	

WORKED EXAMPLE — Same direction

Car (90 km/h) chasing bus (70 km/h), starting 40 km behind

Relative speed	90 − 70 = 20 km/h (car gains at this rate)	
Time to catch	40 ÷ 20	2 hours
	Car travels 2×90=180 km. Bus travels 140+40=180 km. Same position ✓	

3. Average Speed — The Common Misconception

When a journey has two stages at different speeds, the average speed is **NOT** the arithmetic mean of the two speeds. It is total distance divided by total time. The arithmetic mean is correct only if both stages take equal *time*.

WORKED EXAMPLE — Average speed trap

150 km at 60 km/h, then 200 km at 100 km/h

Time 1	150 ÷ 60 = 2.5 h	**2.5 h**
Time 2	200 ÷ 100 = 2 h	**2 h**
Total	350 km ÷ 4.5 h	**77.8 km/h**
Arithmetic mean (wrong)	(60+100)/2	**80 km/h ✗**
	True average 77.8 ≠ 80. The second stage is faster so it contributes less time, pulling the average down.	

4. Mastery Lab — Chapter Eleven

MASTERY LAB · CHAPTER 11 · LEVEL 1

★☆☆☆☆ — D–S–T Identification

Ch. 11

For each: state Given, Wanted, Formula form. No calculation yet.

Ex 1	*Given=___ Wanted=___ Formula=___ Answer=___*

Speed 80 km/h, Time 4.5 h — find Distance	
Ex 2 **Distance 520 km, Time 6.5 h — find Speed**	*Given=___ Wanted=___ Formula=___ Answer=___*
Ex 3 **Distance 360 km, Speed 90 km/h — find Time**	*Given=___ Wanted=___ Formula=___ Answer=___*
Ex 4 **Speed 72 km/h, Time 25 minutes — find Distance (convert time first)**	*Given=___ Wanted=___ Formula=___ Answer=___*

MASTERY LAB · CHAPTER 11 · LEVEL 2

★★☆☆☆ — Unit Conversion

Ch. 11

Convert each, then compute. Show conversion as a separate line.

Ex 1 **36 km/h to m/s**	*Conversion:___ Answer:___*
Ex 2 **15 m/s to km/h**	*Conversion:___ Answer:___*
Ex 3	*Conversion:___ Answer:___*

2 h 45 min to hours (decimal)	
Ex 4 **0.75 hours to minutes**	*Conversion:___ Answer:___*
Ex 5 **Speed = 108 km/h, Time = 30 sec — Distance in metres**	*Conversion:___ Answer:___*

MASTERY LAB · CHAPTER 11 · LEVEL 3

★★★☆☆ — Relative Speed Problems

Ch. 11

State: toward each other (add) or same direction (subtract). Then solve.

1. **Two cyclists 200 km apart, cycling toward each other at 25 and 15 km/h. When do they meet?**

Toward/away: ___ Relative speed: ___ Time: ___ Verify: ___

2. **A truck (60 km/h) and car (90 km/h) start from same point. How far apart after 3.5 h?**

Same direction Relative speed: ___ Gap after 3.5h: ___

3. **Thief on foot (8 km/h) has 30 min head start. Police on cycle at 14 km/h. How long to catch?**

Head start distance: ___ Relative speed: ___ Time: ___

MASTERY LAB · CHAPTER 11 · LEVEL 4

★★★★☆ — Average Speed Traps

Ch. 11

Calculate true average speed (total ÷ total time) and identify where arithmetic mean fails.

1. **100 km at 50 km/h, then 100 km at 100 km/h. Find average speed.**

T1:___ T2:___ Avg speed:___ Arithmetic mean:___ Same? ___

2. **Why is the arithmetic mean wrong for unequal times? One sentence.**

MASTERY LAB · CHAPTER 11 · LEVEL 5

★★★★★ — Full Journey Modelling

Ch. 11

A delivery van leaves City A at 7:00 AM, drives 120 km to City B at 60 km/h, stops 45 minutes, then drives 180 km to City C at 90 km/h.

FULL JOURNEY · FIVE QUESTIONS

1. What time does it arrive at City B?
2. What time does it depart City B?
3. What time does it arrive at City C?
4. Average speed for the journey (excluding the stop)?
5. Average speed including the stop (total distance ÷ total elapsed time)?

♦ ♦ ♦

MENTOR'S SANCTUARY · Chapter Eleven

This Chapter's Purpose

Chapter Eleven introduces the first mathematical model in this book: a formula that describes a physical relationship, not just a computational pattern. D = S × T is not an arithmetic rule — it is a precise statement about how the world behaves. A student who genuinely understands it will apply it correctly in novel situations. A student who has merely memorised it will fail when the problem presents information in an unusual order.

The Tradition on Teaching This

Parāvartya Yojayet — transpose and apply — has appeared throughout this book as a calculation technique. Chapter Eleven shows its deeper meaning: a formula is not a single statement but a family of statements, each obtained by transposing the relationship to isolate a different variable. This is the Vedic view: mathematics is not a collection of rules but a collection of relationships that can be read in multiple directions.

Practical Coaching Note

Unit conversion is the most dangerous omission in this chapter. Before every exercise, require the student to state the units of every quantity. If they are inconsistent, stop. Write the conversion explicitly as its own numbered line.

Diagnostic Flowchart — Chapter Eleven

If student cannot rearrange D=S×T →

Use the triangle every time. Cover the wanted quantity and the remaining two show what to do. Do not permit algebraic rearrangement until the triangle is internalised.

If student makes relative speed errors →

Ask: "Are they moving toward each other or away?" The physical picture always reveals whether to add or subtract.

If student is surprised that arithmetic mean of speeds is wrong →

Sit with the surprise. Ask: "Why does equal-distance averaging give a different result from equal-time averaging?" This is the beginning of genuine mathematical thinking about averages.

♦ ♦ ♦

COMING NEXT · Chapter Twelve — Bills, Budgets and Real-World Arithmetic

Chapter Twelve completes Stage Five with the most directly applied chapter in the book: reading and verifying bills, unit rate comparison shopping, profit and loss percentage, and personal budget design. The arithmetic is the simplest of Stage Five — the challenge is interpretive.

STAGE FIVE · REAL-LIFE MATHEMATICS · CHAPTER TWELVE

Chapter Twelve

Bills, Budgets and Real-World Arithmetic

Reading Bills · Unit Rates · Profit and Loss · Personal Budget Design

"Arithmetic that does not reach the world is incomplete.
The calculation that serves a real decision is the one that matters." — GaṇitSūtram · Stage Five

In 1962, an Indian Railways freight bill was issued for 47 tonnes of coal at ₹1.83 per quintal. The clerk calculated the total as ₹4,047. A supervisor checked it: 47 tonnes = 470 quintals; 470 × 1.83 = 860.10 rupees. The bill was not ₹4,047 but ₹860.10. The error was a factor of roughly 5. The clerk had not checked the units.

Every bill, every budget, every comparison of prices has an arithmetic structure. The structure can be verified. The question is whether the person reading the document knows how to verify it — and whether they do it. Chapter Twelve teaches three arithmetical skills that belong to daily civic life: reading and verifying a bill, calculating unit rates for comparison shopping, and computing profit and loss percentages as a business owner. Level 5 adds the budget design exercise that integrates all three.

The arithmetic in this chapter is the simplest in Stage Five. The challenge is not the calculation. It is the discipline of asking, every time: does this number make sense? Is the unit correct? Does the total

> *match the parts? These questions, asked consistently, are what distinguish a numerate person from someone who can merely calculate.*

Stage Five closes here. Every technique from Stages One through Four meets a real situation in Chapters Ten through Twelve. If any technique feels unclear when applied, return to its chapter and revisit. ***The technique is not the obstacle — the unfamiliar situation is.*** Separate them.

1. Reading and Verifying a Bill

A bill is a structured document. Every line has a meaning, and the total is a claim that can be verified. The arithmetic skill of Chapter Twelve is not just calculation — it is verification: does the total match what the individual lines claim?

FIVE BILL VERIFICATION CHECKS

Check A Each line total = Quantity × Unit price

Check B Subtotal = sum of all line totals

Check C Discount calculation = correct rate on the pre-discount subtotal

Check D GST = correct rate on the post-discount subtotal

Check E Grand total = subtotal − discount + GST

Any one of these five checks failing indicates a billing error. Always perform all five.

WORKED EXAMPLE — Bill Verification

3 notebooks × ₹45 = ₹135; 2 pens × ₹30 = ₹60; 1 ruler × ₹25 = ₹25. Subtotal ₹220. Discount 10%. GST 12% on discounted price.

Check line totals	`3×45=135` ✓ `2×30=60` ✓ `1×25=25` ✓	✓

Subtotal	135+60+25	220 ✓
Discount 10%	220 × 0.10 = 22 Post-discount: 198	₹198
GST 12%	198 × 0.12	23.76
Grand total	198 + 23.76	₹221.76
	All five checks pass. Grand total is ₹221.76.	

2. Unit Rate — The Foundation of Comparison Shopping

A unit rate expresses a quantity per single unit: price per kilogram, earnings per hour, distance per litre. Unit rates allow comparison between options of different sizes. The option with the lower unit rate is the better value.

WORKED EXAMPLE — Unit rate comparison

750g of rice for ₹52.50 or 2 kg for ₹136?

750g option	52.50 ÷ 750 = ₹0.07/g	₹70/kg
2 kg option	136 ÷ 2000 = ₹0.068/g	₹68/kg
Better value	2 kg option (₹68 < ₹70)	2 kg ✓
	Always express unit rates in the same unit (per kg, per litre) before comparing.	

3. Profit, Loss and Break-Even

Every transaction has a cost price (CP) and a selling price (SP). Profit and loss are always calculated as a percentage of the ***cost price***, not the selling price. This is the rule that matters most in Level 3.

PROFIT AND LOSS FORMULAS

Profit % = (SP − CP) ÷ CP × 100

Loss % = (CP − SP) ÷ CP × 100

SP given CP and Profit%: SP = CP × (1 + Profit%/100)

CP given SP and Profit%: CP = SP ÷ (1 + Profit%/100)

⚠ **Profit % is ALWAYS calculated on CP. Using SP as the denominator gives a wrong, smaller percentage.**

WORKED EXAMPLE — Profit %

Shopkeeper buys a fan for ₹1,200 and sells it for ₹1,500

Profit = SP − CP	1500 − 1200	**₹300**
Profit % on CP	300 ÷ 1200 × 100	**25%**
	1200 × 1.25 = 1500 ✓	

WORKED EXAMPLE — Find SP for target profit

CP = ₹2,400. Need 20% profit. What SP?

SP = CP × 1.20	2400 × 1.20	**₹2,880**
	2880 − 2400 = 480 = 20% of 2400 ✓	

WORKED EXAMPLE — Find CP from SP and profit %

Phone sells for ₹18,000 with 25% profit. Find CP.

CP = SP ÷ 1.25	18000 ÷ 1.25	**₹14,400**
	14400 × 1.25 = 18000 ✓	

4. Mastery Lab — Chapter Twelve

MASTERY LAB · CHAPTER 12 · LEVEL 1

★☆☆☆☆ — Bill Verification

Ch. 12

Verify each bill. Identify which (if any) contain errors. Show all five checks.

BILL A · 4 books × ₹120 = ₹480 · 2 pens × ₹35 = ₹75 · Subtotal ₹555 · GST 12% = ₹66.60 · Total ₹621.60

Check A (line totals): ___ Check B (subtotal): ___ Check C (no discount): n/a

Check D (GST): ___ Check E (grand total): ___

Error (if any): ___ Correct total: ___

BILL B · 3 kg sugar × ₹48 = ₹144 · 5 L oil × ₹92 = ₹460 · Subtotal ₹604 · 10% discount = ₹60.40 · GST 5% = ₹30.20 · Total ₹573.80

Check A (line totals): ___ Check B (subtotal): ___ Check C (discount on subtotal): ___

Check D (GST on post-discount): ___ Check E (grand total): ___

Error (if any): ___ Correct total: ___

MASTERY LAB · CHAPTER 12 · LEVEL 2

★★☆☆☆ — Unit Rate Comparisons

Ch. 12

Calculate the unit rate for each option. State which is better value.

Ex 1 **500 ml juice ₹32 vs 1.2 L juice ₹70 — price per ml**	*Option A: ₹___ per unit Option B: ₹___ per unit Better value: ___*
Ex 2 **5 kg rice ₹220 vs 2 kg ₹90 — price per kg**	*Option A: ₹___ per unit Option B: ₹___ per unit Better value: ___*
Ex 3 **₹6,300 for 70 hours work vs ₹4,500 for 48 hours — rate per hour**	*Option A: ₹___ per unit Option B: ₹___ per unit Better value: ___*

MASTERY LAB · CHAPTER 12 · LEVEL 3

★★★☆☆ — Profit and Loss

Ch. 12

Always state whether it is profit or loss. Always divide by CP.

Ex 1 **CP ₹2,400 · SP ₹3,000**	*Profit/Loss: ___ Working: ___ %: ___*
Ex 2 **CP ₹5,000 · SP ₹4,200**	*Profit/Loss: ___ Working: ___ %: ___*
Ex 3 **CP = ? · SP ₹2,640 · Profit 10%**	*Profit/Loss: ___ Working: ___ %: ___*
Ex 4	*Profit/Loss: ___ Working: ___ %: ___*

CP ₹800 · Need 15% profit · Find SP

MASTERY LAB · CHAPTER 12 · LEVEL 4 *Ch. 12*

★★★★☆ — Multi-Step Business Problem

A small food stall buys ingredients for ₹6,000. Makes 200 portions; sells each for ₹40.

1. **Total revenue?**

2. **Profit?**

3. **Profit %?**

4. **If rent ₹1,200 added to costs, what is new profit %?**

5. **At what selling price per portion would profit be exactly 50%?**

MASTERY LAB · CHAPTER 12 · LEVEL 5 *Ch. 12*

★★★★★ — Personal Budget Design

Design a monthly budget for income ₹25,000. At least five categories. All percentages must sum to 100. All amounts must sum to ₹25,000.

MY MONTHLY BUDGET · Income ₹25,000

Rent / Housing: ___% = ₹___

Food: ___% = ₹___

Transport: ___% = ₹___

Education: ___% = ₹___

Savings: ___% = ₹___

Other: ___% = ₹___

Total: ___% = ₹___ *(must equal 100% and ₹25,000)*

Which allocation was hardest to decide? Why?

___

___

___

♦ ♦ ♦

MENTOR'S SANCTUARY · Chapter Twelve

This Chapter's Purpose

Chapter Twelve is Stage Five's closing chapter and the most directly applied in the book. The arithmetic is the simplest of Stage Five: no new techniques are introduced. The challenge is interpretive: does the student read a bill as a sequence of claims to verify, or as a single number to trust? Do they ask "per unit of what?" before comparing

prices? These interpretive habits are what arithmetic education is ultimately building toward.

The Tradition on Teaching This

The Vedic tradition does not separate mathematics from life. Every chapter of this book has been building toward the moment when a student picks up a real bill, reads it, and thinks: "I can verify every line of this." Chapter Twelve is that moment. Do not rush it. If possible, bring a real electricity bill or grocery receipt to the Level 1 exercise.

Practical Coaching Note

The most common error at Level 3 is using SP as the denominator for profit percentage. When a student calculates profit/SP instead of profit/CP, require them to state: "Profit percentage is always calculated on the cost price. Why? Because that is what the seller invested. That is the correct reference."

Level 5 (Budget Design) is the most important exercise in Stage Five for developing financial literacy. Encourage the student to live with their budget allocation for one week and note where reality differs from plan. The reflection question is not about calculation — it is about the relationship between mathematics and actual behaviour.

Diagnostic Flowchart — Chapter Twelve

If student cannot verify a multi-line bill →

Return to the five-check list and work through each check one at a time. The challenge is usually Check D (GST must be applied to the post-discount subtotal, not the original subtotal).

If student skips the unit conversion in unit rate comparison →

Require both rates to be expressed in the same denominator before comparison is permitted. "Price per gram" and "price per kg" cannot be compared directly.

If student disagrees with budget allocation that forces difficult choices →

This is the intended pedagogical moment. The budget forces confrontation with priorities. Mathematics does not resolve that confrontation — it only makes it visible.

♦ ♦ ♦

★ STAGE FIVE COMPLETE ★

◆
◆
◆
◆
◆

Stage Six awaits: Word Problems and the Bridge to Algebra.

GATE REVIEW · PART Three

The Base Engine: Sub-Base, Squaring, and Crosswise — confirm before proceeding to Part Four

Before proceeding, confirm all of the following:

- ☐ The student can compute any percentage mentally using anchor building blocks.
- ☐ The student can solve all four percentage problem types: find, increase, discount, reverse.
- ☐ The student can apply all three forms of D=S×T and convert between km/h, m/s, minutes, and hours.
- ☐ The student can solve relative speed problems (toward each other and same direction).
- ☐ The student can verify a multi-line bill, compute unit rates, and calculate profit/loss %.
- ☐ The student can articulate why profit percentage is computed on CP, not SP.
- ☐ The student has designed and reflected on a personal budget (Level 5).

When all conditions are met: Part Three is complete.

STAGE SIX · THE BRIDGE · CHAPTER THIRTEEN

Chapter Thirteen

Word Problems

The Four-Step Method · SEE · UNDERSTAND · SOLVE · VERIFY

"A word problem is a situation disguised as arithmetic.
The student's first task is not to calculate — it is to see the situation clearly." — GaṇitSūtram · Stage Six

★ STAGE SIX ★

The Bridge

Word Problems decoded · The Four-Step Method · Algebra as recognised arithmetic

Stage Six contains two chapters. Chapter Thirteen decodes word problems using the Four-Step Method. Chapter Fourteen crosses the conceptual bridge into formal algebra — showing that every technique in this book is an algebraic identity the student has been using without the notation. The bridge is crossed in Chapter Fourteen; the approach is cleared in Chapter Thirteen.

A student in an examination reads: "A and B together can finish a job in 4 hours. A alone takes 6 hours. How long for B alone?" She stares at it. She has no formula for "joint work." She writes 4 + 6 = 10 and stops. She has answered a different problem with complete arithmetic accuracy.

The problem was not a calculation problem. It was a situation: two workers, different speeds, combined result. The calculation comes after the situation is understood. What is A's rate of work per hour? What is the combined rate? What, therefore, must B's rate be? The arithmetic is simple: 1/6 + 1/B = 1/4, so 1/B = 1/4 − 1/6 = 1/12. B alone takes 12 hours.

The Four-Step Method exists because mathematical problems in words require two separate skills: understanding the situation (Steps 1 and 2) and applying the technique (Steps 3 and 4). Students who jump from language directly to calculation solve the wrong problem. The method makes the two-phase structure explicit and requires each phase to be completed before the next begins.

1. The Four-Step Method — Full Specification

SEE · UNDERSTAND · SOLVE · VERIFY

STEP 1: SEE Read the problem at least twice. Do not think about numbers on the first reading. Visualise the situation. What is happening? Who is involved? What is moving, changing, or being measured? Draw a diagram if it helps. Do not write any numbers yet.

STEP 2: UNDERSTAND Identify what is GIVEN (known quantities with units) and WANTED (unknown with units). Write these as two explicit lists. Choose the technique or relationship that connects given to wanted. Name it explicitly.

STEP 3: SOLVE Apply the chosen technique. Show every step. Write units at every stage. Do not skip arithmetic steps. A complete solution includes every line of working, not just the final answer.

STEP 4: VERIFY Check the answer. Correct units? Reasonable size? Does substituting back into the original relationship confirm it? Write: VERIFY: [the check]. Do not omit this step.

❖ *"If you cannot complete Step 2, you do not yet understand the problem. Do not proceed to Step 3."*

2. Worked Examples — The Method Applied

WORKED EXAMPLE 1 — PERCENTAGE (Two Steps)

After a 20% discount and 18% GST on the discounted price, a customer pays ₹1,534.32. Find the original price.

▶ **SEE** A price had two changes: discount down, then GST up. Given the final, find the start.

▶ **UNDERSTAND** GIVEN: Final = ₹1,534.32, Discount 20%, GST 18%. WANTED: Original. Technique: reverse percentage twice — first undo GST, then undo discount.

▶ **SOLVE**

```
After 18% GST: Final = Post-disc × 1.18  →  Post-disc = 1534.32 ÷ 1.18 =
₹1,300
After 20% disc: Post-disc = Original × 0.80  →  Original = 1300 ÷ 0.80 =
₹1,625
```

❖ **VERIFY** *1625 × 0.80 = 1300. 1300 × 1.18 = 1534.* ✓

WORKED EXAMPLE 2 — RELATIVE SPEED (Three Conditions)

Train A leaves Mumbai at 6 AM at 80 km/h. Train B leaves same station at 8 AM at 120 km/h. When and where does B overtake A?

▶ **SEE** Two trains, same route, different start times. B is faster and starts later. It will catch A.

▶ **UNDERSTAND** GIVEN: A speed=80, B speed=120, B starts 2h after A. WANTED: Time and place of overtaking. Technique: A's head-start ÷ relative speed = time to close.

▶ **SOLVE**

```
A's head start: 2 h × 80 = 160 km
Relative speed: 120 − 80 = 40 km/h
Time for B to close: 160 ÷ 40 = 4 hours after B's departure
B departs 8 AM + 4 h = 12 noon. Distance: 120 × 4 = 480 km from Mumbai
```

❖ **VERIFY** *A has travelled 80 × 6 (6AM to noon) = 480 km. Same position.* ✓

WORKED EXAMPLE 3 — PROFIT AND UNIT RATE

A shopkeeper mixes 30 kg of tea at ₹120/kg with 20 kg at ₹90/kg. At what price per kg must the mixture be sold to make 25% profit?

▶ **SEE** Two grades of tea mixed. Find cost per kg of mixture, then apply profit percentage.

▶ **UNDERSTAND** GIVEN: 30 kg @ ₹120, 20 kg @ ₹90, Profit 25%. WANTED: SP per kg.

▶ **SOLVE**

```
Total CP: 30×120 + 20×90 = 3600 + 1800 = ₹5,400
Total qty: 50 kg  →  CP per kg: 5400 ÷ 50 = ₹108
SP per kg for 25% profit: 108 × 1.25 = ₹135
```

❖ **VERIFY** *Revenue = 50×135 = 6750. Profit = 6750− 5400 = 1350 = 25% of 5400* ✓

3. Five Common Word Problem Errors

FIVE COMMON ERRORS — RECOGNISE AND AVOID

1. Error 1: Skipping SEE and UNDERSTAND Jumping to numbers before understanding the situation. Result: correct arithmetic applied to the wrong model.

2. Error 2: Unit inconsistency Mixing km/h with minutes without conversion. Result: a number with no valid physical meaning.

3. Error 3: Reverse percentage error Finding p% of the final price and adding it back when asked for the original.

4. Error 4: Wrong denominator in profit % Calculating profit as a % of SP instead of CP. Result: a smaller, incorrect percentage.

5. Error 5: Omitting VERIFY Accepting an answer without checking. Correct or not, the habit leads to undetected errors.

4. Mastery Lab — Chapter Thirteen

MASTERY LAB · CHAPTER 13 · LEVEL 1

★☆☆☆☆ — Step Identification (SEE and UNDERSTAND only)

Ch. 13

Complete Steps 1 and 2 only. Do not calculate.

1. A car travels at 60 km/h for 2.5 hours. How far?

SEE (situation): ___

_____

GIVEN: ___ *WANTED:* ___ *Technique:* ___

_____

2. A shirt costs ₹800 after a 20% discount. What was the original price?

SEE (situation): ___

_____

GIVEN: ___ *WANTED:* ___ *Technique:* ___

_____

3. A and B together finish a job in 4 h. A alone takes 6 h. How long for B alone?

SEE (situation): ___

_____

GIVEN: ___ *WANTED:* ___ *Technique:* ___

_____

MASTERY LAB · CHAPTER 13 · LEVEL 2

★★☆☆☆ — Full Four-Step Solutions

Ch. 13

Apply all four steps explicitly. Label each.

Problem 1: Cyclist travels 45 km at 18 km/h, rests 30 min, then 36 km at 12 km/h. Total time?

SEE:

_____

UNDERSTAND:

SOLVE:

VERIFY:

Problem 2: ₹2,100 covers 35% of monthly expenses. Find total monthly expenses.

SEE:

UNDERSTAND:

SOLVE:

VERIFY:

MASTERY LAB · CHAPTER 13 · LEVEL 3

★★★☆☆ — Multi-Concept Problems

Ch. 13

Each problem requires two or more concepts. Identify all techniques before solving.

1. **A grocer buys 200 kg of apples at ₹30/kg. 10% are spoiled. He sells the rest at ₹45/kg. Profit or loss %?**

   Techniques needed: ___ Sellable qty: ___ Revenue: ___ CP: ___ Profit/Loss %: ___

2. **Two pipes fill a tank in 6 and 8 hours. A drain empties it in 12 hours. All three open: time to fill?**

Rates: 1/6 + 1/8 − 1/12 = ___ Time to fill: ___

MASTERY LAB · CHAPTER 13 · LEVEL 4

★★★★☆ — Error Detection

Ch. 13

Each solution below contains one of the five common errors. Identify the error and write the correct solution.

PROBLEM: After a 25% discount, a TV costs ₹18,000. Find original price.

Student solution: 25% of 18,000 = 4,500. Original = 18,000 + 4,500 = ₹22,500.

Error type: ___ Correct solution: Original = 18000 ÷ ___ = ___

PROBLEM: CP = ₹1,500. SP = ₹2,000. Profit %?

Student solution: Profit = 500. 500 ÷ 2,000 × 100 = 25%.

Error type: ___ Correct denominator: ___ Correct profit %: ___

MASTERY LAB · CHAPTER 13 · LEVEL 5

★★★★★ — The Full Problem Set — Advanced

Ch. 13

Three-step minimum. VERIFY must use a different method from SOLVE. Include one sentence explaining your technique choice.

A train and a car start from opposite ends of a 600 km route at 7 AM. Train at 90 km/h, car at 60 km/h. At what time do they meet, and how far from the train's starting point?

SEE:

UNDERSTAND:

SOLVE (step by step):

VERIFY (different method):

Why I chose this technique:

♦ ♦ ♦

MENTOR'S SANCTUARY · Chapter Thirteen

This Chapter's Purpose

Chapter Thirteen is the most important chapter in Stage Six. The Four-Step Method is not a procedure specific to word problems — it is a general framework for structured reasoning under any form of mathematical ambiguity. A student who genuinely internalises SEE-UNDERSTAND-SOLVE-VERIFY will apply it to every problem they encounter for the rest of their mathematical life.

The Tradition on Teaching This

The Vedic tradition does not proceed to calculation until the object of calculation is clearly seen. The step "SEE" is not a pedagogical convenience — it is a statement about how mathematical thinking actually

works. A problem is not understood because it has been read. It is understood when the student can describe the situation accurately in their own words, without numbers. That is the threshold.

Practical Coaching Note

The most valuable investment in Chapter Thirteen is in Step 2: making the student state GIVEN and WANTED explicitly before any calculation. Students who resist this (finding it "obvious") are the students most likely to solve the wrong problem. Require it every time, without exception, until it becomes reflex.

Diagnostic Flowchart — Chapter Thirteen

If student skips SEE or UNDERSTAND →

Ask: "What is the situation? Tell me in your own words, without numbers." If they cannot, the problem is not yet understood. Do not permit Step 3 until Step 2 is complete.

If student's VERIFY fails →

Do not correct the arithmetic. Ask: "What does the VERIFY tell you?" The student must identify that an error exists. Finding it by returning to UNDERSTAND is the most valuable learning event in the chapter.

If student completes Level 5 →

They are ready for Chapter Fourteen's algebraic word problems. The Four-Step Method applies identically to algebra: SEE the structure, UNDERSTAND what the unknown represents, SOLVE by transposition, VERIFY by substitution.

♦ ♦ ♦

COMING NEXT · Chapter Fourteen — The Bridge to Algebra

"The student who has reached Chapter Fourteen has been doing algebra since Chapter Five. The only thing that has changed is the name." Every technique in this book is an algebraic identity. Chapter Fourteen names them, shows the algebraic form, and crosses the bridge from arithmetic to formal algebra.

STAGE SIX · THE BRIDGE · CHAPTER FOURTEEN

Chapter Fourteen

The Bridge to Algebra

Calanā-Kalanābhyām · Every Technique Unmasked · Area Model Proofs

"The student who has reached Chapter Fourteen has been doing algebra since Chapter Five. The only thing that has changed is the name." — GaṇitSūtram · Stage Six

In 499 CE, Āryabhaṭa completed the Āryabhaṭīyā — a text of 121 verses covering arithmetic, algebra, plane and spherical trigonometry, and the theory of continued fractions. The verses are so compressed that each is essentially an identity: a general statement that can be instantiated by any specific numbers. Āryabhaṭa did not write arithmetic. He wrote algebra.

The student who has worked through this book from Chapter One has been doing algebra throughout. The Base Method used $(a+p)(a+q) = a(a+p+q) + pq$. The Sub-Base Method used $(kB+p)(kB+q) = k^2B^2 + kB(p+q) + pq$. The squaring patterns used $(a+p)^2 = a(a+2p) + p^2$ and $(a+p)(a-p) = a^2 - p^2$. The crosswise method used $(10A+B)(10C+D) = 100AC + 10(AD+BC) + BD$. The techniques were arithmetic. The structures were algebraic.

Chapter Fourteen names the structures. It introduces algebraic notation as the language for statements that are true for all values of the variables, not just specific numbers. It shows that every major technique in this book is an algebraic identity. And it teaches the arithmetic of unknowns — solving linear equations by transposition, and simultaneous equations by addition and subtraction. The bridge is crossed here.

SŪTRA · 9

चलना-कलनाभ्याम् · *Calanā-Kalanābhyām*

"By sequential motion; by differential calculus"

Governs successive differences of polynomial sequences — the discrete analogue of differentiation. Also governs the algebraic bridge: every arithmetic technique seen sequentially (chapter by chapter) now reveals its algebraic identity simultaneously.

1. What Algebra Is

Algebra is arithmetic with unknown quantities. Where arithmetic asks "what is 97 × 103?", algebra asks "what is (a − 3)(a + 3) for any value of a?" The answer — $a^2 - 9$ — is true for every value of a simultaneously. That generality is algebra's distinguishing feature.

The student who has used the difference of squares pattern to compute $97 \times 103 = 100^2 - 3^2 = 9991$ has already used the identity $(a - 3)(a + 3) = a^2 - 9$ with $a = 100$. The algebra was present in the arithmetic. Only the notation was missing.

ARITHMETIC	BRIDGE	ALGEBRA
one specific result	**set a = 100**	*true for all a*

2. Every Technique Unmasked

The following table shows the algebraic identity behind every major technique in this book. Study it. Then close the book and reproduce it from memory. When you can do that, the bridge is crossed.

TECHNIQUE	ALGEBRAIC IDENTITY	SŪTRA	EXAMPLE
Base Method	$(a+p)(a+q) = a(a+p+q) + pq$	*Nīkhilaṃ*	97×98 → a=100, p=−3, q=−2
Sub-Base	$(kB+p)(kB+q) = k^2B^2 + kB(p+q) + pq$	*Ānurūpyeṇa*	48×47 → k=½, B=100
Sq. ending in 5	$(10n+5)^2 = 100n(n+1) + 25$	*Ekādhikena Pūrveṇa*	75^2 → n=7, 7×8|25=5625
Yāvadūnam sq.	$(a+p)^2 = a(a+2p) + p^2$	*Yāvadūnam*	97^2 → a=100, p=−3
Diff. of squares	$(a+p)(a-p) = a^2 - p^2$	*Yāvadūnam*	97×103 → a=100, p=3
Crosswise 2D	$(10A+B)(10C+D) = 100AC + 10(AD+BC) + BD$	*Ūrdhva-Tiryak*	87×56 → A=8,B=7,C=5,D=6
Linear eq.	$ax + b = c \rightarrow x = (c-b)/a$	*Parāvartya*	4x+7=31 → x=6
Simultaneous	add: eliminate one var; subtract: eliminate other	*Saṅkalāna-Vyavakal.*	2x+y=11, x+2y=10

3. Solving Linear Equations — Parāvartya in Algebraic Form

The word *equation* is the algebraic form of the word problem. The Four-Step Method applies directly: SEE the structure, UNDERSTAND what the unknown is, SOLVE by transposition, VERIFY by substitution.

TRANSPOSITION RULE (Parāvartya)

A term moved from one side of an equation to the other changes sign.

Multiplication on one side becomes division on the other.

This is Parāvartya: transpose and apply. The same sūtra that governed D = S×T governs ax + b = c.

WORKED EXAMPLE — Linear equation

3x + 12 = 27

3x = 27 − 12	Transposed +12; sign changed to −12	3x = 15
x = 15 ÷ 3	Transposed ×3; becomes ÷3	x = 5
	3×5 + 12 = 15 + 12 = 27 ✓	

WORKED EXAMPLE — Linear equation (two sides)

5x − 8 = 3x + 14

5x − 3x = 14 + 8	Transpose 3x and −8; signs change	2x = 22
x = 22 ÷ 2		x = 11
	5×11−8 = 47. 3×11+14 = 47 ✓	

WORKED EXAMPLE — Equation with fractions

(x + 3)/4 = 7

x + 3 = 28	Transposed ÷4; becomes ×4	x + 3 = 28
x = 28 − 3		x = 25
	(25+3)/4 = 28/4 = 7 ✓	

4. Area Model Proofs — Geometry as Algebra

Every algebraic identity has a geometric proof using rectangle areas. These proofs follow the Sulbasūtra tradition described in Chapter Four: understand through construction, not through symbol manipulation alone.

THE FOUR AREA MODEL IDENTITIES

1. $(a+b)^2 = a^2 + 2ab + b^2$ — *Square of side (a+b) divided into four regions: a^2, ab, ab, b^2.*
2. $(a-b)^2 = a^2 - 2ab + b^2$ — *Square of side a with L-shaped region removed. L-shape = $2ab - b^2$.*
3. $(a+b)(a-b) = a^2 - b^2$ — *L-shape rearranged into a rectangle of dimensions (a+b) × (a−b).*
4. $(10A+B)(10C+D) = 100AC + 10(AD+BC) + BD$ — *Rectangle divided into four regions: AC, AD, BC, BD (scaled by positional values).*

❖ *"Behōld! Every identity is a statement about area that requires no algebraic proof — only clear sight." — Bhāskara II, Līlāvatī, ~1150 CE*

5. Simultaneous Equations — Saṅkalāna-Vyavakalānabyām

When two linear equations have two unknowns, the sūtra Saṅkalāna-Vyavakalānabyām — by addition and subtraction — provides the elimination method: add the equations to eliminate one variable, subtract to eliminate the other.

WORKED EXAMPLE — Simultaneous equations

$2x + y = 11$ and $x + 2y = 10$

Add	$(2x+y) + (x+2y) = 11+10 \rightarrow 3x+3y=21 \rightarrow x+y=7$	**$x + y = 7$**
Subtract	$(2x+y) - (x+2y) = 11-10 \rightarrow x-y=1$	**$x - y = 1$**
Solve	$x+y=7$ and $x-y=1$: add → $2x=8$ → $x=4$	**$x=4, y=3$**
	$2(4)+3=11$ ✓ $4+2(3)=10$ ✓	

6. Calanā-Kalanābhyām — Successive Differences

The sūtra Calanā-Kalanābhyām governs one final insight: for a polynomial of degree n, the nth-order differences of the sequence are constant. This is the discrete analogue of differentiation — the Vedic anticipation of calculus.

SUCCESSIVE DIFFERENCE ANALYSIS

Sequence of squares: 1, 4, 9, 16, 25, 36

First differences: 3, 5, 7, 9, 11 (odd numbers)

Second differences: 2, 2, 2, 2 (constant → quadratic sequence)

Sequence of cubes: 1, 8, 27, 64, 125

1st differences: 7, 19, 37, 61

2nd differences: 12, 18, 24

3rd differences: 6, 6 (constant = 3! → cubic sequence)

The constant difference equals n! for a sequence generated by a pure nth-degree polynomial. This is what Newton and Leibniz formalised as differentiation in the 17th century. The sūtra had the observation fifteen centuries earlier.

7. Mastery Lab — Chapter Fourteen

MASTERY LAB · CHAPTER 14 · LEVEL 1

★☆☆☆☆ — Identity Recognition

Ch. 14

Match each calculation to its algebraic identity. Name the variables.

Ex 1 **97 × 103**	*Identity: ___ Variables: ___ Chapter: ___*
Ex 2 **(48)² [near 50]**	*Identity: ___ Variables: ___ Chapter: ___*
Ex 3	*Identity: ___ Variables: ___ Chapter: ___*

87 × 56 [crosswise]	_____
Ex 4 **48 × 47 [sub-base]**	*Identity:* ___ *Variables:* ___ *Chapter:* ___ _____
Ex 5 **35^2 [ends in 5]**	*Identity:* ___ *Variables:* ___ *Chapter:* ___ _____

MASTERY LAB · CHAPTER 14 · LEVEL 2

★★☆☆☆ — Linear Equation Solving

Ch. 14

Solve by transposition. Label every transposition step with "Parāvartya". Verify by substitution.

Ex 1 **$4x + 7 = 31$**	*Steps:* ___ *x* = ___ *Verify:* ___ _____
Ex 2 **$6x - 5 = 2x + 19$**	*Steps:* ___ *x* = ___ *Verify:* ___ _____
Ex 3 **$(x - 4)/3 = 8$**	*Steps:* ___ *x* = ___ *Verify:* ___ _____
Ex 4 **$3(x + 2) = 21$**	*Steps:* ___ *x* = ___ *Verify:* ___ _____
Ex 5 **$5x = 2x + 24$**	*Steps:* ___ *x* = ___ *Verify:* ___ _____

MASTERY LAB · CHAPTER 14 · LEVEL 3

★★★☆☆ — Area Model Drawing

Ch. 14

Draw the area model for each identity from memory. Label every region.

AREA MODEL DRAWINGS · Three identities from memory

1. $(a+b)^2 = a^2 + 2ab + b^2$

[Drawing space — draw a square of side (a+b) divided into four labelled regions]

2. $(a+b)(a-b) = a^2 - b^2$

[Drawing space — draw the L-shape rearrangement]

3. $(10A+B)(10C+D)$: the crosswise four regions

[Drawing space — draw the rectangle with four labelled sub-rectangles]

MASTERY LAB · CHAPTER 14 · LEVEL 4

★★★★☆ — Simultaneous Equations

Ch. 14

Apply the addition-subtraction method. Show every step.

Ex 1

$x + y = 15$ and $x - y = 3$

Add:___ Subtract:___ x=___ y=___ Verify:___

Ex 2

$2x + y = 11$ and $x + 2y = 10$

Add:___ Subtract:___ x=___ y=___ Verify:___

Ex 3 **$3x + 2y = 20$ and $x + 2y = 12$**	*Add:___ Subtract:___ x=___ y=___ Verify:___*
Ex 4 **$2x + 3y = 18$ and $2x - y = 6$**	*Add:___ Subtract:___ x=___ y=___ Verify:___*

MASTERY LAB · CHAPTER 14 · LEVEL 5

★★★★★ — The Algebraic Autobiography

Ch. 14

Write a one-page mathematical autobiography. Beginning from Chapter Five, trace every technique you have learned to its algebraic identity. For each: (a) technique name, (b) algebraic identity, (c) one numerical example, (d) the sūtra that governs it. Minimum six entries.

MY ALGEBRAIC AUTOBIOGRAPHY · At least six entries

Technique: ___ Identity: ___ Example: ___ Sūtra: ___

Technique: ___ Identity: ___ Example: ___ Sūtra: ___

Technique: ___ Identity: ___ Example: ___ Sūtra: ___

Technique: ___ Identity: ___ Example: ___ Sūtra: ___

Technique: ___ Identity: ___ Example: ___ Sūtra: ___

Technique: ___ Identity: ___ Example: ___ Sūtra: ___

♦ ♦ ♦

MENTOR'S SANCTUARY · Chapter Fourteen

This Chapter's Purpose

Chapter Fourteen is the intellectual culmination of this book. A student who arrives here having genuinely engaged with every prior chapter will experience something rare: the moment when arithmetic reveals itself as algebra, and every technique they have learned is recognised as an identity they have been using implicitly. This recognition — not any specific algebraic skill — is the primary goal of Chapter Fourteen.

The Tradition on Teaching This

Calanā-Kalanābhyām — by sequential motion — governs the successive differences that reveal polynomial structure. But in Chapter Fourteen, it has a meta-level meaning: the student who has worked through this book chapter by chapter (sequentially) now sees all the techniques simultaneously — as a single algebraic structure. The book's sequential motion has built toward this moment of simultaneous recognition.

Practical Coaching Note

Do not rush Chapter Fourteen. The identity-recognition exercises in Level 1 are more important than the equation-solving in Level 2. A student who can identify which algebraic identity underlies a specific technique has crossed the bridge. That crossing is irreversible.

Diagnostic Flowchart — Chapter Fourteen

If student is surprised they have been "doing algebra" since Chapter Five →

Name it explicitly: "Every technique in this book has an algebraic identity. You have been using them without the algebraic notation. You are not a beginner at algebra — you are an advanced practitioner who now has the formal language to match."

If student finds linear equation solving easy →

This is expected for a student who has completed Chapter Thirteen. The Four-Step Method is the equation-solving method in disguise. Name the connection.

If student cannot draw the area model from memory →

Return to Chapter Seven's geometric proofs. The area models are the same; only the algebraic labelling has been added.

If student completes the Algebraic Autobiography →

Review it carefully. Every entry is a window into how the student has integrated the material. This is the most complete demonstration of learning in the book.

♦ ♦ ♦

★ STAGES FIVE AND SIX COMPLETE ★ · You have crossed the bridge.

◆
◆
◆
◆
◆
◆
◆

One chapter remains: Chapter Fifteen — The Mastery Drill.

GATE REVIEW · PART Four

The Universal Engine: Real-Life Mathematics and the Bridge to Algebra

Before proceeding, confirm all of the following:

- ☐ The student can compute any percentage mentally using anchor building blocks.
- ☐ The student can solve all four percentage problem types: find, increase, discount, reverse.
- ☐ The student can apply all three forms of D=S×T and convert between km/h, m/s, minutes, and hours.
- ☐ The student can solve relative speed problems (toward each other and same direction).
- ☐ The student can verify a multi-line bill, compute unit rates, and calculate profit/loss %.
- ☐ The student applies all four steps of SEE-UNDERSTAND-SOLVE-VERIFY to every word problem.
- ☐ The student can solve any linear equation by transposition and verify by substitution.
- ☐ The student can identify the algebraic identity underlying at least four Vedic techniques.
- ☐ The student can draw the area model proof for at least two identities from memory.
- ☐ The student has completed the Algebraic Autobiography (Chapter Fourteen, Level 5).

When all conditions are met: Part Four is complete.

STAGE SEVEN · MASTERY AND EXCELLENCE · CHAPTER FIFTEEN

Chapter Fifteen

The Mastery Drill

100 Questions · Seven Stages · One Clear Mind

"One hundred questions. One clear mind.
That is all that is required." — GaṇitSūtram · Stage Seven

★ STAGE SEVEN ★

Mastery and Excellence

One hundred questions · Thirty days · The tools you keep forever

In 499 CE, Āryabhaṭa completed the Āryabhaṭīyā at the age of twenty-three. One hundred and eighteen verses. Astronomy, arithmetic, algebra, trigonometry — each compressed to the density of a seed. The great learning centres of Nālandā and Takṣaśilā did not consider a student complete until they could demonstrate fluency across the entire range of their studies. Not individual techniques, but the integration of all techniques. The final examination was not a test of memory. It was a test of understanding.

This chapter is that examination. One hundred questions, arranged in rising complexity, drawn from every stage of this book. They are not arranged by technique but by difficulty and type: foundations first, then base methods, then the interleaved technique questions that require method selection, then

applied real-life problems, then word problems, and finally the questions that invert the learning relationship entirely — where you become the author of the mathematics.

This is not a test. No one is grading it. It is a celebration — a structured act of self-discovery. When you reach Question 100 and count the questions you answered with full confidence, that number will be larger than the number you would have predicted when you opened this book for the first time. That difference is what GaṇitSūtram was designed to create.

BEFORE YOU BEGIN

This chapter is not a test. It is a celebration. One hundred questions drawn from every chapter of GaṇitSūtram, arranged in rising complexity.

Your task is not to finish quickly. Your task is to approach each question with the same calm, pattern-aware attention this entire book has been building. Work at your own pace. **Verify each answer before moving to the next.**

After the drill, count the questions you answered with full confidence. That number will be larger than the number you would have expected when you opened this book for the first time.

⚠ Complete solutions are in Appendix C. Do not consult Appendix C during the drill. Use it afterwards — carefully, to examine every question you found difficult.

"Pencils down until you have read the entire question. Clarity over Speed. Always."

WHY INTERLEAVED RETRIEVAL PRACTICE WORKS

The most reliably effective learning technique identified in cognitive science over the past four decades is ***retrieval practice***: the deliberate act of recalling information from memory rather than re-reading it. ***Interleaved practice*** — mixing problems from different topics —

adds a second benefit: the learner must identify which method applies, not just execute a known procedure. That identification step is where durable mastery is built.

Questions 1–20 activate Stage One skills. Questions 21–40 activate Stage Three. Questions 41–70 interleave Stages Three and Four, requiring method selection. Questions 71–90 require interpretation before calculation. Questions 91–100 invert the learning relationship entirely: the student becomes the author.

❖ *Heritage note: the great learning centres of Nālandā and Takṣaśilā held rigorous comprehensive examinations. A student was not considered complete until they could demonstrate fluency across the full range of their studies. The Mastery Drill stands in this tradition.*

Q 1–20

Number Sense · Estimation · Complements · Patterns · Write only the answer

1) **Round 347 to the nearest hundred.**

Answer: _____

2) **36 + 47 using friendly numbers. Show your anchor.**

Anchor: ___ *Answer:* ___

3) **Estimate 503 × 4 to the nearest hundred.**

Answer: _____

4) **Complement of 73 to 100.**

Answer: _____

5) **Next term: 2, 5, 8, 11, 14, ___**

Answer: _____

6) **89 + 43**

Answer: _____

7) **500 – 287**

Answer: _____

8) **Estimate 198 × 6 (round 198 first).**

Answer: _____

9) **10% of 840**

Answer: _____

10) **25% of 360**

Answer: _____

11) **82 + 29 − 14**

Answer: _____

12) **600 − 383**

Answer: _____

13) **Next term: 100, 91, 82, 73, ___**

Answer: _____

14) **50% of 740**

Answer: _____

15) **75% of 240**

Answer: _____

16) **Round 4,867 to the nearest thousand.**

Answer: _____

17) **Estimate 996 × 4 (round 996 first).**

Answer: _____

18) **15% of 200**

Answer: _____

19) **Complement of 64 to 100.**

Answer: _____

20) **95 + 48**

Answer: _____

Q 21–40

Show gaps, cross-addition, and verify every answer

21) **97 × 98**

Gaps: ___ ___ *Cross:* ___ *Right:* ___ *Answer:* _____ ✓

22) **103 × 104**

Gaps: ___ ___ *Cross:* ___ *Right:* ___ *Answer:* _____ ✓

23) **96 × 94**

Gaps: ___ ___ *Cross:* ___ *Right:* ___ *Answer:* _____ ✓

24) **102 × 98**

Gaps: ___ ___ *Cross:* ___ *Right:* ___ *Answer:* _____ ✓ *(mixed signs)*

25) **108 × 103**

Gaps: ___ ___ *Cross:* ___ *Right:* ___ *Answer:* _____ ✓

26) **9 × 8 (Base 10)**

Gaps: ___ ___ *Cross:* ___ *Right:* ___ *Answer:* _____ ✓

27) **7 × 8 (Base 10)**

Gaps: ___ ___ *Cross:* ___ *Right:* ___ *Answer:* _____ ✓

28) **997 × 998**

Gaps: ___ ___ *Cross:* ___ *Right:* ___ *Answer:* _____ ✓

29) **993 × 996**

Gaps: ___ ___ *Cross:* ___ *Right:* ___ *Answer:* _____ ✓

30) **1003 × 1004**

Gaps: ___ ___ *Cross:* ___ *Right:* ___ *Answer:* _____ ✓

31) **105 × 96 (mixed: one above, one below)**

Gaps: ___ ___ *Cross:* ___ *Right:* ___ *Answer:* _____ ✓

32) **112 × 94**

Gaps: ___ ___ *Cross:* ___ *Right:* ___ *Answer:* _____ ✓

33) **98 × 104**

Gaps: ___ ___ *Cross:* ___ *Right:* ___ *Answer:* _____ ✓

34) **997 × 1003**

Gaps: ___ ___ *Cross:* ___ *Right:* ___ *Answer:* _____ ✓

35) **88 × 92**

Gaps: ___ ___ *Cross:* ___ *Right:* ___ *Answer:* _____ ✓

36) **999 × 1002**

Gaps: ___ ___ *Cross:* ___ *Right:* ___ *Answer:* _____ ✓

37) **89 × 93**

Gaps: ___ ___ *Cross:* ___ *Right:* ___ *Answer:* _____ ✓

38) **107 × 96**

Gaps: ___ ___ *Cross:* ___ *Right:* ___ *Answer:* _____ ✓

39) **99 × 97**

Gaps: ___ ___ *Cross:* ___ *Right:* ___ *Answer:* _____ ✓

40) **1001 × 998**

Gaps: ___ ___ *Cross:* ___ *Right:* ___ *Answer:* _____ ✓

Q 41–55

Ānurūpyeṇa · Yāvadūnam · Ekādhikena · State your method choice for every question

41) **52 × 53**

Method: _____ *Answer:* _____

42) **48 × 49**

Method: _____ *Answer:* _____

43) **26 × 24**

Method: _____ *Answer:* _____

44) **204 × 203**

Method: _____ *Answer:* _____

45) **65^2**

Method: _____ *Answer:* _____

46) **75^2**

Method: _____ *Answer:* _____

47) **45^2**

Method: _____ *Answer:* _____

48) **98^2**

Method: _____ *Answer:* _____

49) **102^2**

Method: _____ *Answer:* _____

50) **51^2**

Method: _____ *Answer:* _____

51) **27 × 23**

Method: _____ *Answer:* _____

52) **49^2**

Method: _____ *Answer:* _____

53) **201 × 198**

Method: _____ *Answer:* _____

54) **55^2**

Method: _____ *Answer:* _____

55) **103 × 97**

Method: _____ *Answer:* _____

Q

Two-digit: show R, M, L. Three-digit: show P5 through P1. Verify all.

56–70

56) **23 × 41**

R: ___ *M:* ___ *L:* ___ *Answer:* _____ ✓

57) **34 × 52**

R: ___ *M:* ___ *L:* ___ *Answer:* _____ ✓

58) **47 × 63**

R: ___ *M:* ___ *L:* ___ *Answer:* _____ ✓

59) **56 × 38**

R: ___ *M:* ___ *L:* ___ *Answer:* _____ ✓

60) **78 × 45**

R: ___ *M:* ___ *L:* ___ *Answer:* _____ ✓

61) **67 × 89**

R: ___ *M:* ___ *L:* ___ *Answer:* _____ ✓

62) **93 × 47**

R: ___ *M:* ___ *L:* ___ *Answer:* _____ ✓

63) **85 × 64**

R: ___ *M:* ___ *L:* ___ *Answer:* _____ ✓

64) **76 × 89**

R: ___ *M:* ___ *L:* ___ *Answer:* _____ ✓

65) **99 × 88**

R: ___ *M:* ___ *L:* ___ *Answer:* _____ ✓

66) **123 × 234**

P5: ___ *P4:* ___ *P3:* ___ *P2:* ___ *P1:* ___ *Answer:* _____ ✓

67) **312 × 213**

P5: ___ *P4:* ___ *P3:* ___ *P2:* ___ *P1:* ___ *Answer:* _____ ✓

68) **143 × 214**

	P5: ___ P4: ___ P3: ___ P2: ___ P1: ___ Answer: _____ ✓
69)	**321 × 132** *P5: ___ P4: ___ P3: ___ P2: ___ P1: ___ Answer: _____ ✓*
70)	**234 × 143** *P5: ___ P4: ___ P3: ___ P2: ___ P1: ___ Answer: _____ ✓*

Q 71–85

Percentages · Distance-Speed-Time · Bills and Budgets · Show working. State units. Verify each.

71)	**35% of ₹480** *₹_____ Verify: ✓*
72)	**₹750 with 20% discount — final price?** *₹_____ Verify: ✓*
73)	**₹360 + 18% GST — total?** *₹_____ Verify: ✓*
74)	**240 km at 60 km/h — time?** *___ hours Verify: ✓*
75)	**₹1,500 shared by 3 people; one pays double the others. How much each?** *Pays double: ₹___ Others: ₹___ Verify: ✓*
76)	**500g for ₹48 vs 750g for ₹69 — which is better value per gram?** *Option A: ₹___/g Option B: ₹___/g Better: ___*
77)	**15% discount on ₹890 — final price?** *₹_____ Verify: ✓*
78)	**5% service charge on ₹640 bill — charge amount?** *₹_____ Verify: ✓*
79)	**180 km at 90 km/h — time?**

___ hours Verify: ✓

80) **Final price ₹560 after 20% discount — find original price.**

₹_____ Verify: ✓

81) **Income ₹35,000: 40% rent, 25% food, 10% savings — remainder in rupees?**

₹_____ Verify: ✓

82) **Two trains 280 km apart, approaching at 60 and 80 km/h. When do they meet?**

___ hours Verify: ✓

83) **Items ₹289 + ₹198 + ₹347 with 5% tax on subtotal. Total?**

₹_____ Verify: ✓

84) **8% of 1,250**

_____ Verify: ✓

85) **120 km at 60 km/h, then 90 km at 45 km/h. Total time?**

___ hours Verify: ✓

Q 86–90

Write GIVEN, WANTED, SOLVE, and VERIFY for each. Do not skip steps.

86) **A factory produces 840 units per day and sells 75%, storing the rest. After 5 days, how many units are stored?**

GIVEN: ___ *WANTED:* ___ *SOLVE:* ___ *Answer:* ___ *VERIFY:* ___

87) **A car travels 360 km. First 120 km at 40 km/h, then 240 km at 60 km/h. Total travel time?**

Segment 1: ___ *Segment 2:* ___ *Total:* ___ *VERIFY:* ___

88) **15 buses, 48 seats each, 72% occupied. How many empty seats in total?**

Total seats: ___ Occupied: ___ Empty: ___ VERIFY: ___

89) **A store has ₹1,24,500 sales. Costs = 40% of sales. GST = 18% of sales. What remains after costs and GST?**

Costs: ___ GST: ___ Remaining: ___ VERIFY: ___

90) **Three siblings share ₹6,720: eldest 40%, middle 35%, youngest the rest. How much each?**

Eldest: ___ Middle: ___ Youngest: ___ Total check: ___

Q 91–100

Compose and solve two problems for each technique. Write them clearly enough that a study partner could solve them.

91–92 — Base Method (Nīkhilaṃ)

Problem A:

Answer: ___ Verify: ✓

Problem B:

Answer: ___ Verify: ✓

93–94 — Sub-Base or Squaring

Problem A:

Answer: ___ Verify: ✓

Problem B:

Answer: ___ Verify: ✓

95–96 — Crosswise Multiplication

Problem A:

Answer: ___ Verify: ✓

Problem B:

Answer: ___ Verify: ✓

97–98 — Percentage or Distance-Speed-Time

Problem A:

Answer: ___ Verify: ✓

Problem B:

Answer: ___ Verify: ✓

99–100 — Word Problem (Four-Step Method)

Problem A:

Answer: ___ Verify: ✓

Problem B:

Answer: ___ Verify: ✓

◆ ◆ ◆

You have completed the Mastery Drill.

Count the questions you answered with full confidence.

That number is larger than the number you would have expected

when you opened this book on the first page.

That difference is what GaṇitSūtram was designed to create.

♦ ♦ ♦

THE 30-DAY PROGRESS JOURNAL

Spaced Repetition · Memory Consolidation · The Rhythm of Mastery

HOW THE 30-DAY JOURNAL WORKS

The 30-Day Journal is a manually scheduled spaced repetition system. Spaced repetition is the most evidence-based method for building long-term memory. The principle: review material ***just before you would naturally forget it.*** Days 1–14 revisit techniques at short intervals. Days 15–21 consolidate crosswise at medium intervals. Days 22–29 integrate real-life applications. Day 30 is the culmination.

The ancient gurukula understood this rhythm intuitively. Students recited, practised, and returned to material in cycles. Modern neuroscience has measured why it works: retrieval at the point of near-forgetting strengthens memory traces more than retrieval at the point of easy recall.

Every review entry you make is a deliberate act of memory consolidation. The act of writing "one thing I found beautiful" is not decorative — it activates emotional encoding, which enhances long-term retention more than factual review alone.

Week One — Days 1–7: Foundations and Base Methods

DAY

1

Ch. 0 & 1

Mindset exercises and pattern spotting. Read Chapter Zero's Stillness Protocol aloud. Then complete 10 gap-spotting exercises from Chapter One.

❖ **One thing I found CLEAR today:**

❖ **What does it feel like to approach a number with curiosity rather than anxiety?**

❖ **One thing I found BEAUTIFUL about the numbers today:**

□ I felt the pattern clearly · □ Getting closer · □ I will return to this tomorrow

"Clarity over Speed. Always."

DAY

2

Chapter 2

Number sense — 10 friendly-number additions, 10 gap-based subtractions. Time yourself, but do not rush. Accuracy before speed.

❖ **One thing I found CLEAR today:**

❖ **One anchor that made addition easier today:**

❖ **One thing I found BEAUTIFUL about the numbers today:**

□ I felt the pattern clearly · □ Getting closer · □ I will return to this tomorrow

"Clarity over Speed. Always."

DAY

3

Chapter 5

Base 100 — 10 problems, both numbers below the base. Write all three steps: gaps, cross-add, right part.

❖ **One thing I found CLEAR today:**

❖ **One moment when the pattern felt obvious rather than calculated:**

❖ **One thing I found BEAUTIFUL about the numbers today:**

□ I felt the pattern clearly · □ Getting closer · □ I will return to this tomorrow

"Clarity over Speed. Always."

DAY

4

Chapter 5

Base 100 — 10 problems, both numbers above. Same format.

❖ **One thing I found CLEAR today:**

❖ **Did the cross-add feel automatic yet? Note where it still requires effort:**

❖ **One thing I found BEAUTIFUL about the numbers today:**

□ I felt the pattern clearly · □ Getting closer · □ I will return to this tomorrow

"Clarity over Speed. Always."

DAY

5

Chapter 5

Base 100 — 10 mixed-sign problems. Apply borrow-and-adjust when the right part is negative.

❖ **One thing I found CLEAR today:**

❖ **The mixed-sign case — in your own words, why does the borrow work?**

❖ **One thing I found BEAUTIFUL about the numbers today:**

□ I felt the pattern clearly · □ Getting closer · □ I will return to this tomorrow

"Clarity over Speed. Always."

DAY

6

Ch. 5 Extended

Base 10 and Base 1,000 — 5 of each. Note how the number of right-part digits changes with the base.

❖ **One thing I found CLEAR today:**

❖ **A pattern in how the base and right-part digits relate:**

❖ **One thing I found BEAUTIFUL about the numbers today:**

□ I felt the pattern clearly · □ Getting closer · □ I will return to this tomorrow

"Clarity over Speed. Always."

DAY

7

Week Review

Base method — 20 mixed problems. Examine any errors: what was the error type? Return to the Diagnostic Flowchart in the relevant Mentor's Sanctuary.

❖ **One thing I found CLEAR today:**

❖ **My most frequent error this week, and its source:**

❖ **One thing I found BEAUTIFUL about the numbers today:**

□ I felt the pattern clearly · □ Getting closer · □ I will return to this tomorrow

"Clarity over Speed. Always."

Week Two — Days 8–14: Sub-Base, Squaring, Crosswise Introduction

DAY

8

Chapter 6

Sub-Base 50 — 10 problems. State sub-base, ratio, and remainder rule aloud before each calculation.

❖ **One thing I found CLEAR today:**

❖ **The remainder rule in your own words:**

❖ **One thing I found BEAUTIFUL about the numbers today:**

□ I felt the pattern clearly · □ Getting closer · □ I will return to this tomorrow

"Clarity over Speed. Always."

DAY

9

Chapter 6

Sub-Base 200 and Sub-Base 25 — 5 of each. Use the decision tree to identify the right sub-base.

❖ **One thing I found CLEAR today:**

❖ **When is Sub-Base 25 preferable to direct crosswise? One sentence:**

❖ **One thing I found BEAUTIFUL about the numbers today:**

□ I felt the pattern clearly · □ Getting closer · □ I will return to this tomorrow

"Clarity over Speed. Always."

DAY

10

Chapter 7

Squares ending in 5 (Pattern One) — 10 problems. Write the identity step before the answer.

❖ **One thing I found CLEAR today:**

❖ **Say the Ekādhikena pattern aloud without looking at the book. Write it here:**

❖ **One thing I found BEAUTIFUL about the numbers today:**

□ I felt the pattern clearly · □ Getting closer · □ I will return to this tomorrow

"Clarity over Speed. Always."

DAY

11

Chapter 7

Yāvadūnam squaring — 8 problems near Base 100, 2 near Base 1,000.

❖ **One thing I found CLEAR today:**

❖ **Connection between Yāvadūnam and the Base Method — in one sentence:**

❖ **One thing I found BEAUTIFUL about the numbers today:**

□ I felt the pattern clearly · □ Getting closer · □ I will return to this tomorrow

"Clarity over Speed. Always."

DAY

Difference of squares — 10 problems. Identify the centre and gap before calculating.

❖ **One thing I found CLEAR today:**

12

Chapter 7

❖ **One multiplication you spotted as a difference of squares situation today:**

❖ **One thing I found BEAUTIFUL about the numbers today:**

□ I felt the pattern clearly · □ Getting closer · □ I will return to this tomorrow

"Clarity over Speed. Always."

DAY

13

Chapter 8

Two-digit crosswise — 10 problems. Name R, M, L before each answer.

❖ **One thing I found CLEAR today:**

❖ **One pair where M (the crosswise middle) was hard to track. What made it hard?**

❖ **One thing I found BEAUTIFUL about the numbers today:**

□ I felt the pattern clearly · □ Getting closer · □ I will return to this tomorrow

"Clarity over Speed. Always."

DAY

14

Week Review

20 problems mixing Sub-Base, Squaring, and Two-digit Crosswise. Use method labels.

❖ **One thing I found CLEAR today:**

❖ **My strongest method this week and my weakest — with reason:**

❖ **One thing I found BEAUTIFUL about the numbers today:**

□ I felt the pattern clearly · □ Getting closer · □ I will return to this tomorrow

"Clarity over Speed. Always."

Week Three — Days 15–21: Three-Digit Crosswise and Verification

DAY 15

Chapter 9

Name P5 through P1 from memory. Then: 5 three-digit crosswise problems, full working.

❖ **One thing I found CLEAR today:**

❖ **Can you name all five products without looking? Drill until yes.**

❖ **One thing I found BEAUTIFUL about the numbers today:**

□ I felt the pattern clearly · □ Getting closer · □ I will return to this tomorrow

"Clarity over Speed. Always."

DAY 16

Chapter 9

5 more three-digit problems. Apply Casting Out Nines after every answer.

❖ **One thing I found CLEAR today:**

❖ **One calculation where Casting Out Nines caught (or confirmed) your answer:**

❖ **One thing I found BEAUTIFUL about the numbers today:**

□ I felt the pattern clearly · □ Getting closer · □ I will return to this tomorrow

"Clarity over Speed. Always."

DAY 17

Chapter 9

Vinculum: convert 87, 96, 79, 98, 88 to base-complement form. Use each in one multiplication.

❖ **One thing I found CLEAR today:**

❖ **The advantage of Vinculum for high-digit numbers, in your own words:**

❖ **One thing I found BEAUTIFUL about the numbers today:**

□ I felt the pattern clearly · □ Getting closer · □ I will return to this tomorrow

"Clarity over Speed. Always."

DAY

18

Review 3

20 mixed three-digit crosswise problems. Apply verification to all twenty. Flag any errors.

❖ **One thing I found CLEAR today:**

❖ **How many passed Casting Out Nines on first attempt? Note the pattern of errors:**

❖ **One thing I found BEAUTIFUL about the numbers today:**

□ I felt the pattern clearly · □ Getting closer · □ I will return to this tomorrow

"Clarity over Speed. Always."

DAY

19

Ch. 3–4

The Sūtra Review. Read each of the sixteen sūtras and its domain. Write the domain without looking.

❖ **One thing I found CLEAR today:**

❖ **Which sūtra do you use most? Which do you find most elegant?**

❖ **One thing I found BEAUTIFUL about the numbers today:**

□ I felt the pattern clearly · □ Getting closer · □ I will return to this tomorrow

"Clarity over Speed. Always."

DAY

Heritage: re-read the Āryabhaṭa passage in Chapter Four. Write two things you did not notice on the first reading.

20

Ch. 3–4

❖ **One thing I found CLEAR today:**

❖ **What strikes you about Āryabhaṭa's precision at astronomical scale?**

❖ **One thing I found BEAUTIFUL about the numbers today:**

□ I felt the pattern clearly · □ Getting closer · □ I will return to this tomorrow

"Clarity over Speed. Always."

DAY

21

Week Review

Crosswise challenge: 5 two-digit pairs and 5 three-digit pairs selected by a friend or mentor. No preparation. Verify all.

❖ **One thing I found CLEAR today:**

❖ **Your confidence level with three-digit crosswise — on a scale of 1–10:**

❖ **One thing I found BEAUTIFUL about the numbers today:**

□ I felt the pattern clearly · □ Getting closer · □ I will return to this tomorrow

"Clarity over Speed. Always."

Week Four — Days 22–29: Real Life, Word Problems, Bridge

DAY

22

Chapter 10

Percentages: 5 anchor-fluency, 3 discount, 2 reverse. Use the strip model for the reverse problems.

❖ **One thing I found CLEAR today:**

❖ **One percentage you can now compute faster than a calculator:**

❖ **One thing I found BEAUTIFUL about the numbers today:**

□ I felt the pattern clearly · □ Getting closer · □ I will return to this tomorrow

"Clarity over Speed. Always."

DAY

23

Chapter 11

D–S–T: 5 problems, one of each type. Write units at every step.

❖ **One thing I found CLEAR today:**

❖ **One situation in daily life where you have used D=S×T without realising it:**

❖ **One thing I found BEAUTIFUL about the numbers today:**

□ I felt the pattern clearly · □ Getting closer · □ I will return to this tomorrow

"Clarity over Speed. Always."

DAY

24

Chapter 12

Bills: verify a real receipt or a sample bill. Unit rates: 3 comparison pairs. Profit%: 2 problems.

❖ **One thing I found CLEAR today:**

❖ **One error you found (or almost made) in a real bill this week:**

❖ **One thing I found BEAUTIFUL about the numbers today:**

□ I felt the pattern clearly · □ Getting closer · □ I will return to this tomorrow

"Clarity over Speed. Always."

DAY

Word problems: 5 problems using the full Four-Step Method. Do not skip VERIFY.

❖ **One thing I found CLEAR today:**

25

Chapter 13

❖ **One problem where VERIFY revealed an error. What was the error?**

❖ **One thing I found BEAUTIFUL about the numbers today:**

□ I felt the pattern clearly · □ Getting closer · □ I will return to this tomorrow

"Clarity over Speed. Always."

DAY

26

Chapter 14

Algebra: solve 5 linear equations. For each, name the identity or sūtra that governs it.

❖ **One thing I found CLEAR today:**

❖ **Which Vedic technique surprised you most when you saw its algebraic form?**

❖ **One thing I found BEAUTIFUL about the numbers today:**

□ I felt the pattern clearly · □ Getting closer · □ I will return to this tomorrow

"Clarity over Speed. Always."

DAY

27

Full Review

30 problems drawn from every chapter. Method labels required. Verify all.

❖ **One thing I found CLEAR today:**

❖ **Total correct on first attempt: ___. Compare to Day 7. The gap is your growth.**

❖ **One thing I found BEAUTIFUL about the numbers today:**

□ I felt the pattern clearly · □ Getting closer · □ I will return to this tomorrow

"Clarity over Speed. Always."

DAY

28

Creative Day

Design 5 original problems — one for each major technique — and exchange with a study partner or solve after a 24-hour gap.

❖ **One thing I found CLEAR today:**

❖ **Writing a good problem: what did you learn about the technique by writing it?**

❖ **One thing I found BEAUTIFUL about the numbers today:**

□ I felt the pattern clearly · □ Getting closer · □ I will return to this tomorrow

"Clarity over Speed. Always."

DAY

29

Chapter 14 Lv 5

Complete or revisit the Algebraic Autobiography. For each technique: identity and one example. Minimum six entries.

❖ **One thing I found CLEAR today:**

❖ **The technique whose algebraic form you find most beautiful:**

❖ **One thing I found BEAUTIFUL about the numbers today:**

□ I felt the pattern clearly · □ Getting closer · □ I will return to this tomorrow

"Clarity over Speed. Always."

DAY 30 · THE CULMINATION

Today's practice: return to the Mastery Drill. Not to redo it in full — but to read through every question and ask yourself: do I understand this? Could I solve this comfortably? Could I explain it to someone else?

For any question where the answer is no: open the relevant chapter, read the Mentor's Sanctuary, and do one example from the Mastery Lab. This is not remediation. It is maintenance. Every skill requires it.

The Reflection

Compare who you were on Day 1 of this journal to who you are today. Not in terms of speed — in terms of how you see numbers. When you see 97 × 103, do you see 9,991 or do you see a calculation to perform?

Write your answer below. There is no correct response. There is only your response.

"Clarity over Speed. Always."

♦ ♦ ♦

THE FOREVER TOOLS

Ten Sūtras as Portable Intelligence · Carry These Always

The following ten tools are distilled from the sixteen sūtras. They are presented not as definitions but as working descriptions — what each tool does, where it applies, and why it belongs in permanent mental inventory. A skilled practitioner reaches for these tools automatically, without consulting a reference. That automaticity is the goal.

♞ 1

The Gap Tool

Nīkhilaṃ Navataścaramaṃ Daśataḥ — *"All from 9, last from 10"*

Domain: Complement calculation. Base Method gaps. Any subtraction from a power of ten.

Found in: *Chapter 5 (Base Method) · Chapter 2 (Complements) · Chapter 9 (Vinculum)*

♞ 2

The Proportion Tool

Ānurūpyeṇa — *"Proportionately"*

Domain: Sub-base multiplication. Any calculation involving a ratio or scaling adjustment.

Found in: *Chapter 6 (Sub-Base Methods) · Chapter 10 (Percentages) · Chapter 12 (Unit Rates)*

♞ 3

The Pattern Tool

Ekādhikena Pūrveṇa — *"By one more than the previous one"*

Domain: Squaring numbers ending in 5. Sequences with constant first differences.

Found in: *Chapter 7 (Squaring Pattern One) · Chapter 1 (Pattern Engine)*

♞ 4

The Extension Tool

Yāvadūnam — *"Whatever the extent of the deficiency"*

Domain: Squaring near any base. Cubing and higher powers near a base.

Found in: *Chapter 7 (Squaring Patterns Two, Three) · Chapter 5 (Base Method connection)*

♞

The Crosswise Tool

5

Ūrdhva-Tiryagbhyām — *"Vertically and crosswise"*

Domain: General multiplication of any two numbers. The universal multiplication algorithm.

Found in: *Chapter 8 (Two-digit) · Chapter 9 (Three-digit) · Chapter 14 (Algebraic identity)*

♞ 6

The Difference Tool

Yāvadūnam (as difference of squares) — *"Whatever the extent — as symmetric gap"*

Domain: Multiplying two numbers equidistant from a convenient centre. $(a+p)(a-p) = a^2 - p^2$.

Found in: *Chapter 7 (Pattern Four) · Chapter 14 (Algebraic identity)*

♞ 7

The Transpose Tool

Parāvartya Yojayet — *"Transpose and apply"*

Domain: Any formula that can be read in multiple directions. Equation solving by transposition.

Found in: *Chapter 11 (D=S×T) · Chapter 13 (Word Problems) · Chapter 14 (Linear Equations)*

♞ 8

The Addition-Subtraction Tool

Saṅkalāna-Vyavakalānabyām — *"By addition and by subtraction"*

Domain: Eliminating one variable from simultaneous equations. Any problem where adding or subtracting two equations reveals structure.

Found in: *Chapter 14 (Simultaneous Equations)*

♞ 9

The Verification Tool

Guṇita-Samuccayaḥ — *"The product of the sum equals the sum of the products"*

Domain: Casting Out Nines. Any modular verification. Checking multiplication by digit sums.

Found in: *Chapter 9 (Casting Out Nines) · Chapter 14 (Modular arithmetic)*

♞ 10

The Difference Tool — Successive

Calanā-Kalanābhyām — *"By sequential motion; by differential calculus"*

Domain: Identifying the degree of a polynomial from constant nth-order differences. The bridge between arithmetic patterns and calculus.

Found in: *Chapter 1 (Pattern Engine) · Chapter 14 (Successive Differences)*

♦ ♦ ♦

THE BEAUTY INDEX

Moments Worth Revisiting · The passages, patterns, and proofs that are worth returning to.

Chapter 1	The Pattern Engine: the moment when a sequence of numbers reveals its hidden rule. The difference between randomness and structure is always a missing observation.
Chapter 2	9,997 × 9,993 = 99,900,021. Four gap digits, one multiplication, one answer. The power of the complement laid bare at large scale.
Chapter 3	The sixteen sūtras as a complete map. Sixteen descriptions of how the mind, when clear, naturally navigates mathematical structure.
Chapter 4	Āryabhaṭa's sidereal year: 365 days, 6 hours, 12 minutes, 30 seconds. The error from the modern value: less than four minutes. Fifth century CE.
Chapter 4	Bhāskara II's single word beneath his geometric proof: Behōld! No algebraic manipulation. Only clear sight.
Chapter 5	The cross-addition: (a+p) + q = (a+q) + p. Two paths to the same number. The symmetry of the Base Method made visible.
Chapter 6	The sub-base ratio k appears in the left part: divide by k to extract the value at base level. Proportion all the way down.
Chapter 7	$(10n+5)^2 = 100n(n+1) + 25$. Three numbers, one second, perfect.
Chapter 7	$(a+p)(a-p) = a^2 - p^2$. The moment you see $97 \times 103 = 100^2 - 9 = 9991$ in a single glance.
Chapter 8	The crosswise method has three partial products computed simultaneously. A GPU in arithmetic: parallel computation, 1958 style.
Chapter 8	(10A+B)(10C+D) = 100·AC + 10·(AD+BC) + BD. The student who sees this sees algebra.
Chapter 9	Casting Out Nines: because $10 \equiv 1 \pmod 9$, every multiplication can be verified from digit sums alone.
Chapter 9	Three-digit crosswise has five products. The middle product P3 contains three terms: A×F + B×E + C×D.
Chapter 10	Reverse percentage: the original is always the longer strip. Divide, never add back. The visual makes the algebraic structure concrete.

Chapter 11	Āryabhaṭa's sidereal year again — this time seen as a D=S×T calculation at astronomical scale.
Chapter 12	A unit rate comparison reveals that the larger package is sometimes worse value. Numbers do not lie.
Chapter 13	The Four-Step Method applied to a hard problem: the answer arrives not from calculation, but from seeing clearly.
Chapter 14	The moment every technique is unmasked as an algebraic identity. The arithmetic was always algebra.
Chapter 14	The successive differences of n^3: 1, 8, 27, 64. First differences: 7, 19, 37. Third differences: 6 = 3! Constant. This is what Newton called calculus.
The whole book	The student who opened with fear and closes with fluency. The distance between those two places. That is the beauty.

MY OWN BEAUTIFUL MOMENTS

Add moments from your own journey below.

_____

_____

_____

_____

_____

_____

_____

_____

_____

_____

♦ ♦ ♦

★ STAGE SEVEN COMPLETE ★

Mastery and Excellence

You have finished.

One hundred questions. Seven stages. Fifteen chapters. Thirty days of practice.

You have moved from fear to calmness, from calmness to foundation, from foundation to technique, from technique to mastery.

- You can multiply two- and three-digit numbers using the crosswise method.
- You can calculate percentages, distances, discounts, and ratios without a calculator.
- You can read a word problem and identify its structure before touching your pencil.
- You can look at 97 × 103 and see 9,991 before the pencil moves.
- You have recognised the algebra inside the arithmetic.

This is not memorisation.

You have understood your way here.

That is the difference. That is what cannot be taken away.

WHAT COMES NEXT

The doors to algebra, geometry, and higher mathematics stand open. The foundation is solid. Every technique in this book is a branch of something larger — a tradition of seeing patterns in the world that stretches back thousands of years and forward into everything you will ever study. The Mastery Drill was not the end. It was the proof that you are ready to begin.

★ ★ ★

★ ★ ★

CERTIFICATE OF COMPLETION

GaṇitSūtram · The Golden Edition

This is to certify that

_____

(name)

has completed all fifteen chapters of GaṇitSūtram · The Golden Edition,
including the Mastery Drill of one hundred questions,
the thirty-day Practice Journal,
and has crossed the Bridge to Algebra.

"Clarity over Speed. Always."

Date: _____

Mentor / Witness: _____

Jawahar R. Mallah · PrathamOne Press

★ ★ ★

★ ★ ★

A WORD FROM THE AUTHOR

You are holding the end of a book that I began writing with one question: what if a student who feared mathematics could be guided, gently and rigorously, to the place where that fear dissolves?

I did not begin with techniques. I began with a student I had seen — many students, really, combined into one — who sat with a pencil in their hand and a formula in their memory, and could do neither: could not apply the formula, could not feel the number, could not find their way in. They were not unintelligent. They were lost in a system that had never shown them what a number actually is.

The Vedic tradition gave me a different starting point. In the gurukula, a student did not begin with procedures. They began with observation. With stillness. With the question: *what do you notice?* The sixteen sūtras are not rules to be memorised — they are descriptions of

patterns that the mind, once shown, will always recognise. The moment a student sees 97×103 and *feels* 9,991 forming rather than calculates it, the fear is gone. Not because the fear was overcome, but because the number has become familiar. Familiar things are not frightening.

GaṇitSūtram was written as a guide from unfamiliarity to familiarity. The seven stages are seven steps of that journey. Stage One asks: can you be still? Can you see the pattern? Stage Seven asks: can you create the problem? The distance between those two questions is the whole of this book.

I have tried to write a book that is honest about difficulty. The three-digit crosswise method is difficult. The reverse percentage requires careful thought. Word problems require reading as well as arithmetic. I have tried not to flatten these difficulties, but to provide enough scaffolding that a patient student — or a patient mentor — can work through them without feeling that difficulty is a sign of failure. Difficulty is a sign that the mind is growing. That is all it is.

To the mentors who will use this book: your calm is the most powerful teaching tool in this room. I wrote that sentence in every Mentor's Sanctuary because I believe it completely. A student who is accompanied by a calm, patient adult will attempt problems that a frightened, pressured student will abandon. The mathematics is secondary. The presence is primary.

To the students who have reached this page: you began this book and you are finishing it. That is not a small thing. Many students begin. Finishing requires a different quality of attention — a willingness to stay with difficulty, to return to what is unclear, to verify rather than guess. You have demonstrated that quality. You will carry it into everything you study next.

The sixteen sūtras were not invented by Tirthaji. They were *noticed.* They are descriptions of how the mind, when it is clear and patient, naturally navigates a mathematical structure. They were waiting to be seen. So was every pattern in this book.

Thank you for seeing them.

Jawahar R. Mallah

PrathamOne Press

"Clarity over Speed. Always."

♦ ♦ ♦

GaṇitSūtram · The Golden Edition

PrathamOne Press · ISBN 978-93-5593-912-8